

# FM-7 (SD card version, Part 5, revised edition)

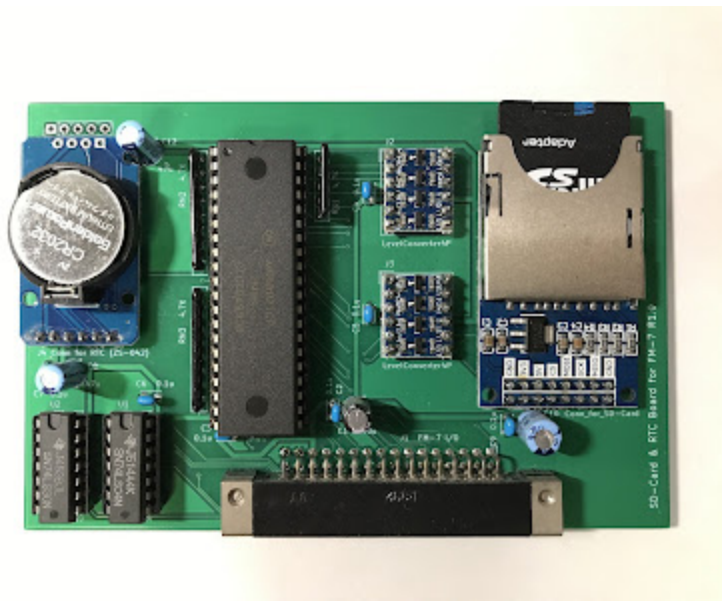
 [flexonsbd.blogspot.com/2023/05/fm-7rtcsdsd.html](https://flexonsbd.blogspot.com/2023/05/fm-7rtcsdsd.html)

Friday, May 19, 2023

## An attempt to connect an RTC and SD card to the FM-7 (SD card version, part 5, revised edition)

I thought SDRWFM was finished, but I found several bugs and have fixed them.

When using it myself, I unconsciously use it in a way that avoids any problems, so I had an acquaintance use it to check for any malfunctions.



This is the same image as last time.

(I've included the same image as last time, but it seems the icon won't display without an image, so I'm doing it to avoid that problem.)

The following three points need to be corrected:

(1) You can now use "-" (hyphen) in filenames.

When retrieving parameters, the code was using all characters less than 0x30 as delimiters, which meant that hyphens were also being treated as delimiters. Since F-BASIC includes hyphens in filenames, the code has been modified to exclude them from being used as delimiters.

(2) Fixed an issue where files saved as binary data using XM7 on Windows, copied to an SD card, and then read by an FM-7 could not be read correctly.

In BASIC programs saved as binary files, the save address is directly written as a link pointer. Therefore, even with the same FM-7, the save address of the BASIC program may differ depending on the device's specifications. If read directly, it may not be saved within the correct range. To solve this, we decided to use a routine provided in F-BASIC to correct the link pointer and read the program while converting the link pointer.

(3) When saving machine code files on the FM-7, a 5-byte footer was not added to the end. Although there was no practical inconvenience, we decided to add the footer for compatibility reasons.

Additionally, although not a correction, it was pointed out that when initializing the SD (and SDHC) card being used, the allocation size needs to be set to 32KB.

```

1      0001      *
2      0002      *   SDRWFM.TXT
3      0003      *
4      0004      *   SD Read/Write for FM-7
5      0005      *
6      0006      *       use RTC & SD Card
7      0007      *
8      0008      *       6821(PIA) Port A -- RTC (I2C Mode)
9      0009      *
10     0010      *           SBC           RTC           Ext IO
11     0011      *   CA2  2  --           --  HV4 LV4 --  I01
12     0012      *   PA0  3  --  SCL  --  HV1 LV1 --  I04
13     0013      *   PA1  4  --           --  HV2 LV2 --  I03
14     0014      *   PA7 10  --  SDA  --  HV3 LV3 --  I02
15     0015      *   GND      --           --           --  GND
16     0016      *           --           --           --  3V3
17     0017      *
18     0018      *
19     0019      *       6821(PIA) Port B -- SD (SPI Mode)
20     0020      *
twenty 0021      *           SBC           SD
one     0022      *   CB2 12  --  HV4 LV4 --  SCK
twenty 0023      *   PB0 13  --  HV1 LV1 --  CS
two     0024      *   PB1 14  --  HV2 LV2 --  MISO
twenty 0025      *   PB7 20  --  HV3 LV3 --  MOSI
three   0026      *   GND 31  --           --  GND
twenty 0027      *   VCC 33  --           --  VCC
four    0028      *
twenty 0029      *       cf. all data line use level converter
five    0030      *
26      0031      *       PB  output mode
27      0032      *       CB2 1bit output
28      0033      *
29      0034      *       READ/WRITE BUFFER  $5E00 - 5FFF
30      0035      *
31      0036      *       Ascii program readable '2023.04.18
32      0037      *       Fix date routine          '2023.04.12
33      0038      *       Implement write cmd      '2023.04.07

```

|    |      |      |                         |     |                                |
|----|------|------|-------------------------|-----|--------------------------------|
| 34 | 0039 | *    | Add command             |     | '2023.03.14                    |
| 35 | 0040 | *    | Change use Port B       |     | '2023.03.07                    |
| 36 | 0041 | *    | Original version        |     | '2022.06.11                    |
| 37 | 0042 | *    | Copyright by Y.Yamamoto |     |                                |
| 38 | 0043 | *    |                         |     |                                |
| 39 | 0044 | 6000 | STTADR                  | EQU | \$6000                         |
| 40 | 0045 |      |                         |     |                                |
| 41 | 0046 | FBB8 | INCHA                   | EQU | \$FBB8 for FM-7                |
| 42 | 0047 | FBBB | OUTCHA                  | EQU | \$FBBB for FM-7                |
| 43 | 0048 | FBF7 | CSR0N                   | EQU | \$FBF7 for FM-7                |
| 44 | 0049 | D807 | LININP                  | EQU | \$D807 for FM-7 (to \$043D- )  |
| 45 | 0050 |      | *DEVN0                  | EQU | \$BF for FM-7                  |
| 46 | 0051 | 043D | LINBUF                  | EQU | \$043D 1 line input buffer for |
| 47 |      |      |                         |     | F-BASIC                        |
| 48 | 0052 | 033C | TXTBUF                  | EQU | \$033C text buffer after       |
| 49 |      |      |                         |     | conversion for F-BASIC         |
| 50 | 0053 | 9162 | RDLIN0                  | EQU | \$9162 read line number for F- |
| 51 |      |      |                         |     | BASIC                          |
| 52 | 0054 | FBF1 | LININT                  | EQU | \$FBF1 (=C28C) one line        |
| 53 |      |      |                         |     | translation for F-BASIC        |
| 54 | 0055 | C730 | FIXPTR                  | EQU | \$C730 fix link pointer of     |
| 55 |      |      |                         |     | binary saved file for F-BASIC  |
| 56 | 0056 | 004B | LIN01                   | EQU | \$4B                           |
| 57 | 0057 | 033A | LIN02                   | EQU | \$33A                          |
| 58 | 0058 | 000D | CR                      | EQU | \$0D                           |
| 59 | 0059 | 000A | LF                      | EQU | \$0A                           |
| 60 | 0060 | 0020 | SPC                     | EQU | \$20                           |
| 61 | 0061 | 001B | ESC                     | EQU | \$1B                           |
| 62 | 0062 |      |                         |     |                                |
| 63 | 0063 | 5E00 | RWBUF                   | EQU | STTADR-\$200                   |
| 64 | 0064 |      |                         |     |                                |
| 65 | 0065 | 02FA | FMDAT                   | EQU | \$02FA date of FM-7            |
| 66 | 0066 | FD9E | PBDAT                   | EQU | \$FD9E for FM-7                |
| 67 | 0067 | FD9F | PBCTL                   | EQU | PBDAT+1                        |
| 68 | 0068 | 0033 | BASADP                  | EQU | \$33 (33,34) pointer for F-    |
| 69 |      |      |                         |     | BASIC                          |
| 70 | 0069 | 0035 | BASAEP                  | EQU | \$35 (35,36) pointer for F-    |
| 71 |      |      |                         |     | BASIC                          |
| 72 | 0070 | 2000 | TXTADR                  | EQU | \$2000 load adr for TEXT       |

|     |      |      |      |        |        |      |                              |
|-----|------|------|------|--------|--------|------|------------------------------|
| 73  | 0071 | 00E5 |      | DELMK  | EQU    | \$E5 | delete mark                  |
| 74  | 0072 |      |      |        |        |      |                              |
| 75  | 0073 | 6000 |      | ORG    | STTADR |      |                              |
| 76  | 0074 |      |      |        |        |      |                              |
| 77  | 0075 | 6000 | 16   | 00A0   | START  | LBRA | MAIN                         |
| 78  | 0076 | 6003 | 01   | VN     | FCB    | 1    | version no                   |
| 79  | 0077 | 6004 | 00   | SCTN0  | FCB    | 0    | top adrs of sct              |
| 80  | 0078 | 6005 | 00   | SCTN1  | FCB    | 0    |                              |
| 81  | 0079 | 6006 | 00   | SCTN2  | FCB    | 0    |                              |
| 82  | 0080 | 6007 | 00   | SCTN3  | FCB    | 0    |                              |
| 83  | 0081 | 6008 | 00   | RESDAT | FCB    | 0    | response data                |
| 84  | 0082 | 6009 | 00   | SDHC   | FCB    | 0    | sd:0,sdhc:1                  |
| 85  | 0083 |      |      |        |        |      |                              |
| 86  | 0084 | 600A | 0000 | SCTBPB | FDB    | 00   | sct no of BPB(4bytes)        |
| 87  | 0085 | 600C | 0000 |        | FDB    | 00   |                              |
| 88  | 0086 | 600E | 0000 | SCTFAT | FDB    | 00   | sct no of                    |
| 89  |      |      |      |        |        |      | FAT#1(4bytes)                |
| 90  | 0087 | 6010 | 0000 |        | FDB    | 00   |                              |
| 91  | 0088 | 6012 | 0000 | SCTDIR | FDB    | 00   | sct no of DIR(4bytes)        |
| 92  | 0089 | 6014 | 0000 |        | FDB    | 00   |                              |
| 93  | 0090 | 6016 | 0000 | SCTDSV | FDB    | 00   | save sct no of               |
| 94  |      |      |      |        |        |      | DIR(4bytes)                  |
| 95  | 0091 | 6018 | 0000 |        | FDB    | 00   |                              |
| 96  | 0092 | 601A | 0000 | SCTCLT | FDB    | 00   | sct no of top                |
| 97  |      |      |      |        |        |      | CLUSTER(4bytes)              |
| 98  | 0093 | 601C | 0000 |        | FDB    | 00   | if SD:#2, SDHC:#3            |
| 99  | 0094 | 601E | 00   | SCTPCL | FCB    | 0    | sct per cluster              |
| 100 | 0095 | 601F | FF   | DIRENT | RMB    | 32   | directory entry              |
| 101 | 0096 | 603F | FF   | WMDY   | RMB    | 3    | mon,day,year                 |
| 102 | 0097 | 6042 | FF   | WFSIZ  | RMB    | 4    | file size(4bytes)            |
| 103 | 0098 | 6046 | FF   | FILSIZ | RMB    | 2    | file size                    |
| 104 | 0099 | 6048 | 0000 | CLSTH  | FDB    | 00   | cluster High                 |
| 105 | 0100 | 604A | 0000 | CLSTL  | FDB    | 00   | cluster Low                  |
| 106 | 0101 | 604C | 00   | CNT    | FCB    | 0    |                              |
| 107 | 0102 | 604D | 00   | MCFLG  | FCB    | 0    | 0:machine code,              |
| 108 |      |      |      |        |        |      | \$FF,\$FE:binary save 1:else |
| 109 | 0103 | 604E | 00   | TPGFLG | FCB    | 0    | 1:top page 0:else            |
| 110 | 0104 | 604F | FF   | FNAME  | RMB    | 11   | file name(8+3)               |
| 111 |      |      |      |        |        |      |                              |

|       |                  |      |         |                      |     |       |                       |
|-------|------------------|------|---------|----------------------|-----|-------|-----------------------|
| 112   | 0105             | 605A | 00      | DPGNO                | FCB | 0     | page no of empty dir  |
| 113   | entry            |      |         |                      |     |       |                       |
| 114   | 0106             | 605B | 0000    | DIROFS               | FDB | 00    | offset of empty dir   |
| 115   | entry            |      |         |                      |     |       |                       |
| 116   | 0107             | 605D | 0000    | BASEXP               | FDB | 00    | basic exe adr         |
| 117   | 0108             | 605F | 0000    | MCADP                | FDB | 00    | machine code start    |
| 118   | adr              |      |         |                      |     |       |                       |
| 119   | 0109             | 6061 | 0000    | MCAEP                | FDB | 00    | machine code end adr  |
| 120   | 0110             | 6063 | 0000    | MCEXP                | FDB | 00    | machine code exec adr |
| 121   | 0111             | 6065 | 0000    | EXSTFL               | FDB | 00    | offset of exist dir   |
| 122   | entry            |      |         |                      |     |       |                       |
| one   | 0112             | 6067 | 0000    | CVTLEN               | FDB | 00    | length of converted   |
| two   | text             |      |         |                      |     |       |                       |
| three | 0113             | 6069 | 0000    | RDTOP                | FDB | 00    | adr of read point     |
| 124   | 0114             | 606B | 0000    | SVTOP                | FDB | 00    | adr of saved point    |
| 125   | 0115             | 606D | 0000    | WK1                  | FDB | 00    |                       |
| 126   | 0116             | 606F | 0000    | WK2                  | FDB | 00    |                       |
| 127   | 0117             | 6071 | 0000    | WK3                  | FDB | 00    |                       |
| 128   | 0118             | 6073 | 0000    | WK4                  | FDB | 00    |                       |
| 129   | 0119             | 6075 | 0000    | WK0                  | FDB | 00    |                       |
| 130   | 0120             |      |         |                      |     |       |                       |
| 131   | 0121             | 6077 | FF      | INPBUF               | RMB | 8     |                       |
| 132   | 0122             | 607F | FF      | CNVBUF               | RMB | 4     |                       |
| 133   | 0123             | 6083 | FF      | FILBUF               | RMB | 32    |                       |
| 134   | 0124             |      |         |                      |     |       |                       |
| 135   | 0125             | 60A3 |         | MAIN                 | EQU | *     |                       |
| 136   | 0126             |      |         | *=====reset pia===== |     |       |                       |
| 137   | 0127             |      |         | *portB:output        |     |       |                       |
| 138   | 0128             | 60A3 | 86 30   |                      | LDA | #\$30 |                       |
| 139   | 0129             | 60A5 | B7 FD9F |                      | STA | PBCTL | select ddra, cb2:0    |
| 140   | 0130             | 60A8 | 86 FD   |                      | LDA | #\$FD | pb0,7:out,pb1:in      |
| 141   | 0131             | 60AA | B7 FD9E |                      | STA | PBDAT | set output mode       |
| 142   | 0132             | 60AD | 86 34   |                      | LDA | #\$34 |                       |
| 143   | 0133             | 60AF | B7 FD9F |                      | STA | PBCTL | select                |
| 144   | peripheral,cb2:0 |      |         |                      |     |       |                       |
| 145   | 0134             |      |         |                      |     |       |                       |
| 146   | 0135             |      |         | *for pia exist check |     |       |                       |
| 147   | 0136             | 60B2 | 86 55   |                      | LDA | #\$55 | test data             |
| 148   | 0137             | 60B4 | B7 FD9E |                      | STA | PBDAT |                       |

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149 0138 60B7 B6 FD9E          LDA      PBDAT
150 0139 60BA 84 FD          ANDA      #$FD          pb1=0
151 0140 60BC 81 55          CMPA      #$55
152 0141 60BE 27 07          BEQ       INIT
153 0142 60C0 8E 707B        LDX      #MERR
154 0143 60C3 17 0D5D        LBSR     PSTRNG
155 0144 60C6 39              RTS       Return F-BASIC
156 0145
157 0146 60C7          INIT    EQU      *
158 0147 60C7 17 0094        LBSR     SDINIT
159 0148
160 0149          *****
161 0150          * MAIN MENU
162 0151          *****
163 0152 60CA          MENU    EQU      *
164 0153 60CA 4F          CLRA
165 0154 60CB 1F 8B          TFR       A,DP
166 0155 60CD 8E 6F90        LDX      #MSMENU1
167 0156 60D0 17 0D50        LBSR     PSTRNG
168 0157 60D3 BD FBF7        JSR      CSRON          cursor on
169 0158 60D6 BD 6F12        JSR      INOUTC
170 0159 60D9 84 DF          ANDA      #$DF          to upper
171 0160
172 0161 60DB 81 46          CMPA      #'F
173 0162 60DD 26 05          BNE      MENU1
174 0163 60DF 17 0324        LBSR     DSPDIR
175 0164 60E2 20 E6          BRA      MENU
176 0165 60E4 81 4C MENU1    CMPA      #'L
177 0166 60E6 26 05          BNE      MENU2
178 0167 60E8 17 03BF        LBSR     LDFILE
179 0168 60EB 20 DD          BRA      MENU
180 0169 60ED 81 42 MENU2    CMPA      #'B
181 0170 60EF 26 05          BNE      MENU3
182 0171 60F1 17 06C5        LBSR     WRBFILE
183 0172 60F4 20 D4          BRA      MENU
184 0173 60F6 81 4D MENU3    CMPA      #'M
185 0174 60F8 26 05          BNE      MENU4
186 0175 60FA 17 0732        LBSR     WRMFILE
187 0176 60FD 20 CB          BRA      MENU

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|     |      |      |    |      |            |      |                |    |
|-----|------|------|----|------|------------|------|----------------|----|
| 188 | 0177 | 60FF | 81 | 4B   | MENU4      | CMPA | #'K            |    |
| 189 | 0178 | 6101 | 26 | 05   |            | BNE  | MENU5          |    |
| 190 | 0179 | 6103 | 17 | 065E |            | LBSR | KILFILE        |    |
| 191 | 0180 | 6106 | 20 | C2   |            | BRA  | MENU           |    |
| 192 | 0181 | 6108 | 81 | 53   | MENU5      | CMPA | #'S            |    |
| 193 | 0182 | 610A | 26 | 04   |            | BNE  | MENU6          |    |
| 194 | 0183 | 610C | 8D | 50   |            | BSR  | SDINIT         |    |
| 195 | 0184 | 610E | 20 | BA   |            | BRA  | MENU           |    |
| 196 | 0185 | 6110 | 81 | 4F   | MENU6      | CMPA | #'O            |    |
| 197 | 0186 | 6112 | 26 | 04   |            | BNE  | MENU7          |    |
| 198 | 0187 | 6114 | 20 | 0A   |            | BRA  | OPMENU         |    |
| 199 | 0188 | 6116 | 20 | B2   |            | BRA  | MENU           |    |
| 200 | 0189 | 6118 | 81 | 51   | MENU7      | CMPA | #'Q            |    |
| 201 | 0190 | 611A | 26 | AE   |            | BNE  | MENU           |    |
| 202 | 0191 | 611C | 17 | 0C49 |            | LBSR | PCRLF          |    |
| 203 | 0192 | 611F | 39 |      |            | RTS  | Return F-BASIC |    |
| 204 | 0193 |      |    |      |            |      |                |    |
| 205 | 0194 |      | *  |      |            |      |                |    |
| 206 | 0195 | 6120 |    |      | OPMENU EQU | *    |                |    |
| 207 | 0196 | 6120 | 8E | 6FE9 |            | LDX  | #MSMENU2       |    |
| 208 | 0197 | 6123 | 17 | 0CFD |            | LBSR | PSTRNG         |    |
| 209 | 0198 | 6126 | BD | 6F12 |            | JSR  | INOUTC         |    |
| 210 | 0199 | 6129 | 80 | 30   |            | SUBA | #\$30          |    |
| 211 | 0200 | 612B | 26 | 02   |            | BNE  | OPMEN6         | 0? |
| 212 | 0201 | 612D | 20 | 9B   |            | BRA  | MENU           |    |
| 213 | 0202 |      |    |      |            |      |                |    |
| 214 | 0203 | 612F | 4A |      | OPMEN6     | DECA | 1?             |    |
| 215 | 0204 | 6130 | 26 | 05   |            | BNE  | OPMEN7         |    |
| 216 | 0205 | 6132 | 17 | 0246 |            | LBSR | RD SCT         |    |
| 217 | 0206 | 6135 | 20 | E9   |            | BRA  | OPMENU         |    |
| 218 | 0207 | 6137 | 4A |      | OPMEN7     | DECA | 2?             |    |
| 219 | 0208 | 6138 | 26 | 05   |            | BNE  | OPMEN8         |    |
| 220 | 0209 | 613A | 17 | 01C1 |            | LBSR | DSPSDDAT       |    |
| 221 | 0210 | 613D | 20 | E1   |            | BRA  | OPMENU         |    |
| 222 | 0211 | 613F | 4A |      | OPMEN8     | DECA | 3?             |    |
| 223 | 0212 | 6140 | 26 | 05   |            | BNE  | OPMEN9         |    |
| 224 | 0213 | 6142 | 17 | 0249 |            | LBSR | DSPMBR         |    |
| 225 | 0214 | 6145 | 20 | D9   |            | BRA  | OPMENU         |    |
| 226 | 0215 | 6147 | 4A |      | OPMEN9     | DECA | 4?             |    |

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227 0216 6148 26 05          BNE      OPMEN10
228 0217 614A 17 0295          LBSR      DSPBPB
229 0218 614D 20 D1          BRA      OPMENU
230 0219 614F 4A      OPMEN10 DECA      5?
231 0220 6150 26 CE          BNE      OPMENU
232 0221 6152 17 029F          LBSR      DSPFAT
233 0222 6155 20 C9          BRA      OPMENU
234 0223
235 0224          *
236 0225 6157          SDERR      EQU      *
237 0226 6157 8E 707B          LDX      #MERR
238 0227 615A 17 0CC6          LBSR      PSTRNG
239 0228 615D 39          RTS          Return F-BASIC
240 0229
241 0230          *****
242 0231          * SD Initialize routine
243 0232          *****
244 0233 615E          SDINIT      EQU      *
245 0234          *----- start -----
246 0235 615E 8E 7064          LDX      #MSINIT
247 0236 6161 17 0CBF          LBSR      PSTRNG
248 0237          * cs:1,di:1,sck>74ms
249 0238 6164 86 FF          LDA      #$FF
250 mosi:b7=H,miso:b1=H(for PIA),cs:b0=H
251 0239 6166 B7 FD9E          STA      PBDAT
252 0240 6169 C6 50          LDB      #80      80times
253 0241 616B 17 0BB5      ISTT      LBSR      SETCKL
254 0242 616E 17 0BBB          LBSR      SETCKH
255 0243 6171 5A          DECB
256 0244 6172 26 F7          BNE      ISTT
257 0245          *
258 0246          *----- cmd 0 -----
259 0247 6174          SCMD0      EQU      *
260 0248 6174 17 0BAC          LBSR      SETCKL
261 0249 6177 C6 06          LDB      #6
262 0250 6179 8E 6F2C          LDX      #CMD0
263 0251 617C A6 80 SC01      LDA      ,X+
264 0252 617E 17 0B34          LBSR      SNDDAT
265 0253 6181 5A          DECB

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|     |      |      |    |                     |      |        |           |
|-----|------|------|----|---------------------|------|--------|-----------|
| 266 | 0254 | 6182 | 26 | F8                  | BNE  | SC01   |           |
| 267 | 0255 |      |    | *get response       |      |        |           |
| 268 | 0256 | 6184 | 5F |                     | CLRB |        |           |
| 269 | 0257 | 6185 |    | GR00                |      |        |           |
| 270 | 0258 | 6185 | 5A |                     | DECB |        |           |
| 271 | 0259 | 6186 | 27 | CF                  | BEQ  | SDERR  |           |
| 272 | 0260 | 6188 | 17 | 0B46                | LBSR | GETDAT |           |
| 273 | 0261 | 618B | B6 | 6008                | LDA  | RESDAT |           |
| 274 | 0262 | 618E | 81 | 01                  | CMPA | #\$01  |           |
| 275 | 0263 | 6190 | 26 | F3                  | BNE  | GR00   |           |
| 276 | 0264 |      |    | *                   |      |        |           |
| 277 | 0265 |      |    | *----- cmd 8 -----  |      |        |           |
| 278 | 0266 | 6192 |    | SCMD8               | EQU  | *      |           |
| 279 | 0267 | 6192 | 17 | 0B8E                | LBSR | SETCKL |           |
| 280 | 0268 | 6195 | 86 | FF                  | LDA  | #\$FF  | dummy cmd |
| 281 | 0269 | 6197 | 17 | 0B1B                | LBSR | SNDDAT |           |
| 282 | 0270 | 619A | C6 | 06                  | LDB  | #6     |           |
| 283 | 0271 | 619C | 8E | 6F44                | LDX  | #CMD8  |           |
| 284 | 0272 | 619F | A6 | 80 SC80             | LDA  | ,X+    |           |
| 285 | 0273 | 61A1 | 17 | 0B11                | LBSR | SNDDAT |           |
| 286 | 0274 | 61A4 | 5A |                     | DECB |        |           |
| 287 | 0275 | 61A5 | 26 | F8                  | BNE  | SC80   |           |
| 288 | 0276 |      |    | *get response       |      |        |           |
| 289 | 0277 | 61A7 |    | GR80                |      |        |           |
| 290 | 0278 | 61A7 | 17 | 0B27                | LBSR | GETDAT |           |
| 291 | 0279 | 61AA | B6 | 6008                | LDA  | RESDAT |           |
| 292 | 0280 |      |    |                     |      |        |           |
| 293 | 0281 | 61AD | 17 | 0B21                | LBSR | GETDAT |           |
| 294 | 0282 | 61B0 | B6 | 6008                | LDA  | RESDAT |           |
| 295 | 0283 | 61B3 | 81 | 01                  | CMPA | #\$01  |           |
| 296 | 0284 | 61B5 | 26 | F0                  | BNE  | GR80   |           |
| 297 | 0285 | 61B7 | C6 | 04                  | LDB  | #4     |           |
| 298 | 0286 | 61B9 | 17 | 0B15 GR81           | LBSR | GETDAT |           |
| 299 | 0287 | 61BC | B6 | 6008                | LDA  | RESDAT |           |
| 300 | 0288 | 61BF | 5A |                     | DECB |        |           |
| 301 | 0289 | 61C0 | 26 | F7                  | BNE  | GR81   |           |
| 302 | 0290 |      |    | *                   |      |        |           |
| 303 | 0291 |      |    | *----- cmd 58 ----- |      |        |           |
| 304 | 0292 | 61C2 |    | SCMD58              | EQU  | *      |           |

|     |      |      |    |                     |      |        |           |
|-----|------|------|----|---------------------|------|--------|-----------|
| 305 | 0293 | 61C2 | 17 | 0B5E                | LBSR | SETCKL |           |
| 306 | 0294 | 61C5 | 86 | FF                  | LDA  | #\$FF  | dummy cmd |
| 307 | 0295 | 61C7 | 17 | 0AEB                | LBSR | SNDDAT |           |
| 308 | 0296 | 61CA | C6 | 06                  | LDB  | #6     |           |
| 309 | 0297 | 61CC | 8E | 6F4A                | LDX  | #CMD58 |           |
| 310 | 0298 | 61CF | A6 | 80 SC581            | LDA  | ,X+    |           |
| 311 | 0299 | 61D1 | 17 | 0AE1                | LBSR | SNDDAT |           |
| 312 | 0300 | 61D4 | 5A |                     | DECB |        |           |
| 313 | 0301 | 61D5 | 26 | F8                  | BNE  | SC581  |           |
| 314 | 0302 |      |    | *get response       |      |        |           |
| 315 | 0303 | 61D7 |    | GR580               |      |        |           |
| 316 | 0304 | 61D7 | 17 | 0AF7                | LBSR | GETDAT |           |
| 317 | 0305 | 61DA | B6 | 6008                | LDA  | RESDAT |           |
| 318 | 0306 | 61DD | 81 | 01                  | CMPA | #\$01  |           |
| 319 | 0307 | 61DF | 26 | F6                  | BNE  | GR580  |           |
| 320 | 0308 | 61E1 | C6 | 04 GR581            | LDB  | #4     |           |
| 321 | 0309 | 61E3 | 17 | 0AEB GR582          | LBSR | GETDAT |           |
| 322 | 0310 | 61E6 | B6 | 6008                | LDA  | RESDAT |           |
| 323 | 0311 | 61E9 | 5A |                     | DECB |        |           |
| 324 | 0312 | 61EA | 26 | F7                  | BNE  | GR582  |           |
| 325 | 0313 | 61EC |    | RPT41 EQU *         |      |        |           |
| 326 | 0314 |      |    | *----- cmd 55 ----- |      |        |           |
| 327 | 0315 | 61EC |    | SCMD55 EQU *        |      |        |           |
| 328 | 0316 | 61EC | 17 | 0B34                | LBSR | SETCKL |           |
| 329 | 0317 | 61EF | 86 | FF                  | LDA  | #\$FF  | dummy cmd |
| 330 | 0318 | 61F1 | 17 | 0AC1                | LBSR | SNDDAT |           |
| 331 | 0319 | 61F4 | C6 | 06                  | LDB  | #6     |           |
| 332 | 0320 | 61F6 | 8E | 6F50                | LDX  | #CMD55 |           |
| 333 | 0321 | 61F9 | A6 | 80 SC551            | LDA  | ,X+    |           |
| 334 | 0322 | 61FB | 17 | 0AB7                | LBSR | SNDDAT |           |
| 335 | 0323 | 61FE | 5A |                     | DECB |        |           |
| 336 | 0324 | 61FF | 26 | F8                  | BNE  | SC551  |           |
| 337 | 0325 |      |    | *get response       |      |        |           |
| 338 | 0326 |      |    | * CLRB              |      |        |           |
| 339 | 0327 | 6201 |    | GR550               |      |        |           |
| 340 | 0328 | 6201 | 17 | 0ACD                | LBSR | GETDAT |           |
| 341 | 0329 | 6204 | B6 | 6008                | LDA  | RESDAT |           |
| 342 | 0330 | 6207 | 81 | 01                  | CMPA | #\$01  |           |
| 343 | 0331 | 6209 | 26 | F6                  | BNE  | GR550  |           |

```

344 0332
345 0333      *----- acmd 41 -----
346 0334 620B      ASCMD41 EQU      *
347 0335 620B 17 0B15      LBSR      SETCKL
348 0336 620E 86 FF      LDA      #$FF      dummy cmd
349 0337 6210 17 0AA2      LBSR      SNDDAT
350 0338 6213 C6 06      LDB      #6
351 0339 6215 8E 6F56      LDX      #ACMD41
352 0340 6218 A6 80 ASC411 LDA      ,X+
353 0341 621A 17 0A98      LBSR      SNDDAT
354 0342 621D 5A      DECB
355 0343 621E 26 F8      BNE      ASC411
356 0344      *get response
357 0345 6220      GR410
358 0346 6220 17 0AAE      LBSR      GETDAT
359 0347 6223 B6 6008      LDA      RESDAT
360 0348 6226 17 0AA8      LBSR      GETDAT
361 0349 6229 B6 6008      LDA      RESDAT
362 0350 622C 81 00      CMPA      #$00
363 0351 622E 26 BC      BNE      RPT41
364 0352
365 0353      *----- second cmd 58 -----
366 0354 6230      SCMD258 EQU      *
367 0355 6230 17 0AF0      LBSR      SETCKL
368 0356 6233 86 FF      LDA      #$FF      dummy cmd
369 0357 6235 17 0A7D      LBSR      SNDDAT
370 0358 6238 C6 06      LDB      #6
371 0359 623A 8E 6F4A      LDX      #CMD58
372 0360 623D A6 80 SC2581 LDA      ,X+
373 0361 623F 17 0A73      LBSR      SNDDAT
374 0362 6242 5A      DECB
375 0363 6243 26 F8      BNE      SC2581
376 0364      *get response
377 0365 6245      GR2580
378 0366 6245 17 0A89      LBSR      GETDAT
379 0367 6248 B6 6008      LDA      RESDAT
380 0368 624B 81 00      CMPA      #$00
381 0369 624D 26 F6      BNE      GR2580
382 0370 624F 17 0A7F      LBSR      GETDAT

```

|     |      |      |      |      |        |      |                |                      |
|-----|------|------|------|------|--------|------|----------------|----------------------|
| 383 | 0371 | 6252 | B6   | 6008 |        | LDA  | RESDAT         |                      |
| 384 | 0372 | 6255 | 48   |      |        | LSLA |                |                      |
| 385 | 0373 | 6256 | 48   |      |        | LSLA |                |                      |
| 386 | 0374 | 6257 | 24   | 03   |        | BCC  | GR2581         |                      |
| 387 | 0375 | 6259 | 7C   | 6009 |        | INC  | SDHC           | if sdhc then 1       |
| 388 | 0376 | 625C | C6   | 03   | GR2581 | LDB  | #3             |                      |
| 389 | 0377 | 625E | 17   | 0A70 | GR2582 | LBSR | GETDAT         |                      |
| 390 | 0378 | 6261 | B6   | 6008 |        | LDA  | RESDAT         |                      |
| 391 | 0379 | 6264 | 5A   |      |        | DECB |                |                      |
| 392 | 0380 | 6265 | 26   | F7   |        | BNE  | GR2582         |                      |
| 393 | 0381 |      |      |      |        |      |                |                      |
| 394 | 0382 | 6267 | 8E   | 704E |        | LDX  | #MSSD0         |                      |
| 395 | 0383 | 626A | B6   | 6009 |        | LDA  | SDHC           |                      |
| 396 | 0384 | 626D | 27   | 03   |        | BEQ  | SD1            |                      |
| 397 | 0385 | 626F | 8E   | 7058 |        | LDX  | #MSSD1         |                      |
| 398 | 0386 | 6272 | 17   | 0BAE | SD1    | LBSR | PSTRNG         |                      |
| 399 | 0387 |      | *    |      |        |      |                |                      |
| 400 | 0388 |      |      |      |        |      |                | *set sct no of BPB   |
| 401 | 0389 | 6275 | 17   | 0134 |        | LBSR | GETMBR         |                      |
| 402 | 0390 | 6278 | 8E   | 5FCA |        | LDX  | #RWBUF+446+8+4 |                      |
| 403 | 0391 | 627B | 108E | 600A |        | LDY  | #SCTBPB        |                      |
| 404 | 0392 | 627F | C6   | 04   |        | LDB  | #4             |                      |
| 405 | 0393 | 6281 | A6   | 82   | IL0P1  | LDA  | , -X           |                      |
| 406 | 0394 | 6283 | A7   | A0   |        | STA  | , Y+           |                      |
| 407 | 0395 | 6285 | 5A   |      |        | DECB |                |                      |
| 408 | 0396 | 6286 | 26   | F9   |        | BNE  | IL0P1          |                      |
| 409 | 0397 |      |      |      |        |      |                |                      |
| 410 | 0398 |      |      |      |        |      |                | *set sct no of FAT#1 |
| 411 | 0399 | 6288 | 17   | 013E |        | LBSR | GETBPB         |                      |
| 412 | 0400 | 628B | 8E   | 600A |        | LDX  | #SCTBPB        |                      |
| 413 | 0401 | 628E | 108E | 600E |        | LDY  | #SCTFAT        |                      |
| 414 | 0402 | 6292 | C6   | 04   |        | LDB  | #4             |                      |
| 415 | 0403 | 6294 | 17   | 0982 |        | LBSR | TRANS          |                      |
| 416 | 0404 | 6297 | 8E   | 600E |        | LDX  | #SCTFAT        |                      |
| 417 | 0405 | 629A | 108E | 5E0E |        | LDY  | #RWBUF+\$0E    | for SD & SDHC        |
| 418 | 0406 | 629E | 17   | 0B51 |        | LBSR | ADD42          |                      |
| 419 | 0407 |      |      |      |        |      |                |                      |
| 420 | 0408 |      |      |      |        |      |                | *set sct no of DIR   |
| 421 | 0409 | 62A1 | 8E   | 600E |        | LDX  | #SCTFAT        |                      |

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422 0410 62A4 108E 6012          LDY      #SCTDIR
423 0411 62A8 C6 04          LDB      #4
424 0412 62AA 17 096C          LBSR      TRANS
425 0413 62AD 8E 5E16          LDX      #RWBUF+$16      for SD
426 0414 62B0 7D 6009          TST      SDHC          0:SD,1:SDHC
427 0415 62B3 27 03          BEQ      SKP1
428 0416 62B5 C6 0E          LDB      #$24-$16
429 0417 62B7 3A          ABX          for SDHC
430 0418 62B8 1F 12 SKP1      TFR      X,Y
431 0419 62BA 8E 6012          LDX      #SCTDIR
432 0420 62BD 17 0B32          LBSR      ADD42
433 0421 62C0 17 0B2F          LBSR      ADD42          skip for FAT#2
434 0422
435 0423          *save no of directory
436 0424 62C3 8E 6012          LDX      #SCTDIR
437 0425 62C6 108E 6016          LDY      #SCTDSV
438 0426 62CA C6 04          LDB      #4
439 0427 62CC 17 094A          LBSR      TRANS
440 0428
441 0429          *set sct per cluster
442 0430 62CF 8E 5E0D          LDX      #RWBUF+$0D
443 0431 62D2 A6 84          LDA      ,X
444 0432 62D4 B7 601E          STA      SCTPCL
445 0433
446 0434          *set sct no of CLUSTER#2
447 0435 62D7 8E 6012          LDX      #SCTDIR
448 0436 62DA 108E 601A          LDY      #SCTCLT
449 0437 62DE C6 04          LDB      #4
450 0438 62E0 17 0936          LBSR      TRANS
451 0439
452 0440 62E3 108E 606F          LDY      #WK1+2
453 0441 62E7 86 00          LDA      #$00
454 0442 62E9 A7 A2          STA      ,-Y
455 0443 62EB 86 20          LDA      #$20          for SD
456 0444 62ED 7D 6009          TST      SDHC
457 0445 62F0 27 03          BEQ      SKP2
458 0446 62F2 B6 601E          LDA      SCTPCL          for SDHC
459 0447 62F5 A7 A2 SKP2      STA      ,-Y
460 0448 62F7 8E 601A          LDX      #SCTCLT

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```

461 0449 62FA 17 0AF5          LBSR    ADD42
462 0450 62FD 39              RTS
463 0451
464 0452          *****
465 0453          * GET SD DATA ROUTINE
466 0454          * use CMD9
467 0455          *****
468 0456 62FE          DSPSDDAT EQU      *
469 0457 62FE 17 0A67          LBSR    PCRLF
470 0458 6301 17 0A1F          LBSR    SETCKL
471 0459 6304 C6 06          LDB      #6
472 0460 6306 8E 6F5C          LDX      #CMD9
473 0461 6309 A6 80 SC91      LDA      ,X+
474 0462 630B 17 09A7          LBSR    SNDDAT
475 0463 630E 5A              DECB
476 0464 630F 26 F8          BNE      SC91
477 0465          *get response
478 0466 6311          GR90
479 0467 6311 17 09BD          LBSR    GETDAT
480 0468 6314 B6 6008          LDA      RESDAT
481 0469 6317 81 00          CMPA     #$00
482 0470 6319 26 F6          BNE      GR90
483 0471
484 0472 631B C6 14          LDB      #16+1+2+1   fe,crcH,crcL,ff
485 0473 631D 17 09B1 GR91    LBSR    GETDAT
486 0474 6320 B6 6008          LDA      RESDAT
487 0475 6323 5A              DECB
488 0476 6324 26 F7          BNE      GR91
489 0477 6326 17 0A3F          LBSR    PCRLF
490 0478
491 0479          *disp offset of MBR,FAT#1,DIR,CLUSTER#2
492 0480 6329 8E 709B          LDX      #MSOFS
493 0481 632C 17 0AF4          LBSR    PSTRNG
494 0482 632F 8E 70AB          LDX      #MSBPB
495 0483 6332 17 0AEE          LBSR    PSTRNG
496 0484 6335 8E 600A          LDX      #SCTBPB
497 0485 6338 17 0A24          LBSR    DSPBI2
498 0486 633B 17 0A35          LBSR    OUTSPC
499 0487 633E 17 0A1E          LBSR    DSPBI2

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500      0488
501      0489  6341  8E  70B4          LDX      #MSFAT
502      0490  6344  17  0ADC          LBSR     PSTRNG
503      0491  6347  8E  600E          LDX      #SCTFAT
504      0492  634A  17  0A12          LBSR     DSPBI2
505      0493  634D  17  0A23          LBSR     OUTSPC
506      0494  6350  17  0A0C          LBSR     DSPBI2
507      0495
508      0496  6353  8E  70BD          LDX      #MSDIR
509      0497  6356  17  0ACA          LBSR     PSTRNG
510      0498  6359  8E  6012          LDX      #SCTDIR
511      0499  635C  17  0A00          LBSR     DSPBI2
512      0500  635F  17  0A11          LBSR     OUTSPC
513      0501  6362  17  09FA          LBSR     DSPBI2
514      0502
515      0503  6365  8E  70C6          LDX      #MSCL2
516      0504  6368  17  0AB8          LBSR     PSTRNG
517      0505  636B  8E  601A          LDX      #SCTCLT
518      0506  636E  17  09EE          LBSR     DSPBI2
519      0507  6371  17  09FF          LBSR     OUTSPC
520      0508  6374  17  09E8          LBSR     DSPBI2
521      0509  6377  17  09EE          LBSR     PCRLF
522      0510  637A  39                RTS
523      0511
524      0512      *
525      0513      *****
526      0514      * READ ROUTINE
527      0515      * use CMD17
528      0516      * set sctno to SCTN2,SCTN3
529      0517      * read data set to RWBUF
530      0518      *****
531      0519  637B      RDSCT  EQU      *
532      0520  637B  17  09EA          LBSR     PCRLF
533      0521  637E  17  096A          LBSR     INSCT      input sct no
534      0522  6381  17  08DD          LBSR     RD1SCT      read 1 sct
535      0523
536      0524      *disp data
537      0525  6384  17  09E1          LBSR     PCRLF
538      0526  6387  17  090D          LBSR     DSPBUF

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539 0527 638A 17 09DB          LBSR    PCRLF
540 0528 638D 39              RTS
541 0529
542 0530          *****
543 0531          * Read MBR & Disp parameter of BPB
544 0532          *****
545 0533 638E          DSPMBR EQU      *
546 0534 638E 17 09D7          LBSR    PCRLF
547 0535 6391 8D 19          BSR      GETMBR
548 0536
549 0537          *disp BPB
550 0538 6393 17 09D2          LBSR    PCRLF
551 0539 6396 8E 5FBE          LDX      #RWBUF+446      1st BPB point
552 0540 6399 C6 10 GET1      LDB      #16
553 0541 639B A6 80 GET2      LDA      ,X+
554 0542 639D 17 0995          LBSR    DSPBIN          disp data
555 0543 63A0 5A              DECB
556 0544 63A1 26 F8          BNE      GET2
557 0545 63A3 17 09C2          LBSR    PCRLF
558 0546 63A6 8C 5FFE          CMPX    #RWBUF+510      446+16*4
559 0547 63A9 25 EE          BLO      GET1
560 0548 63AB 39              RTS
561 0549
562 0550          *-----
563 0551          *read MBR
564 0552 63AC          GETMBR EQU      *
565 0553 63AC 8E 6004          LDX      #SCTN0
566 0554 63AF C6 04          LDB      #4
567 0555 63B1 4F              CLRA
568 0556 63B2 17 086F          LBSR    FILL          set sct no is 0
569 0557 63B5 17 093C          LBSR    SETSCT
570 0558 63B8 17 08A6          LBSR    RD1SCT          read 1 sct
571 0559 63BB 39              RTS
572 0560
573 0561          *-----
574 0562          *set sector no to SCTN0
575 0563          *from X,X+1,X+2,X+3 to SCTN0-SCTN3
576 0564 63BC          SETSN0 EQU      *
577 0565 63BC 108E 6004          LDY      #SCTN0

```

```

578 0566 63C0 C6 04          LDB      #4
579 0567 63C2 17 0854          LBSR     TRANS
580 0568 63C5 17 092C          LBSR     SETSCT
581 0569 63C8 39              RTS
582 0570
583 0571          *-----
584 0572          *read BPB
585 0573 63C9          GETBPB EQU      *
586 0574 63C9 8D E1          BSR      GETMBR
587 0575          *set sct no of BPB
588 0576 63CB 8E 5FCA          LDX      #RWBUF+446+8+4
589 0577 63CE 108E 6004          LDY      #SCTN0
590 0578 63D2 C6 04          LDB      #4
591 0579 63D4 A6 82 GTBLOP LDA      , -X
592 0580 63D6 A7 A0          STA      ,Y+
593 0581 63D8 5A          DECB
594 0582 63D9 26 F9          BNE      GTBLOP
595 0583 63DB 17 0916          LBSR     SETSCT
596 0584 63DE 17 0880          LBSR     RD1SCT      read 1 sct
597 0585 63E1 39              RTS
598 0586
599 0587          *****
600 0588          * Disp BPB(Bios Parameter Block)
601 0589          *****
602 0590 63E2          DSPBPB EQU      *
603 0591 63E2 8E 600A          LDX      #SCTBPB
604 0592 63E5 8D D5          BSR      SETSNO
605 0593 63E7 17 0877          LBSR     RD1SCT      read 1 sct
606 0594
607 0595          *disp BPB
608 0596 63EA 17 097B          LBSR     PCRLF
609 0597 63ED 17 08A7          LBSR     DSPBUF
610 0598 63F0 17 0975          LBSR     PCRLF
611 0599 63F3 39              RTS
612 0600
613 0601          *****
614 0602          * Disp FAT#1
615 0603          *****
616 0604 63F4          DSPFAT EQU      *

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```

617 0605 63F4 8E 600E          LDX      #SCTFAT
618 0606 63F7 8D C3          BSR      SETSNO
619 0607 63F9 17 0865        LBSR      RD1SCT      read 1 sct
620 0608
621 0609          *disp FAT
622 0610 63FC 17 0969        LBSR      PCRLF
623 0611 63FF 17 0895        LBSR      DSPBUF
624 0612 6402 17 0963        LBSR      PCRLF
625 0613 6405 39          RTS
626 0614
627 0615          *****
628 0616          * Disp Directory
629 0617          *****
630 0618 6406          DSPDIR EQU      *
631 0619          *return no of DIR
632 0620 6406 8E 6016          LDX      #SCTDSV
633 0621 6409 108E 6012        LDY      #SCTDIR
634 0622 640D C6 04          LDB      #4
635 0623 640F 17 0807        LBSR      TRANS
636 0624 6412 8E 6012 DSPFL0 LDX      #SCTDIR
637 0625 6415 8D A5          BSR      SETSNO
638 0626 6417 17 0847        LBSR      RD1SCT      read 1 sct
639 0627
640 0628          *disp FILE name
641 0629 641A 17 094B        LBSR      PCRLF
642 0630 641D 17 0948        LBSR      PCRLF
643 0631 6420 8E 5E00          LDX      #RWBUF
644 0632
645 0633 6423 A6 84 DSPFL1 LDA      ,X
646 0634 6425 1027 0080        LBEQ     DSPEND      if top of
647 filename is 0 then end
648 0635 6429 81 E5          CMPA     #$E5      delete mark?
649 0636 642B 27 06          BEQ      DSPCKE
650 0637 642D A6 0B          LDA      11,X      attr
651 0638          * CMPA #$0F long file name?
652 0639          * BEQ DSPCKE
653 0640          * CMPA #$10 directory?
654 0641          * BEQ DSPCKE
655 0642 642F 81 20          CMPA     #$20

```

|     |       |      |      |       |                      |            |                   |
|-----|-------|------|------|-------|----------------------|------------|-------------------|
| 656 | 0643  | 6431 | 27   | 0A    | BEQ                  | DSPFL2     | archive file only |
| 657 | 0644  | 6433 | 30   | 88 20 | DSPCKE LEAX          | 32,X       | next directory    |
| 658 | entry |      |      |       |                      |            |                   |
| 659 | 0645  | 6436 | 8C   | 6000  | CMPX                 | #RWBUF+512 | over buf?         |
| 660 | 0646  | 6439 | 24   | 56    | BHS                  | DSPFL5     |                   |
| 661 | 0647  | 643B | 20   | E6    | BRA                  | DSPFL1     |                   |
| 662 | 0648  |      |      |       |                      |            |                   |
| 663 | 0649  | 643D |      |       | DSPFL2               |            |                   |
| 664 | 0650  |      |      |       | *disp filename       |            |                   |
| 665 | 0651  | 643D | C6   | 08    | LDB                  | #8         |                   |
| 666 | 0652  | 643F | A6   | 80    | DSPFL3 LDA           | ,X+        |                   |
| 667 | 0653  | 6441 | BD   | FBBB  | JSR                  | OUTCHA     |                   |
| 668 | 0654  | 6444 | 5A   |       | DECB                 |            |                   |
| 669 | 0655  | 6445 | 26   | F8    | BNE                  | DSPFL3     |                   |
| 670 | 0656  |      |      |       | *disp ext            |            |                   |
| 671 | 0657  | 6447 | 86   | 2E    | LDA                  | #'.        |                   |
| 672 | 0658  | 6449 | BD   | FBBB  | JSR                  | OUTCHA     |                   |
| 673 | 0659  | 644C | C6   | 03    | LDB                  | #3         |                   |
| 674 | 0660  | 644E | A6   | 80    | DSPFL4 LDA           | ,X+        |                   |
| 675 | 0661  | 6450 | BD   | FBBB  | JSR                  | OUTCHA     |                   |
| 676 | 0662  | 6453 | 5A   |       | DECB                 |            |                   |
| 677 | 0663  | 6454 | 26   | F8    | BNE                  | DSPFL4     |                   |
| 678 | 0664  |      |      |       |                      |            |                   |
| 679 | 0665  |      |      |       | *get FILE time data  |            |                   |
| 680 | 0666  | 6456 | 30   | 07    | LEAX                 | 7,X        |                   |
| 681 | 0667  | 6458 | EC   | 81    | LDD                  | ,X++       |                   |
| 682 | 0668  | 645A | FD   | 606D  | STD                  | WK1        |                   |
| 683 | 0669  |      |      |       |                      |            |                   |
| 684 | 0670  |      |      |       | *get CLUSTER(H) data |            |                   |
| 685 | 0671  | 645D | EC   | 81    | LDD                  | ,X++       |                   |
| 686 | 0672  | 645F | 1E   | 89    | EXG                  | A,B        |                   |
| 687 | 0673  | 6461 | FD   | 6048  | STD                  | CLSTH      |                   |
| 688 | 0674  | 6464 | 30   | 04    | LEAX                 | 4,X        |                   |
| 689 | 0675  | 6466 | EC   | 81    | LDD                  | ,X++       |                   |
| 690 | 0676  | 6468 | 1E   | 89    | EXG                  | A,B        |                   |
| 691 | 0677  | 646A | FD   | 604A  | STD                  | CLSTL      |                   |
| 692 | 0678  |      |      |       |                      |            |                   |
| 693 | 0679  |      |      |       | *get FILE size data  |            |                   |
| 694 | 0680  | 646D | 108E | 6046  | LDY                  | #WFSIZ+4   |                   |

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695 0681 6471 C6 04          LDB      #4
696 0682 6473 A6 80  DSPC1  LDA      ,X+
697 0683 6475 A7 A2          STA      ,-Y
698 0684 6477 5A          DECB
699 0685 6478 26 F9          BNE      DSPC1
700 0686
701 0687          *disp FILE size
702 0688 647A BD 6D73          JSR      OUTSPC
703 0689 647D 17 0A12          LBSR      DSPSIZ      disp FILE size
704 0690
705 0691          *disp FILE time
706 0692 6480 FC 606D          LDD      WK1
707 0693 6483 17 09AD          LBSR      DSPYMD      disp MM-DD-YY
708 0694
709 0695          *disp CLUSTER(Lonly)
710 0696 6486 17 0A5A          LBSR      DSPCLS
711 0697 6489 17 08DC          LBSR      PCRLF
712 0698
713 0699 648C 8C 6000          CMPX      #RWBUF+512
714 0700 648F 25 92          BLO      DSPFL1
715 0701 6491 8E 6012  DSPFL5  LDX      #SCTDIR
716 0702 6494 17 0796          LBSR      INCX4      sct no +1
717 0703 6497 8E 713F          LDX      #MSNXTTP
718 0704 649A 17 0986          LBSR      PSTRNG
719 0705 649D BD FBB8  DSPFL6  JSR      INCHA
720 0706 64A0 81 20          CMPA      #SPC
721 0707 64A2 1027 FF6C          LBEQ      DSPFL0
722 0708 64A6 BD 6D68          JSR      PCRLF
723 0709 64A9 39      DSPEND  RTS
724 0710
725 0711          *****
726 0712          * Load FILE
727 0713          *****
728 0714 64AA          LDFILE  EQU      *
729 0715 64AA 17 0241          LBSR      GETFDT
730 0716 64AD BE 6065          LDX      EXSTFL      if 0 not exist
731 0717 64B0 1027 0125          LBEQ      LDNEND
732 0718
733 0719          *get cluster no

```

|     |      |      |      |      |      |                                     |         |                     |
|-----|------|------|------|------|------|-------------------------------------|---------|---------------------|
| 734 | 0720 | 64B4 | EC   | 88   | 14   | LDD                                 | \$14,X  |                     |
| 735 | 0721 | 64B7 | 1E   | 89   |      | EXG                                 | A,B     |                     |
| 736 | 0722 | 64B9 | FD   | 6048 |      | STD                                 | CLSTH   | high cluster no     |
| 737 | 0723 | 64BC | EC   | 88   | 1A   | LDD                                 | \$1A,X  |                     |
| 738 | 0724 | 64BF | 1E   | 89   |      | EXG                                 | A,B     |                     |
| 739 | 0725 | 64C1 | FD   | 604A |      | STD                                 | CLSTL   | low cluster no      |
| 740 | 0726 |      |      |      |      | *sub start cluster no (SD:2,SDHC:3) |         |                     |
| 741 | 0727 | 64C4 | 83   | 0002 |      | SUBD                                | #2      |                     |
| 742 | 0728 | 64C7 | 7D   | 6009 |      | TST                                 | SDHC    | SD:0,SDHC:1         |
| 743 | 0729 | 64CA | 27   | 03   |      | BEQ                                 | LD3     |                     |
| 744 | 0730 | 64CC | 83   | 0001 |      | SUBD                                | #1      | D:cluster no (start |
| 745 |      |      |      |      |      |                                     |         | 0)                  |
| 746 | 0731 |      | *    |      |      | (A=0,cluster no is only B)          |         |                     |
| 747 | 0732 |      |      |      |      |                                     |         |                     |
| 748 | 0733 | 64CF | 34   | 36   | LD3  | PSHS                                | A,B,X,Y |                     |
| 749 | 0734 | 64D1 | 8E   | 601A |      | LDX                                 | #SCTCLT | top of cluster no   |
| 750 | 0735 | 64D4 | 108E | 606D |      | LDY                                 | #WK1    |                     |
| 751 | 0736 | 64D8 | C6   | 04   |      | LDB                                 | #4      |                     |
| 752 | 0737 | 64DA | 17   | 073C |      | LBSR                                | TRANS   |                     |
| 753 | 0738 | 64DD | 35   | 36   |      | PULS                                | A,B,X,Y |                     |
| 754 | 0739 |      |      |      |      |                                     |         |                     |
| 755 | 0740 |      |      |      |      | *calc start cluster no of file      |         |                     |
| 756 | 0741 |      |      |      |      | *(B)x40h                            |         |                     |
| 757 | 0742 | 64DF | 34   | 30   |      | PSHS                                | X,Y     |                     |
| 758 | 0743 | 64E1 | 8E   | 606D |      | LDX                                 | #WK1    |                     |
| 759 | 0744 | 64E4 | 17   | 0755 |      | LBSR                                | ADD40H  |                     |
| 760 | 0745 | 64E7 | 35   | 30   |      | PULS                                | X,Y     |                     |
| 761 | 0746 |      |      |      |      |                                     |         |                     |
| 762 | 0747 |      |      |      |      | *get file size                      |         |                     |
| 763 | 0748 | 64E9 | 34   | 30   |      | PSHS                                | X,Y     |                     |
| 764 | 0749 | 64EB | 30   | 88   | 20   | LEAX                                | 32,X    |                     |
| 765 | 0750 | 64EE | 108E | 6042 |      | LDY                                 | #WFSIZ  |                     |
| 766 | 0751 | 64F2 | C6   | 04   |      | LDB                                 | #4      |                     |
| 767 | 0752 | 64F4 | A6   | 82   | LD41 | LDA                                 | , -X    |                     |
| 768 | 0753 | 64F6 | A7   | A0   |      | STA                                 | ,Y+     |                     |
| 769 | 0754 | 64F8 | 5A   |      |      | DECB                                |         |                     |
| 770 | 0755 | 64F9 | 26   | F9   |      | BNE                                 | LD41    |                     |
| 771 | 0756 | 64FB | 35   | 30   |      | PULS                                | X,Y     |                     |
| 772 | 0757 |      |      |      |      |                                     |         |                     |

|     |      |      |      |      |     |                                         |         |                    |
|-----|------|------|------|------|-----|-----------------------------------------|---------|--------------------|
| 773 | 0758 | 64FD | EC   | 88   | 1D  | LDD                                     | \$1D,X  | upper of file size |
| 774 | 0759 | 6500 | 1E   | 89   |     | EXG                                     | A,B     |                    |
| 775 | 0760 | 6502 | 44   |      |     | LSRA                                    |         |                    |
| 776 | 0761 | 6503 | 56   |      |     | RORB                                    |         |                    |
| 777 | 0762 | 6504 | C3   | 0001 |     | ADDD                                    | #1      | wfsiz/\$200+1      |
| 778 | 0763 |      |      |      |     |                                         |         |                    |
| 779 | 0764 |      |      |      |     | *limited to 1 cluster(40h sct)          |         |                    |
| 780 | 0765 | 6507 | C1   | 40   |     | CMPB                                    | #\$40   | 40h sct?           |
| 781 | 0766 | 6509 | 23   | 02   |     | BLS                                     | LD5     |                    |
| 782 | 0767 | 650B | C6   | 40   |     | LDB                                     | #\$40   |                    |
| 783 | 0768 | 650D | F7   | 604C | LD5 | STB                                     | CNT     | sct counter        |
| 784 | 0769 | 6510 | F7   | 604E |     | STB                                     | TPGFLG  | set top page       |
| 785 | 0770 |      |      |      |     |                                         |         |                    |
| 786 | 0771 |      |      |      |     | *set start sct no of file               |         |                    |
| 787 | 0772 | 6513 | 34   | 12   |     | PSHS                                    | A,X     |                    |
| 788 | 0773 | 6515 | 8E   | 70FE |     | LDX                                     | #MSLDNG |                    |
| 789 | 0774 | 6518 | 17   | 0908 |     | LBSR                                    | PSTRNG  |                    |
| 790 | 0775 | 651B | 35   | 12   |     | PULS                                    | A,X     |                    |
| 791 | 0776 | 651D |      |      |     | LDL0P                                   |         |                    |
| 792 | 0777 | 651D | 86   | 2E   |     | LDA                                     | #'.     |                    |
| 793 | 0778 | 651F | BD   | FBBB |     | JSR                                     | OUTCHA  |                    |
| 794 | 0779 | 6522 | 8E   | 606D |     | LDX                                     | #WK1    |                    |
| 795 | 0780 | 6525 | 108E | 6004 |     | LDY                                     | #SCTN0  |                    |
| 796 | 0781 | 6529 | C6   | 04   |     | LDB                                     | #4      |                    |
| 797 | 0782 | 652B | 17   | 06EB |     | LBSR                                    | TRANS   |                    |
| 798 | 0783 | 652E | 17   | 07C3 |     | LBSR                                    | SETSCT  |                    |
| 799 | 0784 | 6531 | 17   | 072D |     | LBSR                                    | RD1SCT  | read 1 file data   |
| 800 | 0785 |      |      |      |     |                                         |         |                    |
| 801 | 0786 | 6534 | 7D   | 604E |     | TST                                     | TPGFLG  | top sector page?   |
| 802 | 0787 | 6537 | 27   | 71   |     | BEQ                                     | LD9     |                    |
| 803 | 0788 |      |      |      |     |                                         |         |                    |
| 804 | 0789 |      |      |      |     | *at top sector, ini set                 |         |                    |
| 805 | 0790 |      |      |      |     | *offset 00 data is                      |         |                    |
| 806 | 0791 |      |      |      |     | * \$00 :machine code                    |         |                    |
| 807 | 0792 |      |      |      |     | * \$0D :BASIC PROGRAM (ascii save)      |         |                    |
| 808 | 0793 |      |      |      |     | * \$FF,\$FE:BASIC PROGRAM (binary save) |         |                    |
| 809 | 0794 |      |      |      |     | * else :general text                    |         |                    |
| 810 | 0795 | 6539 | 10BE | 6044 |     | LDY                                     | WFSIZ+2 |                    |
| 811 | 0796 | 653D | 10BF | 6046 |     | STY                                     | FILSIZ  | set file size      |

|     |      |      |      |      |     |      |           |                               |  |
|-----|------|------|------|------|-----|------|-----------|-------------------------------|--|
| 812 | 0797 |      |      |      |     |      |           |                               |  |
| 813 | 0798 | 6541 | 8E   | 5E00 |     | LDX  | #RWBUF    |                               |  |
| 814 | 0799 | 6544 | A6   | 84   |     | LDA  | ,X        |                               |  |
| 815 | 0800 | 6546 | B7   | 604D |     | STA  | MCFLG     |                               |  |
| 816 | 0801 | 6549 | 27   | 04   |     | BEQ  | LD8       | machine code                  |  |
| 817 | 0802 | 654B | 2A   | 4B   |     | BPL  | LD81      | ascii saved basic or          |  |
| 818 | else |      |      |      |     |      |           |                               |  |
| 819 | 0803 | 654D | 2B   | 45   |     | BMI  | LD80      | binary saved basic            |  |
| 820 | 0804 |      |      |      |     |      |           | *at machine code, read header |  |
| 821 | 0805 | 654F | 10AE | 01   | LD8 | LDY  | 1,X       | get file size                 |  |
| 822 | 0806 | 6552 | 10BF | 6046 |     | STY  | FILSIZ    |                               |  |
| 823 | 0807 | 6556 | EE   | 03   |     | LDU  | 3,X       | get store addr                |  |
| 824 | 0808 |      |      |      |     |      |           | *disp store adr, size         |  |
| 825 | 0809 | 6558 | 8E   | 71A5 |     | LDX  | #MSMCAS   |                               |  |
| 826 | 0810 | 655B | 17   | 08C5 |     | LBSR | PSTRNG    |                               |  |
| 827 | 0811 | 655E | 1F   | 30   |     | TFR  | U,D       |                               |  |
| 828 | 0812 | 6560 | 34   | 04   |     | PSHS | B         |                               |  |
| 829 | 0813 | 6562 | 17   | 07D6 |     | LBSR | DSPBI0    |                               |  |
| 830 | 0814 | 6565 | 35   | 04   |     | PULS | B         |                               |  |
| 831 | 0815 | 6567 | 1F   | 98   |     | TFR  | B,A       |                               |  |
| 832 | 0816 | 6569 | 17   | 07CF |     | LBSR | DSPBI0    |                               |  |
| 833 | 0817 | 656C | 86   | 2C   |     | LDA  | #',       |                               |  |
| 834 | 0818 | 656E | BD   | FBBB |     | JSR  | OUTCHA    |                               |  |
| 835 | 0819 | 6571 | 1F   | 20   |     | TFR  | Y,D       |                               |  |
| 836 | 0820 | 6573 | 34   | 04   |     | PSHS | B         |                               |  |
| 837 | 0821 | 6575 | 17   | 07C3 |     | LBSR | DSPBI0    |                               |  |
| 838 | 0822 | 6578 | 35   | 04   |     | PULS | B         |                               |  |
| 839 | 0823 | 657A | 1F   | 98   |     | TFR  | B,A       |                               |  |
| 840 | 0824 | 657C | BD   | 6D3B |     | JSR  | DSPBI0    |                               |  |
| 841 | 0825 |      |      |      |     |      |           |                               |  |
| 842 | 0826 |      |      |      |     |      |           | *offset?                      |  |
| 843 | 0827 | 657F | 8E   | 718C |     | LDX  | #MSMCOF   |                               |  |
| 844 | 0828 | 6582 | 17   | 089E |     | LBSR | PSTRNG    |                               |  |
| 845 | 0829 | 6585 | BD   | 6F12 |     | JSR  | INOUTC    |                               |  |
| 846 | 0830 | 6588 | 84   | DF   |     | ANDA | #\$DF     |                               |  |
| 847 | 0831 | 658A | 81   | 59   |     | CMPA | #'Y       |                               |  |
| 848 | 0832 | 658C | 26   | 1C   |     | BNE  | LD9       |                               |  |
| 849 | 0833 | 658E | 33C9 | E000 |     | LEAU | -\$2000,U | -\$2000 offset                |  |
| 850 | 0834 | 6592 | 20   | 16   |     | BRA  | LD9       |                               |  |

|     |       |      |      |      |                   |      |         |                       |  |
|-----|-------|------|------|------|-------------------|------|---------|-----------------------|--|
| 851 | 0835  |      |      |      |                   |      |         |                       |  |
| 852 | 0836  | 6594 | DE   | 33   | LD80              | LDU  | BASADP  | basic file            |  |
| 853 | 0837  | 6596 | 20   | 12   |                   | BRA  | LD9     |                       |  |
| 854 | 0838  |      |      |      |                   |      |         |                       |  |
| 855 | 0839  | 6598 |      |      | LD81              |      |         |                       |  |
| 856 | 0840  | 6598 | 81   | 0D   |                   | CMPA | #\$0D   | ascii saved BASIC?    |  |
| 857 | 0841  | 659A | 27   | 05   |                   | BEQ  | LD82    |                       |  |
| 858 | 0842  | 659C | CE   | 2000 |                   | LDU  | #TXTADR | (else) not basic      |  |
| 859 | file  |      |      |      |                   |      |         |                       |  |
| 860 | 0843  | 659F | 20   | 09   |                   | BRA  | LD9     |                       |  |
| 861 | 0844  |      |      |      |                   |      |         |                       |  |
| 862 | 0845  | 65A1 |      |      | LD82              |      |         |                       |  |
| 863 | 0846  | 65A1 | DE   | 33   |                   | LDU  | BASADP  |                       |  |
| 864 | 0847  | 65A3 | FF   | 606B |                   | STU  | SVTOP   | set save adr          |  |
| 865 | 0848  | 65A6 | 8D   | 69   |                   | BSR  | MEMSET2 | for ascii saved BASIC |  |
| 866 | 0849  | 65A8 | 20   | 02   |                   | BRA  | LD10    |                       |  |
| 867 | 0850  |      |      |      |                   |      |         |                       |  |
| 868 | 0851  | 65AA |      |      | LD9               |      |         |                       |  |
| 869 | 0852  | 65AA | 8D   | 35   |                   | BSR  | MEMSET  | expand memory from    |  |
| 870 | rwbuf |      |      |      |                   |      |         |                       |  |
| 871 | 0853  |      |      |      |                   |      |         |                       |  |
| 872 | 0854  | 65AC |      |      | LD10              |      |         |                       |  |
| 873 | 0855  | 65AC | 7F   | 604E |                   | CLR  | TPGFLG  | end of top page       |  |
| 874 | 0856  | 65AF | 8E   | 606D |                   | LDX  | #WK1    |                       |  |
| 875 | 0857  | 65B2 | 17   | 0678 |                   | LBSR | INCX4   | sct no +1             |  |
| 876 | 0858  | 65B5 | 7A   | 604C |                   | DEC  | CNT     |                       |  |
| 877 | 0859  | 65B8 | 1026 | FF61 |                   | LBNE | LDL0P   |                       |  |
| 878 | 0860  |      |      |      |                   |      |         |                       |  |
| 879 | 0861  | 65BC |      |      | LDEND             |      |         |                       |  |
| 880 | 0862  | 65BC | 7D   | 604D |                   | TST  | MCFLG   |                       |  |
| 881 | 0863  | 65BF | 27   | 0E   |                   | BEQ  | LDE0    |                       |  |
| 882 | 0864  | 65C1 | 2A   | 0C   |                   | BPL  | LDE0    |                       |  |
| 883 | 0865  |      |      |      | *fix link pointer |      |         |                       |  |
| 884 | 0866  | 65C3 | 9E   | 33   |                   | LDX  | BASADP  |                       |  |
| 885 | 0867  | 65C5 | BD   | C730 |                   | JSR  | FIXPTR  |                       |  |
| 886 | 0868  |      |      |      |                   |      |         |                       |  |
| 887 | 0869  | 65C8 | FC   | 6044 |                   | LDD  | WFSIZ+2 |                       |  |
| 888 | 0870  | 65CB | D3   | 33   |                   | ADDD | BASADP  |                       |  |
| 889 | 0871  | 65CD | DD   | 35   |                   | STD  | BASAEP  | end adrs              |  |

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890 0872 65CF 8E 7116 LDE0 LDX #MSLDED
891 0873 65D2 17 084E LBSR PSTRNG
892 0874 65D5 17 0790 LDE1 LBSR PCRLF
893 0875 65D8 39 RTS
894 0876
895 0877 65D9 8E 7131 LDNEND LDX #MSNOFL
896 0878 65DC 17 0844 LBSR PSTRNG
897 0879 65DF 20 F4 BRA LDE1
898 0880
899 0881 *expand in memory from rwbuf
900 0882 *X:rwbuf, U:memory adr, Y:filesize
901 0883 65E1 MEMSET EQU *
902 0884 65E1 34 30 PSHS X,Y
903 0885 65E3 8E 5E00 LDX #RWBUF
904 0886 65E6 10BE 6046 LDY FILSIZ
905 0887 65EA 7D 604E TST TPGFLG top page?
906 0888 65ED 27 0F BEQ MSLOP
907 0889 65EF 7D 604D TST MCFLG 0,$FF($FE),etc
908 0890 65F2 27 04 BEQ MS0
909 0891 65F4 2A 08 BPL MSLOP etc
910 0892 65F6 2B 04 BMI MS1 basic file
911 0893 * machine code
912 0894 65F8 30 05 MS0 LEAX 5,X skip header
913 0895 65FA 20 02 BRA MSLOP
914 0896 65FC 30 03 MS1 LEAX 3,X skip 3 FFs
915 0897
916 0898 65FE A6 80 MSLOP LDA ,X+
917 0899 6600 A7 C0 STA ,U+
918 0900 6602 31 3F LEAY -1,Y
919 0901 6604 27 05 BEQ MS3
920 0902 6606 8C 6000 CMPX #RWBUF+512
921 0903 6609 25 F3 BLO MSLOP
922 0904 660B 10BF 6046 MS3 STY FILSIZ
923 0905 660F 35 B0 PULS X,Y,PC
924 0906 *
925 0907 *-----
926 0908 *after conversion, expand in memory from rwbuf
927 0909 *X:rwbuf, U:memory adr, Y:filesize
928 0910 6611 MEMSET2 EQU *

```

|     |      |      |      |                                          |      |            |               |
|-----|------|------|------|------------------------------------------|------|------------|---------------|
| 929 | 0911 | 6611 | 34   | 30                                       | PSHS | X,Y        |               |
| 930 | 0912 | 6613 | A6   | 80                                       | LDA  | ,X+        |               |
| 931 | 0913 | 6615 | 81   | 0D                                       | CMPA | #\$0D      | skip top CR   |
| 932 | 0914 | 6617 | 26   | 0D                                       | BNE  | MEM2ED     |               |
| 933 | 0915 | 6619 | A6   | 84                                       | LDA  | ,X         |               |
| 934 | 0916 | 661B | 81   | 0A                                       | CMPA | #\$0A      | skip top LF   |
| 935 | 0917 | 661D | 26   | 02                                       | BNE  | MM21       |               |
| 936 | 0918 | 661F | 30   | 01                                       | LEAX | 1,X        |               |
| 937 | 0919 | 6621 |      | MM21                                     |      |            |               |
| 938 | 0920 | 6621 | BF   | 6069                                     | STX  | RDTOP      | start rwbuf+2 |
| 939 | 0921 | 6624 | 8D   | 0B                                       | BSR  | CNVINT     |               |
| 940 | 0922 | 6626 |      | MEM2ED                                   |      |            |               |
| 941 | 0923 | 6626 | BE   | 606B                                     | LDX  | SVTOP      |               |
| 942 | 0924 | 6629 | 6F   | 80                                       | CLR  | ,X+        |               |
| 943 | 0925 | 662B | 6F   | 80                                       | CLR  | ,X+        |               |
| 944 | 0926 | 662D | 9F   | 35                                       | STX  | BASAEP     |               |
| 945 | 0927 | 662F | 35   | B0                                       | PULS | X,Y,PC     |               |
| 946 | 0928 |      |      |                                          |      |            |               |
| 947 | 0929 |      | *    |                                          |      |            |               |
| 948 | 0930 | 6631 |      | CNVINT EQU *                             |      |            |               |
| 949 | 0931 |      |      | *trans to LINBUF with change CR,LF to 00 |      |            |               |
| 950 | 0932 |      |      | *X:rdtop                                 |      |            |               |
| 951 | 0933 | 6631 |      | CNVLOP                                   |      |            |               |
| 952 | 0934 | 6631 | BE   | 6069                                     | LDX  | RDTOP      |               |
| 953 | 0935 | 6634 | 108E | 043D                                     | LDY  | #LINBUF    |               |
| 954 | 0936 |      |      |                                          |      |            |               |
| 955 | 0937 | 6638 |      | INLOP                                    |      |            |               |
| 956 | 0938 | 6638 | A6   | 80                                       | LDA  | ,X+        |               |
| 957 | 0939 | 663A | A7   | A0                                       | STA  | ,Y+        |               |
| 958 | 0940 |      | *    | -----                                    |      |            |               |
| 959 | 0941 | 663C | 8C   | 6000                                     | CMPX | #RWBUF+512 |               |
| 960 | 0942 | 663F | 25   | 2A                                       | BLO  | INL11      |               |
| 961 | 0943 |      |      |                                          |      |            |               |
| 962 | 0944 | 6641 | 7A   | 604C                                     | DEC  | CNT        |               |
| 963 | 0945 | 6644 | 26   | 01                                       | BNE  | INL10      |               |
| 964 | 0946 | 6646 | 39   |                                          | RTS  |            |               |
| 965 | 0947 |      |      |                                          |      |            |               |
| 966 | 0948 | 6647 |      | INL10                                    |      |            |               |
| 967 | 0949 | 6647 | 34   | 70                                       | PSHS | X,Y,U      |               |

|      |             |      |      |       |       |      |         |                  |  |
|------|-------------|------|------|-------|-------|------|---------|------------------|--|
| 968  | 0950        | 6649 | 86   | 2E    |       | LDA  | #'      | .                |  |
| 969  | 0951        | 664B | BD   | FBBB  |       | JSR  | OUTCHA  |                  |  |
| 970  | 0952        |      |      |       |       |      |         |                  |  |
| 971  | 0953        | 664E | 8E   | 606D  |       | LDX  | #WK1    |                  |  |
| 972  | 0954        | 6651 | 17   | 05D9  |       | LBSR | INCX4   | sct no +1        |  |
| 973  | 0955        | 6654 | 108E | 6004  |       | LDY  | #SCTN0  |                  |  |
| 974  | 0956        | 6658 | C6   | 04    |       | LDB  | #4      |                  |  |
| 975  | 0957        | 665A | 17   | 05BC  |       | LBSR | TRANS   |                  |  |
| 976  | 0958        | 665D | 17   | 0694  |       | LBSR | SETSCT  |                  |  |
| 977  | 0959        | 6660 | 17   | 05FE  |       | LBSR | RD1SCT  | read 1 file data |  |
| 978  | 0960        | 6663 | 35   | 70    |       | PULS | X,Y,U   |                  |  |
| 979  | 0961        |      |      |       |       |      |         |                  |  |
| 980  | 0962        | 6665 | 8E   | 5E00  |       | LDX  | #RWBUF  |                  |  |
| 981  | 0963        | 6668 | BF   | 6069  |       | STX  | RDTOP   |                  |  |
| 982  | 0964        |      | *    | ----- |       |      |         |                  |  |
| 983  | 0965        | 666B | 81   | 0D    | INL11 | CMPA | #\$0D   |                  |  |
| 984  | 0966        | 666D | 26   | C9    |       | BNE  | INLOP   |                  |  |
| 985  | 0967        | 666F | A6   | 80    |       | LDA  | ,X+     |                  |  |
| 986  | 0968        | 6671 | 81   | 0A    |       | CMPA | #\$0A   |                  |  |
| 987  | 0969        | 6673 | 27   | 02    |       | BEQ  | INL2    |                  |  |
| 988  | 0970        | 6675 | 30   | 1F    |       | LEAX | -1,X    |                  |  |
| 989  | 0971        | 6677 | BF   | 6069  | INL2  | STX  | RDTOP   | next read point  |  |
| 990  | 0972        | 667A | 6F   | 3F    |       | CLR  | -1,Y    | set terminate 00 |  |
| 991  | 0973        | 667C | 6F   | A4    |       | CLR  | ,Y      |                  |  |
| 992  | 0974        | 667E | 6F   | 21    |       | CLR  | 1,Y     |                  |  |
| 993  | 0975        |      |      |       |       |      |         |                  |  |
| 994  | 0976        |      |      |       |       |      |         |                  |  |
|      |             |      |      |       |       |      |         |                  |  |
| 995  | 0977        | 6680 | DE   | D9    |       | LDU  | \$D9    |                  |  |
| 996  | 0978        | 6682 | 34   | 40    |       | PSHS | U       |                  |  |
| 997  | 0979        | 6684 | 8E   | 043D  |       | LDX  | #LINBUF |                  |  |
| 998  | 0980        | 6687 | 9F   | D9    |       | STX  | \$D9    |                  |  |
| 999  | 0981        | 6689 | A6   | 84    |       | LDA  | ,X      |                  |  |
| 1000 | 0982        | 668B | 1A   | 01    |       | ORCC | #\$01   |                  |  |
| 1001 | 0983        | 668D | BD   | 9162  |       | JSR  | RDLINO  | read line number |  |
| 1002 | 0984        | 6690 | 9E   | 4B    |       | LDX  | \$4B    |                  |  |
| 1003 | 0985        | 6692 | BF   | 033A  |       | STX  | \$33A   |                  |  |
| 1004 | 0986        | 6695 | 0F   | BF    |       | CLR  | \$BF    | assume keyin     |  |
| 1005 | 0987        | 6697 | BD   | FBF1  |       | JSR  | LININT  | (=C28C) one line |  |
| 1006 | translation |      |      |       |       |      |         |                  |  |

|      |      |      |      |      |             |                                     |         |                    |
|------|------|------|------|------|-------------|-------------------------------------|---------|--------------------|
| 1007 | 0988 | 669A | 35   | 40   |             | PULS                                | U       |                    |
| 1008 | 0989 | 669C | DF   | D9   |             | STU                                 | \$D9    |                    |
| 1009 | 0990 |      |      |      |             |                                     |         |                    |
| 1010 | 0991 | 669E | FD   | 6067 |             | STD                                 | CVTLEN  | store len+1        |
| 1011 | 0992 | 66A1 | 10BE | 6067 |             | LDY                                 | CVTLEN  |                    |
| 1012 | 0993 |      |      |      |             |                                     |         |                    |
| 1013 | 0994 |      |      |      |             | *trans from TXTBUF to BASIC PROGRAM |         |                    |
| 1014 | 0995 | 66A5 | FE   | 606B |             | LDU                                 | SVTOP   |                    |
| 1015 | 0996 | 66A8 | 33   | 42   |             | LEAU                                | 2,U     |                    |
| 1016 | 0997 | 66AA | BE   | 033A |             | LDX                                 | \$33A   |                    |
| 1017 | 0998 | 66AD | AF   | C1   |             | STX                                 | ,U++    |                    |
| 1018 | 0999 | 66AF | 8E   | 033C |             | LDX                                 | #TXTBUF |                    |
| 1019 | 1000 |      |      |      |             |                                     |         |                    |
| 1020 | 1001 |      |      |      |             | *save to bas buf                    |         |                    |
| 1021 | 1002 | 66B2 | A6   | 80   | INL12       | LDA                                 | ,X+     |                    |
| 1022 | 1003 | 66B4 | A7   | C0   |             | STA                                 | ,U+     |                    |
| 1023 | 1004 | 66B6 | 31   | 3F   |             | LEAY                                | -1,Y    |                    |
| 1024 | 1005 | 66B8 | 26   | F8   |             | BNE                                 | INL12   |                    |
| 1025 | 1006 |      |      |      |             |                                     |         |                    |
| 1026 | 1007 | 66BA | BE   | 606B |             | LDX                                 | SVTOP   |                    |
| 1027 | 1008 | 66BD | 1F   | 10   |             | TFR                                 | X,D     |                    |
| 1028 | 1009 | 66BF | F3   | 6067 |             | ADDD                                | CVTLEN  |                    |
| 1029 | 1010 | 66C2 | ED   | 84   |             | STD                                 | ,X      | store next pointer |
| 1030 | 1011 | 66C4 | FD   | 606B |             | STD                                 | SVTOP   | save next point    |
| 1031 | 1012 |      |      |      |             |                                     |         |                    |
| 1032 | 1013 | 66C7 | BE   | 6069 |             | LDX                                 | RDTOP   |                    |
| 1033 | 1014 | 66CA | A6   | 84   |             | LDA                                 | ,X      |                    |
| 1034 | 1015 | 66CC | 1026 | FF61 |             | LBNE                                | CNVLOP  |                    |
| 1035 | 1016 | 66D0 | 39   | CEND | RTS         |                                     |         |                    |
| 1036 | 1017 |      |      |      |             |                                     |         |                    |
| 1037 | 1018 |      |      |      |             | *-----                              |         |                    |
| 1038 | 1019 |      |      |      |             | *X:top of FILE NAME                 |         |                    |
| 1039 | 1020 |      |      |      |             | *return C=1:match,C=0:not match     |         |                    |
| 1040 | 1021 | 66D1 |      |      | CMPFN EQU * |                                     |         |                    |
| 1041 | 1022 | 66D1 | 34   | 10   |             | PSHS                                | X       |                    |
| 1042 | 1023 | 66D3 | 108E | 604F |             | LDY                                 | #FNAME  |                    |
| 1043 | 1024 | 66D7 | C6   | 08   |             | LDB                                 | #8      |                    |
| 1044 | 1025 | 66D9 | A6   | 80   | CMPLOP      | LDA                                 | ,X+     |                    |
| 1045 | 1026 | 66DB | 81   | E5   |             | CMPA                                | #\$E5   | delete mark?       |

|      |      |      |      |      |                                    |       |          |                 |
|------|------|------|------|------|------------------------------------|-------|----------|-----------------|
| 1046 | 1027 | 66DD | 27   | 0B   |                                    | BEQ   | NOTM     |                 |
| 1047 | 1028 | 66DF | A1   | A0   |                                    | CMPA  | ,Y+      |                 |
| 1048 | 1029 | 66E1 | 26   | 07   |                                    | BNE   | NOTM     |                 |
| 1049 | 1030 | 66E3 | 5A   |      |                                    | DECB  |          |                 |
| 1050 | 1031 | 66E4 | 26   | F3   |                                    | BNE   | CMPL0P   |                 |
| 1051 | 1032 | 66E6 | 1A   | 01   | MATCH                              | ORCC  | #\$01    | C=1             |
| 1052 | 1033 | 66E8 | 20   | 02   |                                    | BRA   | CMPEND   |                 |
| 1053 | 1034 | 66EA | 1C   | FE   | NOTM                               | ANDCC | #\$FE    | C=0             |
| 1054 | 1035 | 66EC | 35   | 90   | CMPEND                             | PULS  | X,PC     |                 |
| 1055 | 1036 |      |      |      |                                    |       |          |                 |
| 1056 | 1037 |      |      |      | *-----                             |       |          |                 |
| 1057 | 1038 |      |      |      | *input filename                    |       |          |                 |
| 1058 | 1039 |      |      |      | *output:                           |       |          |                 |
| 1059 | 1040 |      |      |      | * if exist EXSTFL(directory entry) |       |          |                 |
| 1060 | 1041 |      |      |      | * not exist EXSTFL(=0)             |       |          |                 |
| 1061 | 1042 | 66EE |      |      | GETFDT EQU                         | *     |          |                 |
| 1062 | 1043 | 66EE | 34   | 02   |                                    | PSHS  | A        |                 |
| 1063 | 1044 | 66F0 | 8E   | 0000 |                                    | LDX   | #0       |                 |
| 1064 | 1045 | 66F3 | BF   | 6065 |                                    | STX   | EXSTFL   |                 |
| 1065 | 1046 |      |      |      | *return no of DIR                  |       |          |                 |
| 1066 | 1047 | 66F6 | 8E   | 6016 |                                    | LDX   | #SCTDSV  |                 |
| 1067 | 1048 | 66F9 | 108E | 6012 |                                    | LDY   | #SCTDIR  |                 |
| 1068 | 1049 | 66FD | C6   | 04   |                                    | LDB   | #4       |                 |
| 1069 | 1050 | 66FF | 17   | 0517 |                                    | LBSR  | TRANS    |                 |
| 1070 | 1051 |      |      |      |                                    |       |          |                 |
| 1071 | 1052 | 6702 | 35   | 02   |                                    | PULS  | A        |                 |
| 1072 | 1053 | 6704 | 81   | 4D   |                                    | CMPA  | #'M      |                 |
| 1073 | 1054 | 6706 | 26   | 05   |                                    | BNE   | GTF1     |                 |
| 1074 | 1055 | 6708 | 8E   | 70E6 |                                    | LDX   | #MSMSV   |                 |
| 1075 | 1056 | 670B | 20   | 03   |                                    | BRA   | GTF2     |                 |
| 1076 | 1057 | 670D | 8E   | 70D6 | GTF1                               | LDX   | #MSINFN  |                 |
| 1077 | 1058 | 6710 | 17   | 0710 | GTF2                               | LBSR  | PSTRNG   |                 |
| 1078 | 1059 | 6713 | 17   | 0166 |                                    | LBSR  | INPFLDAT | input file name |
| 1079 | 1060 |      |      |      |                                    |       |          |                 |
| 1080 | 1061 |      |      |      | *set filename                      |       |          |                 |
| 1081 | 1062 | 6716 | 8E   | 6083 |                                    | LDX   | #FILBUF  |                 |
| 1082 | 1063 | 6719 | 17   | 00E9 |                                    | LBSR  | SETFNM   |                 |
| 1083 | 1064 |      |      |      |                                    |       |          |                 |
| 1084 | 1065 | 671C | 8E   | 6012 | LDL0                               | LDX   | #SCTDIR  |                 |

|      |       |          |      |       |       |      |            |                               |
|------|-------|----------|------|-------|-------|------|------------|-------------------------------|
| 1085 | 1066  | 671F     | 17   | FC9A  |       | LBSR | SETSNO     |                               |
| 1086 | 1067  | 6722     | 17   | 053C  |       | LBSR | RD1SCT     | read 1 sct                    |
| 1087 | 1068  |          |      |       |       |      |            |                               |
| 1088 | 1069  | 6725     | 8E   | 5E00  |       | LDX  | #RWBUF     |                               |
| 1089 | 1070  | 6728     | A6   | 84    | LDFL1 | LDA  | ,X         |                               |
| 1090 | 1071  | 672A     | 27   | 37    |       | BEQ  | LDFLE      | if top of filename is         |
| 1091 | 0     | then end |      |       |       |      |            |                               |
| 1092 | 1072  | 672C     | 81   | E5    |       | CMPA | #\$E5      | delete mark?                  |
| 1093 | 1073  | 672E     | 26   | 10    |       | BNE  | LD1        |                               |
| 1094 | 1074  | 6730     | A6   | 0B    |       | LDA  | 11,X       | attr                          |
| 1095 | 1075  | 6732     | 81   | 20    |       | CMPA | #\$20      | archive file only             |
| 1096 | 1076  | 6734     | 27   | 0A    |       | BEQ  | LD1        |                               |
| 1097 | 1077  | 6736     | 30   | 88 20 |       | LEAX | 32,X       | next directory                |
| 1098 | entry |          |      |       |       |      |            |                               |
| 1099 | 1078  | 6739     | 8C   | 6000  |       | CMPX | #RWBUF+512 | over buf?                     |
| 1100 | 1079  | 673C     | 24   | 0E    |       | BHS  | LDFL5      |                               |
| 1101 | 1080  | 673E     | 20   | E8    |       | BRA  | LDFL1      |                               |
| 1102 | 1081  |          |      |       |       |      |            |                               |
| 1103 | 1082  |          |      |       |       |      |            | *check file name              |
| 1104 | 1083  | 6740     |      |       | LD1   |      |            |                               |
| 1105 | 1084  | 6740     | 8D   | 8F    |       | BSR  | CMPFN      | C=1:match                     |
| 1106 | 1085  | 6742     | 25   | 10    |       | BCS  | LD2        |                               |
| 1107 | 1086  | 6744     | 30   | 88 20 |       | LEAX | 32,X       |                               |
| 1108 | 1087  | 6747     | 8C   | 6000  |       | CMPX | #RWBUF+512 |                               |
| 1109 | 1088  | 674A     | 25   | DC    |       | BLO  | LDFL1      |                               |
| 1110 | 1089  |          |      |       |       |      |            |                               |
| 1111 | 1090  | 674C     |      |       | LDFL5 |      |            |                               |
| 1112 | 1091  | 674C     | 8E   | 6012  |       | LDX  | #SCTDIR    |                               |
| 1113 | 1092  | 674F     | 17   | 04DB  |       | LBSR | INCX4      | sct no +1                     |
| 1114 | 1093  | 6752     | 20   | C8    |       | BRA  | LDFL0      |                               |
| 1115 | 1094  |          |      |       |       |      |            |                               |
| 1116 | 1095  |          |      |       |       |      |            | *hit file, get dirctory entry |
| 1117 | 1096  | 6754     | BF   | 6065  | LD2   | STX  | EXSTFL     | adr of dir entry              |
| 1118 | 1097  | 6757     | 108E | 601F  |       | LDY  | #DIRENT    |                               |
| 1119 | 1098  | 675B     | C6   | 20    |       | LDB  | #32        |                               |
| 1120 | 1099  | 675D     | 17   | 04B9  |       | LBSR | TRANS      |                               |
| 1121 | 1100  | 6760     | 8E   | 601F  |       | LDX  | #DIRENT    |                               |
| 1122 | 1101  | 6763     | 39   |       | LDFLE | RTS  |            |                               |
| 1123 | 1102  |          |      |       |       |      |            |                               |

```

1124 1103 *
1125 1104 *****
1126 1105 * KILL FILE
1127 1106 *****
1128 1107 6764 KILFILE EQU *
1129 1108 6764 8D 88 BSR GETFDT
1130 1109 6766 BE 6065 LDX EXSTFL if 0 not exist
1131 1110 6769 27 5B BEQ NOFEND
1132 1111
1133 1112 676B 8E 715C LDX #MSKILL
1134 1113 676E 17 06B2 LBSR PSTRNG
1135 1114 6771 BD 6F12 JSR INOUTC
1136 1115 6774 84 DF ANDA # $DF
1137 1116 6776 81 59 CMPA #'Y
1138 1117 6778 26 3E BNE KILEND
1139 1118 677A 8D 01 BSR KILSUB
1140 1119 677C 39 RTS
1141 1120
1142 1121 *-----
1143 1122 677D KILSUB EQU *
1144 1123 *write dir
1145 1124 677D BE 6065 LDX EXSTFL top adr of dir
1146 entry
1147 1125 6780 EC 88 1A LDD $1A,X get clstld
1148 1126 6783 1E 89 EXG A,B B:cluster
1149 1127 6785 FD 604A STD CLSTL
1150 1128 6788 86 E5 LDA # $E5 delete mark
1151 1129 678A A7 84 STA ,X
1152 1130 678C 17 03C4 LBSR WRTSD1
1153 1131
1154 1132 *get fat sct & write
1155 1133 678F 8E 600E LDX #SCTFAT
1156 1134 6792 17 FC27 LBSR SETSNO
1157 1135 6795 17 04C9 LBSR RD1SCT read 1 sct
1158 1136
1159 1137 6798 FC 604A LDD CLSTL B:cluster
1160 1138 679B 58 LSLB
1161 1139 679C 7D 6009 TST SDHC sd:0,sdhc:1
1162 1140 679F 27 01 BEQ KL1

```

```

1163 1141 67A1 58          LSLB
1164 1142 67A2 8E 5E00 KL1    LDX    #RWBUF
1165 1143 67A5 30 85          LEAX    B,X
1166 1144 67A7 C6 02          LDB     #2
1167 1145 67A9 7D 6009          TST     SDHC          sd:0,sdhc:1
1168 1146 67AC 27 02          BEQ     KL2
1169 1147 67AE C6 04          LDB     #4
1170 1148 67B0 6F 80 KL2    CLR     ,X+
1171 1149 67B2 5A          DECB
1172 1150 67B3 26 FB          BNE     KL2
1173 1151 67B5 17 039B          LBSR    WRTSD1
1174 1152 67B8          KILEND
1175 1153 67B8 39          RTS
1176 1154
1177 1155          *
1178 1156          *****
1179 1157          * Write BASIC FILE
1180 1158          *****
1181 1159 67B9          WRBFILE EQU    *
1182 1160          *basic file exist check
1183 1161 67B9 DC 35          LDD     BASAEP
1184 1162 67BB 93 33          SUBD    BASADP
1185 1163 67BD 1083 0002          CMPD    #2
1186 1164 67C1 22 0A          BHI     WRB0          exist
1187 1165 67C3 17 05A2          LBSR    PCRLF
1188 1166 67C6 8E 7131 NOFEND LDX     #MSNOFL
1189 1167 67C9 17 0657          LBSR    PSTRNG
1190 1168 67CC 39          RTS
1191 1169
1192 1170          *save BASIC program file(binary saved)
1193 1171 67CD 86 FF WRB0    LDA     #$FF
1194 1172 67CF B7 604D          STA     MCFLG          set basic file
1195 1173          *input file name
1196 1174 67D2 17 FF19          LBSR    GETFDT
1197 1175 67D5 BE 6065          LDX     EXSTFL          if 0 not exist
1198 1176 67D8 27 11          BEQ     WRB1
1199 1177
1200 1178          *confirm over write
1201 1179 67DA 8E 716F          LDX     #MSEXST

```

|      |      |      |      |            |      |           |                        |
|------|------|------|------|------------|------|-----------|------------------------|
| 1202 | 1180 | 67DD | 17   | 0643       | LBSR | PSTRNG    |                        |
| 1203 | 1181 | 67E0 | BD   | 6F12       | JSR  | INOUTC    |                        |
| 1204 | 1182 | 67E3 | 84   | DF         | ANDA | #\$DF     |                        |
| 1205 | 1183 | 67E5 | 81   | 59         | CMPA | #'Y       |                        |
| 1206 | 1184 | 67E7 | 26   | 1B         | BNE  | WRBEND    |                        |
| 1207 | 1185 |      |      |            |      |           | *delete exist file     |
| 1208 | 1186 | 67E9 | 8D   | 92         | BSR  | KILSUB    |                        |
| 1209 | 1187 |      |      |            |      |           |                        |
| 1210 | 1188 | 67EB |      | WRB1       |      |           |                        |
| 1211 | 1189 |      |      |            |      |           | *set filename          |
| 1212 | 1190 | 67EB | 8E   | 6083       | LDX  | #FILBUF   |                        |
| 1213 | 1191 | 67EE | 8D   | 15         | BSR  | SETFNM    |                        |
| 1214 | 1192 |      |      |            |      |           |                        |
| 1215 | 1193 |      |      |            |      |           | *adrs                  |
| 1216 | 1194 |      |      |            |      |           | * BASADP start adr     |
| 1217 | 1195 |      |      |            |      |           | * BASAEP end adr       |
| 1218 | 1196 |      |      |            |      |           | * BASEXP exec adr      |
| 1219 | 1197 |      |      |            |      |           |                        |
| 1220 | 1198 |      |      |            |      |           | *main of write routine |
| 1221 | 1199 |      |      |            |      |           | *clear dir entry       |
| 1222 | 1200 | 67F0 | 8D   | 2F         | BSR  | CLRDIR    |                        |
| 1223 | 1201 |      |      |            |      |           |                        |
| 1224 | 1202 | 67F2 | 17   | 0104       | LBSR | SRCFAT    | search empty           |
| 1225 |      |      |      |            |      |           | cluster on FAT         |
| 1226 | 1203 | 67F5 | 17   | 0159       | LBSR | SRCDIR    | search empty DIR       |
| 1227 |      |      |      |            |      |           | entry                  |
| 1228 | 1204 | 67F8 | 17   | 0195       | LBSR | MKFLDT    | make file data         |
| 1229 | 1205 | 67FB | 17   | 021B       | LBSR | WRTFIL    | write file             |
| 1230 | 1206 | 67FE | 17   | 03AF       | LBSR | WRTDIR    | write DIR entry        |
| 1231 | 1207 | 6801 | 17   | 03D9       | LBSR | WRTFAT    | set FAT                |
| 1232 | 1208 | 6804 |      | WRBEND     |      |           |                        |
| 1233 | 1209 | 6804 | 39   |            | RTS  |           |                        |
| 1234 | 1210 |      |      |            |      |           |                        |
| 1235 | 1211 |      |      |            |      |           | *-----                 |
| 1236 | 1212 |      |      |            |      |           | *set filename          |
| 1237 | 1213 | 6805 |      | SETFNM EQU | *    |           |                        |
| 1238 | 1214 |      |      |            |      |           | *clear buf             |
| 1239 | 1215 | 6805 | 108E | 605A       | LDY  | #FNAME+11 |                        |
| 1240 | 1216 | 6809 | C6   | 0B         | LDB  | #11       |                        |

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1241 1217 680B 86 20          LDA      #$20
1242 1218 680D A7 A2 WRFLP2 STA      , -Y
1243 1219 680F 5A          DECB
1244 1220 6810 26 FB          BNE      WRFLP2
1245 1221
1246 1222          *store filename
1247 1223 6812 A6 80 WRFL1  LDA      ,X+
1248 1224 6814 81 2D          CMPA     #' -
1249 1225 6816 27 04          BEQ      WRFL11
1250 1226 6818 81 30          CMPA     #$30
1251 1227 681A 25 04          BLO      WRFL2      until delimiter
1252 1228 681C A7 A0 WRFL11 STA      ,Y+
1253 1229 681E 20 F2          BRA      WRFL1
1254 1230 6820          WRFL2
1255 1231 6820 39          RTS
1256 1232
1257 1233          *-----
1258 1234          *clear dir entry
1259 1235 6821          CLRDIR EQU      *
1260 1236 6821 8E 601F          LDX      #DIRENT
1261 1237 6824 C6 20          LDB      #32
1262 1238 6826 4F          CLRA
1263 1239 6827 17 03FA          LBSR      FILL
1264 1240 682A 86 20          LDA      #$20      set attr
1265 1241 682C A7 0B          STA      $0B,X
1266 1242 682E 39          RTS
1267 1243
1268 1244          *
1269 1245          *****
1270 1246          * Write MACHINE FILE
1271 1247          *****
1272 1248 682F          WRMFILE EQU      *
1273 1249          *save machine code
1274 1250          *filename,topadr,endadr,exeadr ...machine file
1275 1251 682F 17 FEBC          LBSR      GETFDT
1276 1252 6832 BE 6065          LDX      EXSTFL      if 0 not exist
1277 1253 6835 27 12          BEQ      WRM1
1278 1254
1279 1255          *confirm over write

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|      |                |      |      |      |      |         |                        |
|------|----------------|------|------|------|------|---------|------------------------|
| 1280 | 1256           | 6837 | 8E   | 716F | LDX  | #MSEXST |                        |
| 1281 | 1257           | 683A | 17   | 05E6 | LBSR | PSTRNG  |                        |
| 1282 | 1258           | 683D | BD   | 6F12 | JSR  | INOUTC  |                        |
| 1283 | 1259           | 6840 | 84   | DF   | ANDA | #\$DF   |                        |
| 1284 | 1260           | 6842 | 81   | 59   | CMPA | #'Y     |                        |
| 1285 | 1261           | 6844 | 26   | 35   | BNE  | WRMEND  |                        |
| 1286 | 1262           |      |      |      |      |         | *delete exist file     |
| 1287 | 1263           | 6846 | 17   | FF34 | LBSR | KILSUB  |                        |
| 1288 | 1264           |      |      |      |      |         |                        |
| 1289 | 1265           | 6849 |      | WRM1 |      |         |                        |
| 1290 | 1266           | 6849 | 7F   | 604D | CLR  | MCFLG   | set machine            |
| 1291 | code(set 0)    |      |      |      |      |         |                        |
| 1292 | 1267           |      |      |      |      |         |                        |
| 1293 | 1268           |      |      |      |      |         | *set filename          |
| 1294 | 1269           | 684C | 8E   | 6083 | LDX  | #FILBUF |                        |
| 1295 | 1270           | 684F | 8D   | B4   | BSR  | SETFNM  |                        |
| 1296 | 1271           |      |      |      |      |         |                        |
| 1297 | 1272           |      |      |      |      |         | *set top adr           |
| 1298 | 1273           | 6851 | 8D   | 43   | BSR  | ASC2BIN |                        |
| 1299 | 1274           | 6853 | 10BF | 605F | STY  | MCADP   | start adr              |
| 1300 | 1275           | 6857 | 8D   | 30   | BSR  | SRCDLM  |                        |
| 1301 | 1276           | 6859 | 8D   | 3B   | BSR  | ASC2BIN |                        |
| 1302 | 1277           | 685B | 10BF | 6061 | STY  | MCAEP   | end adr                |
| 1303 | 1278           | 685F | 8D   | 28   | BSR  | SRCDLM  |                        |
| 1304 | 1279           | 6861 | 8D   | 33   | BSR  | ASC2BIN |                        |
| 1305 | 1280           | 6863 | 10BF | 6063 | STY  | MCEXP   | exec adr               |
| 1306 | 1281           |      |      |      |      |         |                        |
| 1307 | 1282           |      |      |      |      |         | *main of write routine |
| 1308 | 1283           |      |      |      |      |         | *clear dir entry       |
| 1309 | 1284           | 6867 | 8D   | B8   | BSR  | CLRDIR  |                        |
| 1310 | 1285           |      |      |      |      |         |                        |
| 1311 | 1286           | 6869 | 17   | 008D | LBSR | SRCFAT  | search empty           |
| 1312 | cluster on FAT |      |      |      |      |         |                        |
| 1313 | 1287           | 686C | 17   | 00E2 | LBSR | SRCDIR  | search empty DIR       |
| 1314 | entry          |      |      |      |      |         |                        |
| 1315 | 1288           | 686F | 17   | 011E | LBSR | MKFLDT  | make file data         |
| 1316 | 1289           | 6872 | 17   | 01A4 | LBSR | WRTFIL  | write file             |
| 1317 | 1290           | 6875 | 17   | 0338 | LBSR | WRTDIR  | write DIR entry        |
| 1318 | 1291           | 6878 | 17   | 0362 | LBSR | WRTFAT  | set FAT                |

|      |      |      |              |                                                |         |  |
|------|------|------|--------------|------------------------------------------------|---------|--|
| 1319 | 1292 | 687B |              | WRMEND                                         |         |  |
| 1320 | 1293 | 687B | 39           |                                                | RTS     |  |
| 1321 | 1294 |      |              |                                                |         |  |
| 1322 | 1295 |      |              | *-----                                         |         |  |
| 1323 | 1296 |      |              | *input filename & parameter to filbuf          |         |  |
| 1324 | 1297 | 687C |              | INPFLDAT EQU *                                 |         |  |
| 1325 | 1298 | 687C | 34 24        | PSHS                                           | B,Y     |  |
| 1326 | 1299 | 687E | 108E 6083    | LDY                                            | #FILBUF |  |
| 1327 | 1300 | 6882 | C6 1E        | LDB                                            | #30     |  |
| 1328 | 1301 | 6884 | 17 0692      | LBSR                                           | LININ   |  |
| 1329 | 1302 | 6887 | 35 A4        | PULS                                           | B,Y,PC  |  |
| 1330 | 1303 |      |              |                                                |         |  |
| 1331 | 1304 |      |              | *-----                                         |         |  |
| 1332 | 1305 |      |              | *search delimiter                              |         |  |
| 1333 | 1306 |      |              | *delimiter:less than \$30, excluding '-'(\$2D) |         |  |
| 1334 | 1307 |      |              | *return X:point next char of delimiter         |         |  |
| 1335 | 1308 | 6889 |              | SRCDLM EQU *                                   |         |  |
| 1336 | 1309 | 6889 | 30 1F        | LEAX                                           | -1,X    |  |
| 1337 | 1310 | 688B | A6 80 SRDLM1 | LDA                                            | ,X+     |  |
| 1338 | 1311 | 688D | 81 2D        | CMPA                                           | #' -    |  |
| 1339 | 1312 | 688F | 27 FA        | BEQ                                            | SRDLM1  |  |
| 1340 | 1313 | 6891 | 81 30        | CMPA                                           | #\$30   |  |
| 1341 | 1314 | 6893 | 24 F6        | BHS                                            | SRDLM1  |  |
| 1342 | 1315 | 6895 | 39           |                                                | RTS     |  |
| 1343 | 1316 |      |              |                                                |         |  |
| 1344 | 1317 |      |              | *-----                                         |         |  |
| 1345 | 1318 |      |              | *X - X+3, return Y                             |         |  |
| 1346 | 1319 | 6896 |              | ASC2BIN EQU *                                  |         |  |
| 1347 | 1320 |      |              | *clr buf                                       |         |  |
| 1348 | 1321 | 6896 | 34 10        | PSHS                                           | X       |  |
| 1349 | 1322 | 6898 | 8E 607F      | LDX                                            | #CNVBUF |  |
| 1350 | 1323 | 689B | 86 30        | LDA                                            | #\$30   |  |
| 1351 | 1324 | 689D | C6 04        | LDB                                            | #4      |  |
| 1352 | 1325 | 689F | 17 0382      | LBSR                                           | FILL    |  |
| 1353 | 1326 | 68A2 | 35 10        | PULS                                           | X       |  |
| 1354 | 1327 |      |              |                                                |         |  |
| 1355 | 1328 |      |              | *search next delimiter                         |         |  |
| 1356 | 1329 |      |              | *return B:char count                           |         |  |
| 1357 | 1330 | 68A4 | 34 10        | PSHS                                           | X       |  |

|      |      |      |      |                 |        |         |                 |
|------|------|------|------|-----------------|--------|---------|-----------------|
| 1358 | 1331 | 68A6 | 5F   |                 | CLRB   |         |                 |
| 1359 | 1332 | 68A7 | 5C   | SRL0P1          | INCB   |         |                 |
| 1360 | 1333 | 68A8 | A6   | 80              | LDA    | ,X+     |                 |
| 1361 | 1334 | 68AA | 81   | 30              | CMPA   | #\$30   | count over \$30 |
| 1362 | 1335 | 68AC | 24   | F9              | BHS    | SRL0P1  |                 |
| 1363 | 1336 | 68AE | 5A   |                 | DECB   |         |                 |
| 1364 | 1337 | 68AF | 35   | 10              | PULS   | X       |                 |
| 1365 | 1338 |      |      |                 |        |         |                 |
| 1366 | 1339 |      |      | *trans          |        |         |                 |
| 1367 | 1340 | 68B1 | 108E | 607F            | LDY    | #CNVBUF |                 |
| 1368 | 1341 | 68B5 | 34   | 04              | PSHS   | B       |                 |
| 1369 | 1342 | 68B7 | C6   | 04              | LDB    | #4      |                 |
| 1370 | 1343 | 68B9 | E0   | E4              | SUBB   | ,S      |                 |
| 1371 | 1344 | 68BB | 31   | A5              | LEAY   | B,Y     |                 |
| 1372 | 1345 | 68BD | 35   | 04              | PULS   | B       |                 |
| 1373 | 1346 | 68BF | A6   | 80              | SRL0P2 | LDA     | ,X+             |
| 1374 | 1347 | 68C1 | A7   | A0              | STA    | ,Y+     |                 |
| 1375 | 1348 | 68C3 | 5A   |                 | DECB   |         |                 |
| 1376 | 1349 | 68C4 | 26   | F9              | BNE    | SRL0P2  |                 |
| 1377 | 1350 |      |      |                 |        |         |                 |
| 1378 | 1351 |      |      | *convert to bin |        |         |                 |
| 1379 | 1352 | 68C6 | 34   | 10              | PSHS   | X       |                 |
| 1380 | 1353 | 68C8 | 8E   | 607F            | LDX    | #CNVBUF |                 |
| 1381 | 1354 | 68CB | EC   | 81              | LDD    | ,X++    |                 |
| 1382 | 1355 | 68CD | 8D   | 0F              | BSR    | CNVSUB  |                 |
| 1383 | 1356 | 68CF | 34   | 02              | PSHS   | A       |                 |
| 1384 | 1357 | 68D1 | EC   | 81              | LDD    | ,X++    |                 |
| 1385 | 1358 | 68D3 | 8D   | 09              | BSR    | CNVSUB  |                 |
| 1386 | 1359 | 68D5 | 1F   | 89              | TFR    | A,B     |                 |
| 1387 | 1360 | 68D7 | 35   | 02              | PULS   | A       |                 |
| 1388 | 1361 | 68D9 | 1F   | 02              | TFR    | D,Y     |                 |
| 1389 | 1362 | 68DB | 35   | 10              | PULS   | X       |                 |
| 1390 | 1363 | 68DD | 39   |                 | RTS    |         |                 |
| 1391 | 1364 |      |      |                 |        |         |                 |
| 1392 | 1365 |      |      | *cnv sub        |        |         |                 |
| 1393 | 1366 |      |      | *return A       |        |         |                 |
| 1394 | 1367 | 68DE |      | CNVSUB          | EQU    | *       |                 |
| 1395 | 1368 | 68DE | 80   | 30              | SUBA   | #\$30   |                 |
| 1396 | 1369 | 68E0 | 81   | 11              | CMPA   | #\$11   |                 |

|      |      |      |    |      |                            |      |         |                      |
|------|------|------|----|------|----------------------------|------|---------|----------------------|
| 1397 | 1370 | 68E2 | 25 | 02   |                            | BLO  | CNV1    |                      |
| 1398 | 1371 | 68E4 | 80 | 07   |                            | SUBA | #7      |                      |
| 1399 | 1372 | 68E6 | 48 |      | CNV1                       | LSLA |         |                      |
| 1400 | 1373 | 68E7 | 48 |      |                            | LSLA |         |                      |
| 1401 | 1374 | 68E8 | 48 |      |                            | LSLA |         |                      |
| 1402 | 1375 | 68E9 | 48 |      |                            | LSLA |         |                      |
| 1403 | 1376 | 68EA | C0 | 30   |                            | SUBB | #\$30   |                      |
| 1404 | 1377 | 68EC | C1 | 11   |                            | CMPB | #\$11   |                      |
| 1405 | 1378 | 68EE | 25 | 02   |                            | BLO  | CNV2    |                      |
| 1406 | 1379 | 68F0 | C0 | 07   |                            | SUBB | #7      |                      |
| 1407 | 1380 | 68F2 | C4 | 0F   | CNV2                       | ANDB | #\$0F   |                      |
| 1408 | 1381 | 68F4 | 34 | 04   |                            | PSHS | B       |                      |
| 1409 | 1382 | 68F6 | AB | E0   |                            | ADDA | ,S+     |                      |
| 1410 | 1383 | 68F8 | 39 |      |                            | RTS  |         |                      |
| 1411 | 1384 |      |    |      |                            |      |         |                      |
| 1412 | 1385 |      |    |      | *-----                     |      |         |                      |
| 1413 | 1386 |      |    |      | * search empty area in FAT |      |         |                      |
| 1414 | 1387 | 68F9 |    |      | SRCFAT EQU *               |      |         |                      |
| 1415 | 1388 | 68F9 | 8E | 600E |                            | LDX  | #SCTFAT |                      |
| 1416 | 1389 | 68FC | 17 | FABD |                            | LBSR | SETSNO  |                      |
| 1417 | 1390 | 68FF | 17 | 035F |                            | LBSR | RD1SCT  | read 1 sct           |
| 1418 | 1391 |      |    |      |                            |      |         |                      |
| 1419 | 1392 | 6902 | 8E | 5E00 |                            | LDX  | #RWBUF  |                      |
| 1420 | 1393 |      |    |      | *search \$00               |      |         |                      |
| 1421 | 1394 | 6905 |    |      | SRC0                       |      |         |                      |
| 1422 | 1395 | 6905 | C6 | 02   |                            | LDB  | #2      | SD                   |
| 1423 | 1396 | 6907 | 7D | 6009 |                            | TST  | SDHC    | 0:SD,1:SDHC          |
| 1424 | 1397 | 690A | 27 | 02   |                            | BEQ  | SRCL0P  |                      |
| 1425 | 1398 | 690C | C6 | 04   |                            | LDB  | #4      | SDHC                 |
| 1426 | 1399 | 690E | 8D | 32   | SRCL0P                     | BSR  | SRC00   | search 1st 00 ...1st |
| 1427 | 00   |      |    |      |                            |      |         |                      |
| 1428 | 1400 | 6910 | 24 | 2F   |                            | BCC  | SFATED  | not exist empty area |
| 1429 | 1401 | 6912 | A6 | 80   |                            | LDA  | ,X+     | next is 00? ..2nd 00 |
| 1430 | 1402 | 6914 | 26 | EF   |                            | BNE  | SRC0    | restart              |
| 1431 | 1403 | 6916 | 7D | 6009 |                            | TST  | SDHC    | 0:SD,1:SDHC          |
| 1432 | 1404 | 6919 | 27 | 0A   |                            | BEQ  | SETX    |                      |
| 1433 | 1405 | 691B | A6 | 80   |                            | LDA  | ,X+     | next is 00? ..3rd 00 |
| 1434 | 1406 | 691D | 26 | E6   |                            | BNE  | SRC0    | restart              |
| 1435 | 1407 | 691F | A6 | 80   |                            | LDA  | ,X+     | next is 00? ..4th 00 |

|      |        |                          |      |       |        |       |            |                  |
|------|--------|--------------------------|------|-------|--------|-------|------------|------------------|
| 1436 | 1408   | 6921                     | 26   | E2    |        | BNE   | SRC0       | restart          |
| 1437 | 1409   | 6923                     | 30   | 1E    |        | LEAX  | -2,X       |                  |
| 1438 | 1410   | 6925                     |      |       | SETX   |       |            |                  |
| 1439 | 1411   | 6925                     | 30   | 1E    |        | LEAX  | -2,X       | pos in RWBUF     |
| 1440 | 1412   | 6927                     | 3089 | A200  |        | LEAX  | -RWBUF,X   | relative pos     |
| 1441 | 1413   | 692B                     | 1F   | 10    |        | TFR   | X,D        |                  |
| 1442 | 1414   | 692D                     | 44   |       |        | LSRA  |            |                  |
| 1443 | 1415   | 692E                     | 56   |       |        | RORB  | 1/2        |                  |
| 1444 | 1416   | 692F                     | 7D   | 6009  |        | TST   | SDHC       | 0:SD,1:SDHC      |
| 1445 | 1417   | 6932                     | 27   | 02    |        | BEQ   | SETCL      |                  |
| 1446 | 1418   | 6934                     | 44   |       |        | LSRA  |            |                  |
| 1447 | 1419   | 6935                     | 56   |       |        | RORB  | 1/2        |                  |
| 1448 | 1420   | 6936                     | FD   | 604A  | SETCL  | STD   | CLSTL      |                  |
| 1449 | 1421   | 6939                     | 1E   | 89    |        | EXG   | A,B        |                  |
| 1450 | 1422   | 693B                     | 8E   | 601F  |        | LDX   | #DIRENT    |                  |
| 1451 | 1423   | 693E                     | ED   | 88 1A |        | STD   | \$1A,X     | set cluster L to |
| 1452 | dirent |                          |      |       |        |       |            |                  |
| 1453 | 1424   | 6941                     | 39   |       | SFATED | RTS   |            |                  |
| 1454 | 1425   |                          |      |       |        |       |            |                  |
| 1455 | 1426   | *search 00 in FAT        |      |       |        |       |            |                  |
| 1456 | 1427   | 6942                     | 8C   | 6000  | SRC00  | CMPX  | #RWBUF+512 |                  |
| 1457 | 1428   | 6945                     | 24   | 07    |        | BHS   | NOEXST     |                  |
| 1458 | 1429   | 6947                     | A6   | 80    |        | LDA   | ,X+        |                  |
| 1459 | 1430   | 6949                     | 26   | F7    |        | BNE   | SRC00      |                  |
| 1460 | 1431   | 694B                     | 1A   | 01    | EXIST  | ORCC  | #\$01      | set C            |
| 1461 | 1432   | 694D                     | 39   |       |        | RTS   |            |                  |
| 1462 | 1433   | 694E                     | 1C   | FE    | NOEXST | ANDCC | #\$FE      | reset C          |
| 1463 | 1434   | 6950                     | 39   |       |        | RTS   |            |                  |
| 1464 | 1435   |                          |      |       |        |       |            |                  |
| 1465 | 1436   | *-----                   |      |       |        |       |            |                  |
| 1466 | 1437   | * search empty DIR entry |      |       |        |       |            |                  |
| 1467 | 1438   | * return:DPGNO,DIROFS    |      |       |        |       |            |                  |
| 1468 | 1439   | 6951                     |      |       | SRCDIR | EQU   | *          |                  |
| 1469 | 1440   | 6951                     | 7F   | 605A  |        | CLR   | DPGNO      |                  |
| 1470 | 1441   | *return no of DIR        |      |       |        |       |            |                  |
| 1471 | 1442   | 6954                     | 8E   | 6016  |        | LDX   | #SCTDSV    |                  |
| 1472 | 1443   | 6957                     | 108E | 6012  |        | LDY   | #SCTDIR    |                  |
| 1473 | 1444   | 695B                     | C6   | 04    |        | LDB   | #4         |                  |
| 1474 | 1445   | 695D                     | 17   | 02B9  |        | LBSR  | TRANS      |                  |

|      |      |      |      |       |            |      |            |                |
|------|------|------|------|-------|------------|------|------------|----------------|
| 1475 | 1446 | 6960 | 8E   | 6012  | SRCFL0     | LDX  | #SCTDIR    |                |
| 1476 | 1447 | 6963 | 17   | FA56  | SRCFL1     | LBSR | SETSNO     |                |
| 1477 | 1448 | 6966 | 17   | 02F8  |            | LBSR | RD1SCT     | read 1 sct     |
| 1478 | 1449 |      |      |       |            |      |            |                |
| 1479 | 1450 | 6969 | 8E   | 5E00  |            | LDX  | #RWBUF     |                |
| 1480 | 1451 | 696C | 8C   | 6000  | DL0P1      | CMPX | #RWBUF+512 |                |
| 1481 | 1452 | 696F | 24   | 0D    |            | BHS  | SRCFL5     |                |
| 1482 | 1453 | 6971 | A6   | 80    |            | LDA  | ,X+        |                |
| 1483 | 1454 | 6973 | 27   | 15    |            | BEQ  | DIRGET     |                |
| 1484 | 1455 | 6975 | 81   | E5    |            | CMPA | #\$E5      | delete mark?   |
| 1485 | 1456 | 6977 | 27   | 11    |            | BEQ  | DIRGET     |                |
| 1486 | 1457 | 6979 | 30   | 88 1F |            | LEAX | 32-1,X     |                |
| 1487 | 1458 | 697C | 20   | EE    |            | BRA  | DL0P1      |                |
| 1488 | 1459 |      |      |       |            |      |            |                |
| 1489 | 1460 | 697E | 7C   | 605A  | SRCFL5     | INC  | DPGNO      |                |
| 1490 | 1461 | 6981 | 8E   | 6012  |            | LDX  | #SCTDIR    |                |
| 1491 | 1462 | 6984 | 17   | 02A6  |            | LBSR | INCX4      | sct no +1      |
| 1492 | 1463 | 6987 | 20   | DA    |            | BRA  | SRCFL1     |                |
| 1493 | 1464 | 6989 |      | DSPED |            |      |            |                |
| 1494 | 1465 | 6989 | 39   |       |            | RTS  |            |                |
| 1495 | 1466 |      |      |       |            |      |            |                |
| 1496 | 1467 | 698A | 30   | 1F    | DIRGET     | LEAX | -1,X       |                |
| 1497 | 1468 | 698C | BF   | 605B  |            | STX  | DIROFS     |                |
| 1498 | 1469 | 698F | 39   |       |            | RTS  |            |                |
| 1499 | 1470 |      |      |       |            |      |            |                |
| 1500 | 1471 |      |      |       |            |      |            |                |
| 1501 | 1472 | 6990 |      |       | MKFLDT EQU | *    |            |                |
| 1502 | 1473 |      |      |       |            |      |            |                |
| 1503 | 1474 |      |      |       |            |      |            |                |
| 1504 | 1475 | 6990 | 8E   | 604F  |            | LDX  | #FNAME     |                |
| 1505 | 1476 | 6993 | 108E | 601F  |            | LDY  | #DIRENT    |                |
| 1506 | 1477 | 6997 | C6   | 0B    |            | LDB  | #11        |                |
| 1507 | 1478 | 6999 | 17   | 027D  |            | LBSR | TRANS      | set file name  |
| 1508 | 1479 |      |      |       |            |      |            |                |
| 1509 | 1480 |      |      |       |            |      |            |                |
| 1510 | 1481 | 699C | 7D   | 604D  |            | TST  | MCFLG      | 00,FF(FE),else |
| 1511 | 1482 | 699F | 2B   | 08    |            | BMI  | MKF1       |                |
| 1512 | 1483 | 69A1 | FC   | 6061  |            | LDD  | MCAEP      |                |
| 1513 | 1484 | 69A4 | B3   | 605F  |            | SUBD | MCADP      |                |

|      |      |      |      |                    |                      |      |         |              |
|------|------|------|------|--------------------|----------------------|------|---------|--------------|
| 1514 | 1485 | 69A7 | 20   | 04                 |                      | BRA  | MKF2    |              |
| 1515 | 1486 | 69A9 |      |                    | MKF1                 |      |         |              |
| 1516 | 1487 | 69A9 | DC   | 35                 |                      | LDD  | BASAEP  |              |
| 1517 | 1488 | 69AB | 93   | 33                 |                      | SUBD | BASADP  |              |
| 1518 | 1489 | 69AD |      |                    | MKF2                 |      |         |              |
| 1519 | 1490 | 69AD | C3   | 0001               |                      | ADDD | #1      |              |
| 1520 | 1491 | 69B0 | FD   | 6046               |                      | STD  | FILSIZ  |              |
| 1521 | 1492 | 69B3 | FD   | 6075               |                      | STD  | WK0     |              |
| 1522 | 1493 |      | *    | EXG                | A,B                  |      |         |              |
| 1523 | 1494 |      | *    | LDX                | #DIRENT              |      |         |              |
| 1524 | 1495 |      | *    | STD                | \$1C,X               |      |         |              |
| 1525 | 1496 |      |      |                    |                      |      |         |              |
| 1526 | 1497 |      |      |                    | *set date (for FM-7) |      |         |              |
| 1527 | 1498 | 69B6 | 8D   | 12                 |                      | BSR  | STDATE  |              |
| 1528 | 1499 | 69B8 | FC   | 603F               |                      | LDD  | WMDY    |              |
| 1529 | 1500 | 69BB | 1E   | 89                 |                      | EXG  | A,B     |              |
| 1530 | 1501 |      |      |                    |                      |      |         |              |
| 1531 | 1502 |      |      |                    | *year,month,day      |      |         |              |
| 1532 | 1503 | 69BD | 8E   | 601F               |                      | LDX  | #DIRENT |              |
| 1533 | 1504 | 69C0 | ED   | 88 18              |                      | STD  | \$18,X  |              |
| 1534 | 1505 | 69C3 | ED   | 88 10              |                      | STD  | \$10,X  |              |
| 1535 | 1506 | 69C6 | ED   | 88 12              |                      | STD  | \$12,X  |              |
| 1536 | 1507 | 69C9 | 39   |                    |                      | RTS  |         |              |
| 1537 | 1508 |      |      |                    |                      |      |         |              |
| 1538 | 1509 |      | *    | -----              |                      |      |         |              |
| 1539 | 1510 |      | *    | FM day to SD       |                      |      |         |              |
| 1540 | 1511 |      | *    | retuen WMDY,WMDY+1 |                      |      |         |              |
| 1541 | 1512 | 69CA |      | STDATE             | EQU                  | *    |         |              |
| 1542 | 1513 | 69CA | 8E   | 02FA               |                      | LDX  | #FMDAT  | date of FM-7 |
| 1543 | 1514 | 69CD | 108E | 603F               |                      | LDY  | #WMDY   |              |
| 1544 | 1515 | 69D1 | C6   | 03                 |                      | LDB  | #3      |              |
| 1545 | 1516 | 69D3 | 34   | 04                 | ABLOP                | PSHS | B       |              |
| 1546 | 1517 | 69D5 | EC   | 81                 |                      | LDD  | ,X++    |              |
| 1547 | 1518 | 69D7 | 8D   | 2E                 |                      | BSR  | APLUSB  |              |
| 1548 | 1519 | 69D9 | A7   | A0                 |                      | STA  | ,Y+     |              |
| 1549 | 1520 | 69DB | 35   | 04                 |                      | PULS | B       |              |
| 1550 | 1521 | 69DD | 26   | F4                 |                      | BNE  | ABLOP   |              |
| 1551 | 1522 |      | *    | year               |                      |      |         |              |
| 1552 | 1523 | 69DF | B6   | 603F               |                      | LDA  | WMDY    |              |

|      |      |      |                                 |      |            |        |               |
|------|------|------|---------------------------------|------|------------|--------|---------------|
| 1553 | 1524 | 69E2 | 8B                              | 14   | ADDA       | #20    | ---2000-1980  |
| 1554 | 1525 | 69E4 | 48                              |      | LSLA       |        |               |
| 1555 | 1526 | 69E5 | B7                              | 603F | STA        | WMDY   |               |
| 1556 | 1527 |      | *month                          |      |            |        |               |
| 1557 | 1528 | 69E8 | B6                              | 6040 | LDA        | WMDY+1 |               |
| 1558 | 1529 | 69EB | 48                              |      | LSLA       |        |               |
| 1559 | 1530 | 69EC | 48                              |      | LSLA       |        |               |
| 1560 | 1531 | 69ED | 48                              |      | LSLA       |        |               |
| 1561 | 1532 | 69EE | 48                              |      | LSLA       |        |               |
| 1562 | 1533 | 69EF | 48                              |      | ASLA       |        |               |
| 1563 | 1534 | 69F0 | B7                              | 6040 | STA        | WMDY+1 |               |
| 1564 | 1535 | 69F3 | F6                              | 603F | LDB        | WMDY   |               |
| 1565 | 1536 | 69F6 | C9                              | 00   | ADCB       | #0     | plus C        |
| 1566 | 1537 | 69F8 | F7                              | 603F | STB        | WMDY   |               |
| 1567 | 1538 |      | *day                            |      |            |        |               |
| 1568 | 1539 | 69FB | B6                              | 6041 | LDA        | WMDY+2 |               |
| 1569 | 1540 | 69FE | 84                              | 1F   | ANDA       | #\$1F  |               |
| 1570 | 1541 | 6A00 | BB                              | 6040 | ADDA       | WMDY+1 |               |
| 1571 | 1542 | 6A03 | B7                              | 6040 | STA        | WMDY+1 |               |
| 1572 | 1543 | 6A06 | 39                              |      | RTS        |        |               |
| 1573 | 1544 |      |                                 |      |            |        |               |
| 1574 | 1545 |      | *lower nibble of A,B to A+B     |      |            |        |               |
| 1575 | 1546 | 6A07 |                                 |      | APLUSB EQU | *      |               |
| 1576 | 1547 | 6A07 | 34                              | 10   | PSHS       | X      |               |
| 1577 | 1548 | 6A09 | 34                              | 04   | PSHS       | B      |               |
| 1578 | 1549 | 6A0B | C6                              | 0A   | LDB        | #10    |               |
| 1579 | 1550 | 6A0D | 3D                              |      | MUL        |        |               |
| 1580 | 1551 | 6A0E | 1F                              | 01   | TFR        | D,X    |               |
| 1581 | 1552 | 6A10 | 35                              | 04   | PULS       | B      |               |
| 1582 | 1553 | 6A12 | 3A                              |      | ABX        |        |               |
| 1583 | 1554 | 6A13 | 1F                              | 10   | TFR        | X,D    |               |
| 1584 | 1555 | 6A15 | 1E                              | 89   | EXG        | A,B    | A:data        |
| 1585 | 1556 | 6A17 | 35                              | 90   | PULS       | X,PC   |               |
| 1586 | 1557 |      | *                               |      |            |        |               |
| 1587 | 1558 |      | *-----                          |      |            |        |               |
| 1588 | 1559 | 6A19 |                                 |      | WRTFIL EQU | *      |               |
| 1589 | 1560 |      | *calc file size for write to sd |      |            |        |               |
| 1590 | 1561 | 6A19 | FC                              | 6075 | LDD        | WK0    |               |
| 1591 | 1562 | 6A1C | 7D                              | 604D | TST        | MCFLG  | 0,FF(FE),else |

|      |                  |                                   |    |       |        |      |          |                     |
|------|------------------|-----------------------------------|----|-------|--------|------|----------|---------------------|
| 1592 | 1563             | 6A1F                              | 27 | 04    |        | BEQ  | WRTF0    | machine code        |
| 1593 | 1564             | 6A21                              | 2B | 07    |        | BMI  | WRTF1    | basic(binary)       |
| 1594 | 1565             | 6A23                              | 2A | 08    |        | BPL  | WRTF2    | else                |
| 1595 | 1566             | 6A25                              | C3 | 000A  | WRTF0  | ADDD | #10      | add header(&footer) |
| 1596 | 1567             | 6A28                              | 20 | 03    |        | BRA  | WRTF2    |                     |
| 1597 | 1568             | 6A2A                              | C3 | 0003  | WRTF1  | ADDD | #3       | add header          |
| 1598 | 1569             | 6A2D                              |    |       | WRTF2  |      |          |                     |
| 1599 | 1570             | 6A2D                              | FD | 6075  |        | STD  | WK0      | number of bytes     |
| 1600 | actually written |                                   |    |       |        |      |          |                     |
| 1601 | 1571             | 6A30                              | 1E | 89    |        | EXG  | A,B      |                     |
| 1602 | 1572             | 6A32                              | 8E | 601F  |        | LDX  | #DIRENT  |                     |
| 1603 | 1573             | 6A35                              | ED | 88 1C |        | STD  | \$1C,X   |                     |
| 1604 | 1574             | 6A38                              | 1E | 89    |        | EXG  | A,B      |                     |
| 1605 | 1575             | *calc counter (CNT:512byte units) |    |       |        |      |          |                     |
| 1606 | 1576             | 6A3A                              | 7F | 604C  |        | CLR  | CNT      |                     |
| 1607 | 1577             | 6A3D                              | 7C | 604C  | CALLOP | INC  | CNT      |                     |
| 1608 | 1578             | 6A40                              | 83 | 0200  |        | SUBD | #512     |                     |
| 1609 | 1579             | 6A43                              | 24 | F8    |        | BHS  | CALLOP   |                     |
| 1610 | 1580             |                                   |    |       |        |      |          |                     |
| 1611 | 1581             | 6A45                              | 86 | 01    |        | LDA  | #1       |                     |
| 1612 | 1582             | 6A47                              | B7 | 604E  |        | STA  | TPGFLG   | set top page        |
| 1613 | 1583             |                                   |    |       |        |      |          |                     |
| 1614 | 1584             | 6A4A                              | 7D | 604D  |        | TST  | MCFLG    | 0,FF(FE),else       |
| 1615 | 1585             | 6A4D                              | 2B | 05    |        | BMI  | WRTF3    |                     |
| 1616 | 1586             | 6A4F                              | BE | 605F  |        | LDX  | MCADP    | saved top adr       |
| 1617 | 1587             | 6A52                              | 20 | 02    |        | BRA  | WRTF4    |                     |
| 1618 | 1588             | 6A54                              | 9E | 33    | WRTF3  | LDX  | BASADP   | saved top adr       |
| 1619 | 1589             | 6A56                              | BF | 606D  | WRTF4  | STX  | WK1      | src topadr          |
| 1620 | 1590             | 6A59                              | CE | 5E00  |        | LDU  | #RWBUF   |                     |
| 1621 | 1591             |                                   |    |       |        |      |          |                     |
| 1622 | 1592             | *get clusterL                     |    |       |        |      |          |                     |
| 1623 | 1593             | 6A5C                              | 8E | 601F  |        | LDX  | #DIRENT  |                     |
| 1624 | 1594             | 6A5F                              | EC | 88 1A |        | LDD  | \$1A,X   |                     |
| 1625 | 1595             | 6A62                              | 1E | 89    |        | EXG  | A,B      | B:clusterL          |
| 1626 | 1596             |                                   |    |       |        |      |          |                     |
| 1627 | 1597             | 6A64                              | BE | 601A  |        | LDX  | SCTCLT   | sct no of top       |
| 1628 | cluster          |                                   |    |       |        |      |          |                     |
| 1629 | 1598             | 6A67                              | BF | 606F  |        | STX  | WK2      |                     |
| 1630 | 1599             | 6A6A                              | BE | 601C  |        | LDX  | SCTCLT+2 |                     |

|      |      |      |      |       |      |         |                             |
|------|------|------|------|-------|------|---------|-----------------------------|
| 1631 | 1600 | 6A6D | BF   | 6071  | STX  | WK2+2   |                             |
| 1632 | 1601 |      |      |       |      |         |                             |
| 1633 | 1602 | 6A70 | C0   | 02    | SUBB | #2      | for SD                      |
| 1634 | 1603 | 6A72 | 7D   | 6009  | TST  | SDHC    | 0:SD,1:SDHC                 |
| 1635 | 1604 | 6A75 | 27   | 01    | BEQ  | WR2     |                             |
| 1636 | 1605 | 6A77 | 5A   |       | DECB |         |                             |
| 1637 | 1606 | 6A78 |      | WR2   |      |         |                             |
| 1638 | 1607 |      |      |       |      |         |                             |
| 1639 | 1608 |      |      |       |      |         | *(X)..(X+3)+(B)x40h         |
| 1640 | 1609 | 6A78 | 8E   | 606F  | LDX  | #WK2    |                             |
| 1641 | 1610 | 6A7B | 17   | 01BE  | LBSR | ADD40H  | cal of sct no               |
| 1642 | 1611 |      |      |       |      |         |                             |
| 1643 | 1612 |      |      |       |      |         | *set start sct no of file   |
| 1644 | 1613 | 6A7E | 34   | 10    | PSHS | X       |                             |
| 1645 | 1614 | 6A80 | 8E   | 710A  | LDX  | #MSWTNG |                             |
| 1646 | 1615 | 6A83 | 17   | 039D  | LBSR | PSTRNG  |                             |
| 1647 | 1616 | 6A86 | 35   | 10    | PULS | X       |                             |
| 1648 | 1617 |      |      |       |      |         |                             |
| 1649 | 1618 | 6A88 |      | WRL0P |      |         |                             |
| 1650 | 1619 | 6A88 | 86   | 2E    | LDA  | #'.     |                             |
| 1651 | 1620 | 6A8A | BD   | FBBB  | JSR  | OUTCHA  |                             |
| 1652 | 1621 | 6A8D | FC   | 6075  | LDD  | WK0     |                             |
| 1653 | 1622 | 6A90 | 1F   | 02    | TFR  | D,Y     |                             |
| 1654 | 1623 | 6A92 | 1083 | 0200  | CPMD | #512    |                             |
| 1655 | 1624 | 6A96 | 23   | 0A    | BLS  | WR3     |                             |
| 1656 | 1625 | 6A98 | 83   | 0200  | SUBD | #512    |                             |
| 1657 | 1626 | 6A9B | FD   | 6075  | STD  | WK0     |                             |
| 1658 | 1627 | 6A9E | 108E | 0200  | LDY  | #512    |                             |
| 1659 | 1628 | 6AA2 |      | WR3   |      |         |                             |
| 1660 | 1629 |      |      |       |      |         |                             |
| 1661 | 1630 |      |      |       |      |         | *write data to rwbuf        |
| 1662 | 1631 |      |      |       |      |         | * (X)->(U:RWBUF)            |
| 1663 | 1632 |      |      |       |      |         | * (Y):SIZE 512bytes or Less |
| 1664 | 1633 | 6AA2 | BE   | 606D  | LDX  | WK1     |                             |
| 1665 | 1634 | 6AA5 | CE   | 5E00  | LDU  | #RWBUF  |                             |
| 1666 | 1635 | 6AA8 | 8D   | 44    | BSR  | WRSUB   |                             |
| 1667 | 1636 |      |      |       |      |         |                             |
| 1668 | 1637 |      |      |       |      |         | *set to sctn0-3             |
| 1669 | 1638 | 6AAA | 34   | 30    | PSHS | X,Y     |                             |

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1670 1639 6AAC 8E 606F          LDX      #WK2
1671 1640 6AAF 108E 6004          LDY      #SCTN0
1672 1641 6AB3 C6 04          LDB      #4
1673 1642 6AB5 17 0161          LBSR     TRANS
1674 1643 6AB8 35 30          PULS     X,Y
1675 1644
1676 1645          *write data to sd
1677 1646 6ABA 17 0093          LBSR     WRTSD      ...not wrtsd1
1678 1647
1679 1648          *inc sctn0-3
1680 1649 6ABD 8E 606F          LDX      #WK2
1681 1650 6AC0 17 016A          LBSR     INCX4
1682 1651 6AC3 BE 606D          LDX      WK1      src topadr
1683 1652 6AC6 3089 0200          LEAX     512,X
1684 1653          *check top page
1685 1654 6ACA 7D 604E          TST      TPGFLG      top page?
1686 1655 6ACD 27 0D          BEQ      WR6      no
1687 1656 6ACF 7D 604D          TST      MCFLG
1688 1657 6AD2 27 04          BEQ      WR4      machine code
1689 1658 6AD4 2B 04          BMI      WR5      basic(binary)
1690 1659 6AD6 2A 04          BPL      WR6
1691 1660 6AD8 30 1E WR4      LEAX     -2,X      rewind 3or5
1692 1661 6ADA 30 1D WR5      LEAX     -3,X
1693 1662 6ADC          WR6
1694 1663 6ADC 7F 604E          CLR      TPGFLG
1695 1664 6ADF BF 606D          STX      WK1
1696 1665 6AE2 7A 604C          DEC      CNT
1697 1666 6AE5 26 A1          BNE      WRLOP
1698 1667
1699 1668 6AE7 8E 7123 WREND    LDX      #MSWTED
1700 1669 6AEA 17 0336          LBSR     PSTRNG
1701 1670 6AED 39          RTS
1702 1671
1703 1672          *-----
1704 1673          *write data to rwbuf
1705 1674          * (X)->(U:RWBUF)
1706 1675          * (Y):SIZE 512bytes or Less
1707 1676          *clear rwbuf
1708 1677 6AEE          WRSUB    EQU      *

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|      |      |      |      |      |                             |        |                   |
|------|------|------|------|------|-----------------------------|--------|-------------------|
| 1709 | 1678 | 6AEE | 34   | 30   | PSHS                        | X,Y    |                   |
| 1710 | 1679 | 6AF0 | 4F   |      | CLRA                        |        |                   |
| 1711 | 1680 | 6AF1 | 8E   | 5E00 | LDX                         | #RWBUF |                   |
| 1712 | 1681 | 6AF4 | 108E | 0200 | LDY                         | #512   |                   |
| 1713 | 1682 | 6AF8 | A7   | 80   | WWCLOP                      | STA    | ,X+               |
| 1714 | 1683 | 6AFA | 31   | 3F   | LEAY                        | -1,Y   |                   |
| 1715 | 1684 | 6AFC | 26   | FA   | BNE                         | WWCLOP |                   |
| 1716 | 1685 | 6AFE | 35   | 30   | PULS                        | X,Y    |                   |
| 1717 | 1686 |      |      |      |                             |        |                   |
| 1718 | 1687 |      |      |      | *set data to rwbuf          |        |                   |
| 1719 | 1688 | 6B00 | CE   | 5E00 | LDU                         | #RWBUF |                   |
| 1720 | 1689 |      |      |      |                             |        |                   |
| 1721 | 1690 | 6B03 | 7D   | 604E | TST                         | TPGFLG | 1:top page,0:else |
| 1722 | 1691 | 6B06 | 27   | 16   | BEQ                         | WRS01  | no                |
| 1723 | 1692 | 6B08 | 7D   | 604D | TST                         | MCFLG  | 0,\$FF(\$FE),etc  |
| 1724 | 1693 | 6B0B | 2B   | 32   | BMI                         | WRS1   |                   |
| 1725 | 1694 |      |      |      |                             |        |                   |
| 1726 | 1695 |      |      |      | *add header of machine code |        |                   |
| 1727 | 1696 | 6B0D | 34   | 10   | PSHS                        | X      |                   |
| 1728 | 1697 | 6B0F | 4F   |      | CLRA                        |        |                   |
| 1729 | 1698 | 6B10 | A7   | C0   | STA                         | ,U+    | \$00              |
| 1730 | 1699 | 6B12 | BE   | 6046 | LDX                         | FILSIZ |                   |
| 1731 | 1700 | 6B15 | AF   | C1   | STX                         | ,U++   | file size         |
| 1732 | 1701 | 6B17 | BE   | 605F | LDX                         | MCADP  |                   |
| 1733 | 1702 | 6B1A | AF   | C1   | STX                         | ,U++   | save adr          |
| 1734 | 1703 | 6B1C | 35   | 10   | PULS                        | X      |                   |
| 1735 | 1704 |      |      |      |                             |        |                   |
| 1736 | 1705 | 6B1E |      |      |                             | WRS01  |                   |
| 1737 | 1706 |      |      |      | *check last page            |        |                   |
| 1738 | 1707 | 6B1E | B6   | 604C | LDA                         | CNT    |                   |
| 1739 | 1708 | 6B21 | 81   | 01   | CMPA                        | #1     |                   |
| 1740 | 1709 | 6B23 | 26   | 02   | BNE                         | WRS2   |                   |
| 1741 | 1710 | 6B25 | 31   | 3B   | LEAY                        | -5,Y   |                   |
| 1742 | 1711 |      |      |      | *set write data             |        |                   |
| 1743 | 1712 | 6B27 | 8D   | 1E   | WRS2                        | BSR    | WRDSET            |
| 1744 | 1713 |      |      |      | *check last page            |        |                   |
| 1745 | 1714 | 6B29 | B6   | 604C | LDA                         | CNT    |                   |
| 1746 | 1715 | 6B2C | 81   | 01   | CMPA                        | #1     |                   |
| 1747 | 1716 | 6B2E | 26   | 0E   | BNE                         | WRS3   |                   |

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1748 1717
1749 1718      *add footer
1750 1719 6B30      WRS211
1751 1720      * CMPU #RWBUF+512-5
1752 1721      * BLS WRS21
1753 1722      * RTS
1754 1723 6B30 86 FF WRS21 LDA    #$FF
1755 1724 6B32 A7 C0      STA    ,U+
1756 1725 6B34 4F      CLRA
1757 1726 6B35 A7 C0      STA    ,U+
1758 1727 6B37 A7 C0      STA    ,U+
1759 1728 6B39 BE 6063      LDX    MCEXP
1760 1729 6B3C AF C1      STX    ,U++
1761 1730 6B3E 39      WRS3    RTS
1762 1731
1763 1732 6B3F      WRS1
1764 1733      *add header of basic file
1765 1734 6B3F 86 FF      LDA    #$FF
1766 1735 6B41 A7 C0      STA    ,U+
1767 1736 6B43 A7 C0      STA    ,U+
1768 1737 6B45 A7 C0      STA    ,U+
1769 1738
1770 1739      *data set
1771 1740 6B47 A6 80 WRDSET LDA    ,X+
1772 1741 6B49 A7 C0      STA    ,U+
1773 1742 6B4B 31 3F      LEAY    -1,Y
1774 1743 6B4D 26 F8      BNE     WRDSET
1775 1744 6B4F 39      RTS
1776 1745
1777 1746      *write data to sd
1778 1747 6B50      WRTSD EQU    *
1779 1748 6B50 17 01A1      LBSR    SETSCT      *set SCT No
1780 1749      *copy sct no to cmd param
1781 1750 6B53 8E 6004 WRTSD1 LDX    #SCTN0
1782 1751 6B56 108E 6F3F      LDY    #CMD24+1
1783 1752 6B5A C6 04      LDB     #4
1784 1753 6B5C 17 00BA      LBSR    TRANS
1785 1754      *send cmd
1786 1755 6B5F 17 01C1      LBSR    SETCKL

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1787 1756 6B62 C6 06          LDB      #6
1788 1757 6B64 8E 6F3E          LDX      #CMD24
1789 1758 6B67 A6 80 SDW1      LDA      ,X+
1790 1759 6B69 17 0149          LBSR      SNDDAT
1791 1760 6B6C 5A              DECB
1792 1761 6B6D 26 F8          BNE      SDW1
1793 1762          *get cmd response
1794 1763 6B6F          SDW2
1795 1764 6B6F 17 015F          LBSR      GETDAT
1796 1765 6B72 B6 6008          LDA      RESDAT
1797 1766 6B75 81 00          CMPA      #$00
1798 1767 6B77 26 F6          BNE      SDW2
1799 1768          *1byte skip
1800 1769 6B79 86 FF          LDA      #$FF
1801 1770 6B7B 17 0137          LBSR      SNDDAT      1byte skip
1802 1771
1803 1772          *send data packet
1804 1773 6B7E 8E 5E00          LDX      #RWBUF      <-temp -1=""
1805 008d="" 0093="" 00b4="" 00b6="" 00ce="" 011a="" 011d="" 0126=""
1806 0129="" 012b="" 0133="" 01="" 0200="" 02="" 03="" 04="" 05="" 06=""
1807 08="" 0:sd="" 0f="" 108e="" 10be="" 12="" 16="" 1774="" 1775=""
1808 1776="" 1777="" 1778="" 1779="" 1780="" 1781="" 1782="" 1783=""
1809 1784="" 1785="" 1786="" 1787="" 1788="" 1789="" 1790="" 1791=""
1810 1792="" 1793="" 1794="" 1795="" 1796="" 1797="" 1798="" 1799="" 17=""
1811 1800="" 1801="" 1802="" 1803="" 1804="" 1805="" 1806="" 1807=""
1812 1808="" 1809="" 1810="" 1811="" 1812="" 1813="" 1814="" 1815=""
1813 1816="" 1817="" 1818="" 1819="" 1820="" 1821="" 1822="" 1823=""
1814 1824="" 1825="" 1826="" 1827="" 1828="" 1829="" 1830="" 1831=""
1815 1832="" 1833="" 1834="" 1835="" 1836="" 1837="" 1838="" 1839=""
1816 1840="" 1841="" 1842="" 1843="" 1844="" 1845="" 1846="" 1847=""
1817 1848="" 1849="" 1850="" 1851="" 1852="" 1853="" 1854="" 1855=""
1818 1856="" 1857="" 1858="" 1859="" 1860="" 1861="" 1862="" 1863=""
1819 1864="" 1865="" 1866="" 1867="" 1868="" 1869="" 1870="" 1871=""
1820 1872="" 1873="" 1874="" 1875="" 1876="" 1877="" 1878="" 1879=""
1821 1880="" 1881="" 1882="" 1883="" 1884="" 1885="" 1886="" 1887=""
1822 1888="" 1889="" 1890="" 1891="" 1892="" 1893="" 1894="" 1895=""
1823 1896="" 1897="" 1898="" 1899="" 1900="" 1901="" 1902="" 1903=""
1824 1904="" 1905="" 1906="" 1907="" 1908="" 1909="" 1910="" 1911=""
1825 1912="" 1913="" 1914="" 1915="" 1916="" 1917="" 1918="" 1919=""

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1826 1920="" 1921="" 1922="" 1923="" 1924="" 1925="" 1926="" 1927=""  
1827 1928="" 1929="" 1930="" 1931="" 1932="" 1933="" 1934="" 1935=""  
1828 1936="" 1937="" 1938="" 1939="" 1940="" 1941="" 1942="" 1943=""  
1829 1944="" 1945="" 1946="" 1947="" 1948="" 1949="" 1950="" 1951=""  
1830 1952="" 1953="" 1954="" 1955="" 1956="" 1957="" 1958="" 1959=""  
1831 1960="" 1961="" 1962="" 1963="" 1964="" 1965="" 1966="" 1967=""  
1832 1968="" 1969="" 1970="" 1971="" 1972="" 1973="" 1974="" 1975=""  
1833 1976="" 1977="" 1978="" 1979="" 1980="" 1981="" 1982="" 1983=""  
1834 1984="" 1985="" 1986="" 1987="" 1988="" 1989="" 1990="" 1991=""  
1835 1992="" 1993="" 1994="" 1995="" 1996="" 1997="" 1998="" 1="" 1a=""  
1836 1e="" 1f="" 20="" 24="" 26="" 27="" 2="" 31="" 34="" 35="" 36=""  
1837 39="" 3="" 3a="" 3d="" 3f="" 40="" 44="" 48="" 4c="" 50="" 54=""  
1838 5a="" 5c="" 5e00="" 5e="" 6004="" 6008="" 6009="" 600e="" 6012=""  
1839 6016="" 601f="" 605a="" 605b="" 60="" 68="" 6b81="" 6b85="" 6b87=""  
1840 6b8a="" 6b8c="" 6b8f="" 6b91="" 6b93="" 6b95="" 6b98="" 6b9b=""  
1841 6b9e="" 6ba1="" 6ba3="" 6ba5="" 6ba8="" 6bab="" 6bad="" 6baf=""  
1842 6bb0="" 6bb3="" 6bb7="" 6bb9="" 6bbb="" 6bbe="" 6bc1="" 6bc3=""  
1843 6bc5="" 6bc6="" 6bc8="" 6bcb="" 6bce="" 6bd1="" 6bd5="" 6bd7=""  
1844 6bd9="" 6bdc="" 6bdd="" 6be0="" 6be3="" 6be5="" 6be7="" 6be9=""  
1845 6bec="" 6bef="" 6bf1="" 6bf3="" 6bf5="" 6bf8="" 6bfa="" 6bfc=""  
1846 6bff="" 6c00="" 6c02="" 6c04="" 6c06="" 6c08="" 6c0a="" 6c0d=""  
1847 6c0f="" 6c11="" 6c13="" 6c15="" 6c18="" 6c19="" 6c1b="" 6c1d=""  
1848 6c1f="" 6c20="" 6c22="" 6c24="" 6c26="" 6c28="" 6c29="" 6c2b=""  
1849 6c2d="" 6c2f="" 6c31="" 6c33="" 6c35="" 6c37="" 6c39="" 6c3b=""  
1850 6c3c="" 6c3e="" 6c40="" 6c42="" 6c44="" 6c46="" 6c48="" 6c4a=""  
1851 6c4c="" 6c4e="" 6c50="" 6c52="" 6c54="" 6c56="" 6c58="" 6c5a=""  
1852 6c5c="" 6c5d="" 6c5f="" 6c61="" 6c64="" 6c68="" 6c6a="" 6c6c=""  
1853 6c6f="" 6c71="" 6c74="" 6c76="" 6c78="" 6c79="" 6c7b="" 6c7d=""  
1854 6c80="" 6c82="" 6c84="" 6c87="" 6c8b="" 6c8d="" 6c90="" 6c92=""  
1855 6c94="" 6c96="" 6c97="" 6c9a="" 6c9d="" 6c="" 6ca1="" 6ca3="" 6ca5=""  
1856 6ca8="" 6caa="" 6cac="" 6cad="" 6caf="" 6cb2="" 6cb4="" 6cb5=""  
1857 6cb7="" 6cba="" 6cbc="" 6cbe="" 6cc0="" 6cc2="" 6cc5="" 6cc7=""  
1858 6cc9="" 6cca="" 6ccc="" 6ccd="" 6ccf="" 6cd1="" 6cd3="" 6cd5=""  
1859 6cd7="" 6cd8="" 6cd9="" 6cda="" 6cdc="" 6cdf="" 6ce0="" 6f38=""  
1860 6f39="" 70="" 7d="" 7f="" 80="" 81="" 84="" 86="" 88="" 89="" 8a=""  
1861 8b="" 8d="" 8e="" 96="" a0="" a6="" a7="" a:fill="" a="" abx="" ad=""  
1862 add0="" add1="" add2="" add3="" add40h="" adda="" addb="" adr=""  
1863 adx0="" adx1="" adx2="" adx3="" anda="" b1-="" b6="" b7="" b:cnt=""  
1864 bcc="" beq="" bne="" bra="" bsr="" buf="" busy="" c6="" cb="" ckrs=""

```

1865 clr="" cluster="" cmd="" cmpa="" copy="" crch="" crcl="" cs:b0="L"
1866 d="" data="" decb="" df="" dir="" dirofs="" disp="" dpgno="" dspb1=""
1867 dspb2="" dspb3="" dspbin="" dspbuf="" e5="" ec="" ed="" equ="" exg=""
1868 f4="" f5="" f6="" f7="" f7cd="" f7f1="" f9="" fb="" fc="" fd9e=""
1869 fe="" ff3b="" ff77="" ff="" fill="" filop="" get="" getd1=""
1870 getdat="" inc="" incx4="" lbsr="" lda="" ldb="" ldd="" ldx="" ldy=""
1871 leay="" ls1a="" lsra="" miso:b1="H(for" no="" nop="" of="" ora=""
1872 param="" pbdatt="" pcr1f="" pia="" pshs="" puls="" rd1="" rd1sct=""
1873 rd2="" rd3="" read="" resdat="" response="" return="" rts="" save=""
1874 sct="" sd:2="" sdhc:4="" sdhc="" send="" setckh="" setckl=""
1875 setsno="" sndd1="" snddat="" snddt1="" src="" sta="" temp="" tfr=""
1876 to="" token="" trans="" trlop="" tst="" until="" wait="" wrdlop=""
1877 wrdskp="" wrf1="" wrf2="" wrf3="" wrtdir="" wrtfat="" wrtsd1=""
1878 wspfl0="" x40h="" y:dst="" y="">C
1879 1999 6CE1 79 6008 ROL RESDAT C->b0
1880 2000 6CE4 5A DECB
1881 2001 6CE5 26 EE BNE GETD1
1882 2002 6CE7 8D 3A BSR SETCKL
1883 2003 6CE9 35 86 PULS A,B,PC
1884 2004 *
1885 2005 *-----
1886 2006 *input SCT No
1887 2007 6CEB INSCT EQU *
1888 2008 6CEB 8E 7046 LDX #MSSCT
1889 2009 6CEE 17 0132 LBSR PSTRNG
1890 2010 6CF1 17 0085 LBSR GET8CH get 8chr, convert
1891 4bin
1892 2011
1893 2012 6CF4 SETSCT EQU *
1894 2013 6CF4 34 32 PSHS A,X,Y
1895 2014 6CF6 7D 6009 TST SDHC 0:SD,1:SDHC
1896 2015 6CF9 26 1A BNE CALCHC
1897 2016 *SD card
1898 2017 6CFB 8E 6005 LDX #SCTN1
1899 2018 6CFE 108E 6004 LDY #SCTN0
1900 2019 6D02 C6 03 LDB #3
1901 2020 6D04 17 FF12 LBSR TRANS
1902 2021 6D07 7F 6007 CLR SCTN3
1903 2022 *calc top adrs of sct

```

|      |      |      |      |                         |     |      |          |       |
|------|------|------|------|-------------------------|-----|------|----------|-------|
| 1904 | 2023 | 6D0A |      | CALCSD                  | EQU | *    |          |       |
| 1905 | 2024 | 6D0A | 78   | 6007                    |     | LSL  | SCTN3    |       |
| 1906 | 2025 | 6D0D | 79   | 6006                    |     | ROL  | SCTN2    |       |
| 1907 | 2026 | 6D10 | 79   | 6005                    |     | ROL  | SCTN1    |       |
| 1908 | 2027 | 6D13 | 35   | B2                      |     | PULS | A,X,Y,PC |       |
| 1909 | 2028 |      |      |                         |     |      |          |       |
| 1910 | 2029 |      |      | *SDHC card              |     |      |          |       |
| 1911 | 2030 |      |      | *calc top adrs of sct   |     |      |          |       |
| 1912 | 2031 | 6D15 |      | CALCHC                  | EQU | *    |          |       |
| 1913 | 2032 | 6D15 | 8E   | 6004                    |     | LDX  | #SCTN0   |       |
| 1914 | 2033 | 6D18 | 108E | 6F39                    |     | LDY  | #CMD17+1 |       |
| 1915 | 2034 | 6D1C | C6   | 04                      |     | LDB  | #4       |       |
| 1916 | 2035 | 6D1E | 17   | FEF8                    |     | LBSR | TRANS    |       |
| 1917 | 2036 | 6D21 | 35   | B2                      |     | PULS | A,X,Y,PC |       |
| 1918 | 2037 |      |      |                         |     |      |          |       |
| 1919 | 2038 |      |      | *                       |     |      |          |       |
| 1920 | 2039 |      |      | *set sck:L              |     |      |          |       |
| 1921 | 2040 | 6D23 |      | SETCKL                  | EQU | *    |          |       |
| 1922 | 2041 | 6D23 | 34   | 02                      |     | PSHS | A        |       |
| 1923 | 2042 | 6D25 | 86   | 34                      |     | LDA  | #\$34    | ca2=L |
| 1924 | 2043 | 6D27 | B7   | FD9F                    |     | STA  | PBCTL    |       |
| 1925 | 2044 | 6D2A | 35   | 82                      |     | PULS | A,PC     |       |
| 1926 | 2045 |      |      |                         |     |      |          |       |
| 1927 | 2046 |      |      | *set sck:H              |     |      |          |       |
| 1928 | 2047 | 6D2C |      | SETCKH                  | EQU | *    |          |       |
| 1929 | 2048 | 6D2C | 34   | 02                      |     | PSHS | A        |       |
| 1930 | 2049 | 6D2E | 86   | 3C                      |     | LDA  | #\$3C    | ca2=H |
| 1931 | 2050 | 6D30 | B7   | FD9F                    |     | STA  | PBCTL    |       |
| 1932 | 2051 | 6D33 | 35   | 82                      |     | PULS | A,PC     |       |
| 1933 | 2052 |      |      |                         |     |      |          |       |
| 1934 | 2053 |      |      | *binary(A)->ascii(A)(B) |     |      |          |       |
| 1935 | 2054 | 6D35 |      | DSPBIN                  | EQU | *    |          |       |
| 1936 | 2055 | 6D35 | 8D   | 04                      |     | BSR  | DSPBI0   |       |
| 1937 | 2056 | 6D37 | BD   | 6D73                    |     | JSR  | OUTSPC   |       |
| 1938 | 2057 | 6D3A | 39   |                         |     | RTS  |          |       |
| 1939 | 2058 |      |      | *                       |     |      |          |       |
| 1940 | 2059 | 6D3B |      | DSPBI0                  | EQU | *    |          |       |
| 1941 | 2060 | 6D3B | 34   | 06                      |     | PSHS | A,B      |       |
| 1942 | 2061 | 6D3D | 1F   | 89                      |     | TFR  | A,B      |       |

|      |      |      |       |        |      |        |
|------|------|------|-------|--------|------|--------|
| 1943 | 2062 | 6D3F | 44    |        | LSRA |        |
| 1944 | 2063 | 6D40 | 44    |        | LSRA |        |
| 1945 | 2064 | 6D41 | 44    |        | LSRA |        |
| 1946 | 2065 | 6D42 | 44    |        | LSRA |        |
| 1947 | 2066 | 6D43 | 8A    | 30     | ORA  | #\$30  |
| 1948 | 2067 | 6D45 | C4    | 0F     | ANDB | #\$0F  |
| 1949 | 2068 | 6D47 | CA    | 30     | ORB  | #\$30  |
| 1950 | 2069 | 6D49 | 81    | 39     | CMPA | #\$39  |
| 1951 | 2070 | 6D4B | 23    | 02     | BLS  | BI21   |
| 1952 | 2071 | 6D4D | 8B    | 07     | ADDA | #\$07  |
| 1953 | 2072 | 6D4F | C1    | 39     | BI21 | CMPB   |
| 1954 | 2073 | 6D51 | 23    | 02     | BLS  | BI22   |
| 1955 | 2074 | 6D53 | CB    | 07     | ADDB | #\$07  |
| 1956 | 2075 | 6D55 | BD    | FBBB   | BI22 | JSR    |
| 1957 | 2076 | 6D58 | 1F    | 98     | TFR  | B,A    |
| 1958 | 2077 | 6D5A | BD    | FBBB   | JSR  | OUTCHA |
| 1959 | 2078 | 6D5D | 35    | 86     | PULS | A,B,PC |
| 1960 | 2079 |      |       |        |      |        |
| 1961 | 2080 |      | *     |        |      |        |
| 1962 | 2081 | 6D5F |       | DSPBI2 | EQU  | *      |
| 1963 | 2082 | 6D5F | A6    | 80     | LDA  | ,X+    |
| 1964 | 2083 | 6D61 | 8D    | D8     | BSR  | DSPBI0 |
| 1965 | 2084 | 6D63 | A6    | 80     | LDA  | ,X+    |
| 1966 | 2085 | 6D65 | 8D    | D4     | BSR  | DSPBI0 |
| 1967 | 2086 | 6D67 | 39    |        | RTS  |        |
| 1968 | 2087 |      |       |        |      |        |
| 1969 | 2088 |      | *     |        |      |        |
| 1970 | 2089 |      | *crlf |        |      |        |
| 1971 | 2090 | 6D68 |       | PCRLF  | EQU  | *      |
| 1972 | 2091 | 6D68 | 86    | 0D     | LDA  | #CR    |
| 1973 | 2092 | 6D6A | BD    | FBBB   | JSR  | OUTCHA |
| 1974 | 2093 | 6D6D | 86    | 0A     | LDA  | #LF    |
| 1975 | 2094 | 6D6F | BD    | FBBB   | JSR  | OUTCHA |
| 1976 | 2095 | 6D72 | 39    |        | RTS  |        |
| 1977 | 2096 |      |       |        |      |        |
| 1978 | 2097 |      | *out  | spc    |      |        |
| 1979 | 2098 | 6D73 |       | OUTSPC | EQU  | *      |
| 1980 | 2099 | 6D73 | 86    | 20     | LDA  | #SPC   |
| 1981 | 2100 | 6D75 | BD    | FBBB   | JSR  | OUTCHA |

|      |      |      |                 |        |       |      |           |
|------|------|------|-----------------|--------|-------|------|-----------|
| 1982 | 2101 | 6D78 | 39              |        | RTS   |      |           |
| 1983 | 2102 |      |                 |        |       |      |           |
| 1984 | 2103 |      | *               |        |       |      |           |
| 1985 | 2104 | 6D79 |                 | GET8CH | EQU   | *    |           |
| 1986 | 2105 |      | *input chr      |        |       |      |           |
| 1987 | 2106 | 6D79 | 8E              | 6077   |       | LDX  | #INPBUF   |
| 1988 | 2107 | 6D7C | 5F              |        |       | CLRB |           |
| 1989 | 2108 | 6D7D | BD              | 6F12   | CLOP  | JSR  | INOUTC    |
| 1990 | 2109 | 6D80 | 81              | 0D     |       | CMPA | #CR       |
| 1991 | 2110 | 6D82 | 27              | 07     |       | BEQ  | INEND     |
| 1992 | 2111 | 6D84 | A7              | 85     |       | STA  | B,X       |
| 1993 | 2112 | 6D86 | 5C              |        |       | INCB |           |
| 1994 | 2113 | 6D87 | C1              | 08     |       | CMPB | #8        |
| 1995 | 2114 | 6D89 | 25              | F2     |       | BLO  | CLOP      |
| 1996 | 2115 |      |                 |        |       |      |           |
| 1997 | 2116 | 6D8B |                 | INEND  |       |      |           |
| 1998 | 2117 |      | *shift          |        |       |      |           |
| 1999 | 2118 | 6D8B | 5D              |        |       | TSTB |           |
| 2000 | 2119 | 6D8C | 27              | 21     |       | BEQ  | CLEND     |
| 2001 | 2120 | 6D8E | 8E              | 6077   |       | LDX  | #INPBUF   |
| 2002 | 2121 | 6D91 | 3A              |        |       | ABX  |           |
| 2003 | 2122 | 6D92 | 108E            | 607F   |       | LDY  | #INPBUF+8 |
| 2004 | 2123 | 6D96 | 34              | 04     |       | PSHS | B         |
| 2005 | 2124 | 6D98 | A6              | 82     | SFLOP | LDA  | , -X      |
| 2006 | 2125 | 6D9A | A7              | A2     |       | STA  | , -Y      |
| 2007 | 2126 | 6D9C | 5A              |        |       | DECB |           |
| 2008 | 2127 | 6D9D | 26              | F9     |       | BNE  | SFLOP     |
| 2009 | 2128 |      | *clr            |        |       |      |           |
| 2010 | 2129 | 6D9F | C6              | 08     | CL1   | LDB  | #8        |
| 2011 | 2130 | 6DA1 | E0              | E0     |       | SUBB | ,S+       |
| 2012 | 2131 | 6DA3 | 27              | 0A     |       | BEQ  | CLEND     |
| 2013 | 2132 | 6DA5 | 8E              | 6077   |       | LDX  | #INPBUF   |
| 2014 | 2133 | 6DA8 | 86              | 30     |       | LDA  | #'0       |
| 2015 | 2134 | 6DAA | A7              | 80     | CLLOP | STA  | ,X+       |
| 2016 | 2135 | 6DAC | 5A              |        |       | DECB |           |
| 2017 | 2136 | 6DAD | 26              | FB     |       | BNE  | CLLOP     |
| 2018 | 2137 | 6DAF |                 | CLEND  |       |      |           |
| 2019 | 2138 |      |                 |        |       |      |           |
| 2020 | 2139 |      | *convert to bin |        |       |      |           |

|      |      |      |      |        |        |      |         |
|------|------|------|------|--------|--------|------|---------|
| 2021 | 2140 | 6DAF | C6   | 04     |        | LDB  | #4      |
| 2022 | 2141 | 6DB1 | 8E   | 6077   |        | LDX  | #INPBUF |
| 2023 | 2142 | 6DB4 | 108E | 6004   |        | LDY  | #SCTN0  |
| 2024 | 2143 | 6DB8 |      | CVT1   |        |      |         |
| 2025 | 2144 | 6DB8 | A6   | 80     |        | LDA  | ,X+     |
| 2026 | 2145 | 6DBA | 8D   | 1E     |        | BSR  | CVBIN   |
| 2027 | 2146 | 6DBC | 25   | 16     |        | BCS  | EREND1  |
| 2028 | 2147 | 6DBE | 48   |        |        | LSLA |         |
| 2029 | 2148 | 6DBF | 48   |        |        | LSLA |         |
| 2030 | 2149 | 6DC0 | 48   |        |        | LSLA |         |
| 2031 | 2150 | 6DC1 | 48   |        |        | LSLA |         |
| 2032 | 2151 | 6DC2 | 34   | 02     |        | PSHS | A       |
| 2033 | 2152 | 6DC4 | A6   | 80     |        | LDA  | ,X+     |
| 2034 | 2153 | 6DC6 | 8D   | 12     |        | BSR  | CVBIN   |
| 2035 | 2154 | 6DC8 | 25   | 08     |        | BCS  | EREND0  |
| 2036 | 2155 | 6DCA | AB   | E0     |        | ADDA | ,S+     |
| 2037 | 2156 | 6DCC | A7   | A0     |        | STA  | ,Y+     |
| 2038 | 2157 | 6DCE | 5A   |        |        | DECB |         |
| 2039 | 2158 | 6DCF | 26   | E7     |        | BNE  | CVT1    |
| 2040 | 2159 | 6DD1 | 39   |        |        | RTS  |         |
| 2041 | 2160 |      |      |        |        |      |         |
| 2042 | 2161 | 6DD2 | 35   | 02     | EREND0 | PULS | A       |
| 2043 | 2162 | 6DD4 | 8E   | 708C   | EREND1 | LDX  | #MSERR  |
| 2044 | 2163 | 6DD7 | 8D   | 4A     |        | BSR  | PSTRNG  |
| 2045 | 2164 | 6DD9 | 39   | EEND   |        | RTS  |         |
| 2046 | 2165 |      |      |        |        |      |         |
| 2047 | 2166 |      |      |        |        |      |         |
| 2048 | 2167 |      |      |        |        |      |         |
| 2049 | 2168 | 6DDA |      | CVBIN  | EQU    | *    |         |
| 2050 | 2169 | 6DDA | 80   | 30     |        | SUBA | #'0     |
| 2051 | 2170 | 6DDC | 2B   | 11     |        | BMI  | CVERR   |
| 2052 | 2171 | 6DDE | 81   | 09     |        | CMPA | #9      |
| 2053 | 2172 | 6DE0 | 23   | 0A     |        | BLS  | CVBEND  |
| 2054 | 2173 | 6DE2 | 81   | 11     |        | CMPA | #\$11   |
| 2055 | 2174 | 6DE4 | 25   | 09     |        | BLO  | CVERR   |
| 2056 | 2175 | 6DE6 | 81   | 16     |        | CMPA | #\$16   |
| 2057 | 2176 | 6DE8 | 22   | 05     |        | BHI  | CVERR   |
| 2058 | 2177 | 6DEA | 80   | 07     |        | SUBA | #7      |
| 2059 | 2178 | 6DEC |      | CVBEND |        |      |         |

|      |      |      |    |    |       |                           |         |         |
|------|------|------|----|----|-------|---------------------------|---------|---------|
| 2060 | 2179 | 6DEC | 1C | FE |       | ANDCC                     | #\$FE   | reset C |
| 2061 | 2180 | 6DEE | 39 |    |       | RTS                       |         |         |
| 2062 | 2181 | 6DEF | 1A | 01 | CVERR | ORCC                      | #\$01   | set C   |
| 2063 | 2182 | 6DF1 | 39 |    |       | RTS                       |         |         |
| 2064 | 2183 |      |    |    |       |                           |         |         |
| 2065 | 2184 |      |    |    |       | *-----                    |         |         |
| 2066 | 2185 |      |    |    |       | *add 2bytes to 4bytes     |         |         |
| 2067 | 2186 |      |    |    |       | * X,X+1,X+2,X+3 + Y,Y+1   |         |         |
| 2068 | 2187 |      |    |    |       | * data X is big endian    |         |         |
| 2069 | 2188 |      |    |    |       | * data Y is little endian |         |         |
| 2070 | 2189 | 6DF2 |    |    | ADD42 | EQU                       | *       |         |
| 2071 | 2190 | 6DF2 | 34 | 36 |       | PSHS                      | A,B,X,Y |         |
| 2072 | 2191 | 6DF4 | 32 | 7C |       | LEAS                      | -4,S    |         |
| 2073 | 2192 | 6DF6 | C6 | 04 |       | LDB                       | #4      |         |
| 2074 | 2193 | 6DF8 | A6 | 80 | AD1   | LDA                       | ,X+     |         |
| 2075 | 2194 | 6DFA | A7 | E0 |       | STA                       | ,S+     |         |
| 2076 | 2195 | 6DFC | 5A |    |       | DECB                      |         |         |
| 2077 | 2196 | 6DFD | 26 | F9 |       | BNE                       | AD1     |         |
| 2078 | 2197 |      |    |    |       |                           |         |         |
| 2079 | 2198 |      |    |    |       | *add (lower)              |         |         |
| 2080 | 2199 | 6DFF | A6 | A0 |       | LDA                       | ,Y+     |         |
| 2081 | 2200 | 6E01 | AB | E2 |       | ADDA                      | , -S    |         |
| 2082 | 2201 | 6E03 | A7 | E4 |       | STA                       | ,S      |         |
| 2083 | 2202 | 6E05 | 24 | 02 |       | BCC                       | AD2     |         |
| 2084 | 2203 | 6E07 | 6C | 7F |       | INC                       | -1,S    |         |
| 2085 | 2204 |      |    |    |       | *add (upper)              |         |         |
| 2086 | 2205 | 6E09 | A6 | A4 | AD2   | LDA                       | ,Y      |         |
| 2087 | 2206 | 6E0B | AB | E2 |       | ADDA                      | , -S    |         |
| 2088 | 2207 | 6E0D | A7 | E4 |       | STA                       | ,S      |         |
| 2089 | 2208 | 6E0F | 24 | 02 |       | BCC                       | AD3     |         |
| 2090 | 2209 | 6E11 | 6C | 7F |       | INC                       | -1,S    |         |
| 2091 | 2210 | 6E13 | 30 | 1C | AD3   | LEAX                      | -4,X    |         |
| 2092 | 2211 | 6E15 | 32 | 7E |       | LEAS                      | -2,S    |         |
| 2093 | 2212 | 6E17 | C6 | 04 |       | LDB                       | #4      |         |
| 2094 | 2213 | 6E19 | A6 | E0 | AD4   | LDA                       | ,S+     |         |
| 2095 | 2214 | 6E1B | A7 | 80 |       | STA                       | ,X+     |         |
| 2096 | 2215 | 6E1D | 5A |    |       | DECB                      |         |         |
| 2097 | 2216 | 6E1E | 26 | F9 |       | BNE                       | AD4     |         |
| 2098 | 2217 | 6E20 | 35 | 36 |       | PULS                      | A,B,X,Y |         |

|      |      |      |    |        |                        |        |      |    |
|------|------|------|----|--------|------------------------|--------|------|----|
| 2099 | 2218 | 6E22 | 39 |        | RTS                    |        |      |    |
| 2100 | 2219 |      |    |        |                        |        |      |    |
| 2101 | 2220 |      |    |        | *-----                 |        |      |    |
| 2102 | 2221 |      |    |        | * pstring              |        |      |    |
| 2103 | 2222 | 6E23 |    | PSTRNG | EQU                    | *      |      |    |
| 2104 | 2223 | 6E23 | 34 | 02     | PSHS                   | A      |      |    |
| 2105 | 2224 | 6E25 | A6 | 80     | PSLOP                  | LDA    | ,X+  |    |
| 2106 | 2225 | 6E27 | 81 | 04     | CMPA                   | #4     |      |    |
| 2107 | 2226 | 6E29 | 27 | 05     | BEQ                    | PSEND  |      |    |
| 2108 | 2227 | 6E2B | BD | FBBB   | JSR                    | OUTCHA |      |    |
| 2109 | 2228 | 6E2E | 20 | F5     | BRA                    | PSLOP  |      |    |
| 2110 | 2229 | 6E30 | 35 | 02     | PSEND                  | PULS   | A    |    |
| 2111 | 2230 | 6E32 | 39 |        | RTS                    |        |      |    |
| 2112 | 2231 |      |    |        |                        |        |      |    |
| 2113 | 2232 |      |    |        | *-----                 |        |      |    |
| 2114 | 2233 |      |    |        | *disp MM-DD-YY         |        |      |    |
| 2115 | 2234 |      |    |        | *indata A(Low),B(High) |        |      |    |
| 2116 | 2235 | 6E33 |    | DSPYMD | EQU                    | *      |      |    |
| 2117 | 2236 | 6E33 | 1E | 89     | EXG                    | A,B    |      |    |
| 2118 | 2237 | 6E35 | 34 | 10     | PSHS                   | X      |      |    |
| 2119 | 2238 | 6E37 | 34 | 06     | PSHS                   | A,B    |      |    |
| 2120 | 2239 | 6E39 | C4 | E0     | ANDB                   | #\$E0  |      |    |
| 2121 | 2240 | 6E3B | 54 |        | LSRB                   |        |      |    |
| 2122 | 2241 | 6E3C | 54 |        | LSRB                   |        |      |    |
| 2123 | 2242 | 6E3D | 54 |        | LSRB                   |        |      |    |
| 2124 | 2243 | 6E3E | 54 |        | LSRB                   |        |      |    |
| 2125 | 2244 | 6E3F | 54 |        | LSRB                   |        |      |    |
| 2126 | 2245 | 6E40 | 47 |        | ASRA                   |        |      |    |
| 2127 | 2246 | 6E41 | 24 | 02     | BCC                    | DYMD1  |      |    |
| 2128 | 2247 | 6E43 | CB | 08     | ADDB                   | #8     |      |    |
| 2129 | 2248 | 6E45 | F7 | 603F   | DYMD1                  | STB    | WMDY | MM |
| 2130 | 2249 | 6E48 | 35 | 06     | PULS                   | A,B    |      |    |
| 2131 | 2250 | 6E4A | 34 | 04     | PSHS                   | B      |      |    |
| 2132 | 2251 | 6E4C | C4 | 1F     | ANDB                   | #\$1F  |      |    |
| 2133 | 2252 | 6E4E | F7 | 6040   | STB                    | WMDY+1 | DD   |    |
| 2134 | 2253 | 6E51 | 35 | 04     | PULS                   | B      |      |    |
| 2135 | 2254 |      |    |        |                        |        |      |    |
| 2136 | 2255 | 6E53 | 44 |        | LSRA                   |        |      |    |
| 2137 | 2256 |      |    |        | * ASLB                 |        |      |    |

|      |      |      |    |      |                                      |          |  |             |
|------|------|------|----|------|--------------------------------------|----------|--|-------------|
| 2138 | 2257 |      |    |      | * BCC                                | DYMD2    |  |             |
| 2139 | 2258 |      |    |      | * INCA                               |          |  |             |
| 2140 | 2259 | 6E54 |    |      |                                      | DYMD2    |  |             |
| 2141 | 2260 | 6E54 | 80 | 14   | SUBA                                 | #20      |  | A+1980-2000 |
| 2142 | 2261 | 6E56 | B7 | 6041 | STA                                  | WMDY+2   |  | YY          |
| 2143 | 2262 |      |    |      |                                      |          |  |             |
| 2144 | 2263 |      |    |      | *disp                                | MM-DD-YY |  |             |
| 2145 | 2264 | 6E59 | BD | 6D73 | JSR                                  | OUTSPC   |  |             |
| 2146 | 2265 | 6E5C | 8E | 603F | LDX                                  | #WMDY    |  |             |
| 2147 | 2266 | 6E5F | A6 | 80   | LDA                                  | ,X+      |  |             |
| 2148 | 2267 | 6E61 | 8D | 14   | BSR                                  | DB2ASC   |  | MM          |
| 2149 | 2268 | 6E63 | 86 | 2D   | LDA                                  | #' -     |  |             |
| 2150 | 2269 | 6E65 | BD | FBBB | JSR                                  | OUTCHA   |  |             |
| 2151 | 2270 | 6E68 | A6 | 80   | LDA                                  | ,X+      |  |             |
| 2152 | 2271 | 6E6A | 8D | 0B   | BSR                                  | DB2ASC   |  | DD          |
| 2153 | 2272 | 6E6C | 86 | 2D   | LDA                                  | #' -     |  |             |
| 2154 | 2273 | 6E6E | BD | FBBB | JSR                                  | OUTCHA   |  |             |
| 2155 | 2274 | 6E71 | A6 | 80   | LDA                                  | ,X+      |  |             |
| 2156 | 2275 | 6E73 | 8D | 02   | BSR                                  | DB2ASC   |  | YY          |
| 2157 | 2276 | 6E75 | 35 | 90   | PULS                                 | X,PC     |  |             |
| 2158 | 2277 |      |    |      | *                                    |          |  |             |
| 2159 | 2278 |      |    |      | *-----                               |          |  |             |
| 2160 | 2279 |      |    |      | *convert decimal value to ascii code |          |  |             |
| 2161 | 2280 |      |    |      | *(A:BIN)->ASCII                      |          |  |             |
| 2162 | 2281 | 6E77 |    |      | DB2ASC EQU                           | *        |  |             |
| 2163 | 2282 | 6E77 | 34 | 06   | PSHS                                 | A,B      |  |             |
| 2164 | 2283 | 6E79 | 1F | 89   | TFR                                  | A,B      |  |             |
| 2165 | 2284 | 6E7B | 86 | 30   | LDA                                  | #\$30    |  |             |
| 2166 | 2285 | 6E7D | C1 | 0A   | DBLOP CMPB                           | #10      |  |             |
| 2167 | 2286 | 6E7F | 25 | 05   | BLO                                  | DB2      |  |             |
| 2168 | 2287 | 6E81 | C0 | 0A   | SUBB                                 | #10      |  |             |
| 2169 | 2288 | 6E83 | 4C |      | INCA                                 |          |  |             |
| 2170 | 2289 | 6E84 | 20 | F7   | BRA                                  | DBLOP    |  |             |
| 2171 | 2290 | 6E86 | BD | FBBB | DB2 JSR                              | OUTCHA   |  |             |
| 2172 | 2291 | 6E89 | 1F | 98   | TFR                                  | B,A      |  |             |
| 2173 | 2292 | 6E8B | 8B | 30   | ADDA                                 | #\$30    |  |             |
| 2174 | 2293 | 6E8D | BD | FBBB | JSR                                  | OUTCHA   |  |             |
| 2175 | 2294 | 6E90 | 35 | 86   | PULS                                 | A,B,PC   |  |             |
| 2176 | 2295 |      |    |      |                                      |          |  |             |

|      |      |      |    |      |                 |      |         |              |
|------|------|------|----|------|-----------------|------|---------|--------------|
| 2177 | 2296 |      |    |      | *disp FILE size |      |         |              |
| 2178 | 2297 | 6E92 |    |      | DSPSIZ EQU      | *    |         |              |
| 2179 | 2298 | 6E92 | 34 | 10   | PSHS            |      | X       |              |
| 2180 | 2299 | 6E94 | BE | 6042 | LDX             |      | WFSIZ   |              |
| 2181 | 2300 | 6E97 | 27 | 07   | BEQ             |      | DSS1    | if 64k over? |
| 2182 | 2301 | 6E99 | 8E | 7157 | LDX             |      | #MSOVR  |              |
| 2183 | 2302 | 6E9C | 8D | 85   | BSR             |      | PSTRNG  |              |
| 2184 | 2303 | 6E9E | 20 | 3E   | BRA             |      | DSEND   |              |
| 2185 | 2304 |      |    |      |                 |      |         |              |
| 2186 | 2305 | 6EA0 |    |      | DSS1            |      |         |              |
| 2187 | 2306 |      |    |      | * /1024(400h)   |      |         |              |
| 2188 | 2307 | 6EA0 | B6 | 6044 | LDA             |      | WFSIZ+2 | under 64k    |
| 2189 | 2308 | 6EA3 | 44 |      | LSRA            |      |         |              |
| 2190 | 2309 | 6EA4 | 44 |      | LSRA            |      |         |              |
| 2191 | 2310 |      |    |      | *disp dec       |      |         |              |
| 2192 | 2311 | 6EA5 | 1F | 89   | TFR             |      | A,B     |              |
| 2193 | 2312 | 6EA7 | 81 | 64   | CMPA            |      | #100    |              |
| 2194 | 2313 | 6EA9 | 23 | 10   | BLS             |      | DSS13   |              |
| 2195 | 2314 | 6EAB | 86 | 30   | LDA             |      | #'0     |              |
| 2196 | 2315 | 6EAD | C1 | 64   | DSL0P1          | CMPB | #100    |              |
| 2197 | 2316 | 6EAF | 25 | 05   | BLO             |      | DSS12   |              |
| 2198 | 2317 | 6EB1 | C0 | 64   | SUBB            |      | #100    |              |
| 2199 | 2318 | 6EB3 | 4C |      | INCA            |      |         |              |
| 2200 | 2319 | 6EB4 | 20 | F7   | BRA             |      | DSL0P1  |              |
| 2201 | 2320 | 6EB6 | BD | FBBB | DSS12           | JSR  | OUTCHA  |              |
| 2202 | 2321 | 6EB9 | 20 | 09   | BRA             |      | DSS14   |              |
| 2203 | 2322 |      |    |      |                 |      |         |              |
| 2204 | 2323 |      |    |      | *under 100      |      |         |              |
| 2205 | 2324 | 6EBB | BD | 6D73 | DSS13           | JSR  | OUTSPC  |              |
| 2206 | 2325 | 6EBE | 86 | 20   | LDA             |      | #SPC    |              |
| 2207 | 2326 | 6EC0 | C1 | 0A   | CMPB            |      | #10     |              |
| 2208 | 2327 | 6EC2 | 25 | 0B   | BLO             |      | DSS2    |              |
| 2209 | 2328 |      |    |      |                 |      |         |              |
| 2210 | 2329 | 6EC4 | 86 | 30   | DSS14           | LDA  | #'0     |              |
| 2211 | 2330 | 6EC6 | C1 | 0A   | DSL0P2          | CMPB | #10     |              |
| 2212 | 2331 | 6EC8 | 25 | 05   | BLO             |      | DSS2    |              |
| 2213 | 2332 | 6ECA | C0 | 0A   | SUBB            |      | #10     |              |
| 2214 | 2333 | 6ECC | 4C |      | INCA            |      |         |              |
| 2215 | 2334 | 6ECD | 20 | F7   | BRA             |      | DSL0P2  |              |

|      |      |      |      |      |        |      |           |                  |
|------|------|------|------|------|--------|------|-----------|------------------|
| 2216 | 2335 |      |      |      |        |      |           |                  |
| 2217 | 2336 | 6ECF | BD   | FBBB | DSS2   | JSR  | OUTCHA    | 10's place       |
| 2218 | 2337 |      |      |      |        |      |           |                  |
|      |      |      |      |      |        |      |           | *under 10        |
| 2219 | 2338 | 6ED2 | 1F   | 98   | DSS21  | TFR  | B,A       |                  |
| 2220 | 2339 | 6ED4 | 8B   | 31   |        | ADDA | #\$31     | '1               |
| 2221 | 2340 | 6ED6 | BD   | FBBB | DSS3   | JSR  | OUTCHA    | 1's place        |
| 2222 | 2341 | 6ED9 | 86   | 6B   |        | LDA  | #'k       |                  |
| 2223 | 2342 | 6EDB | BD   | FBBB |        | JSR  | OUTCHA    |                  |
| 2224 | 2343 | 6EDE | BD   | 6D73 | DSEND  | JSR  | OUTSPC    |                  |
| 2225 | 2344 | 6EE1 | 35   | 90   |        | PULS | X,PC      |                  |
| 2226 | 2345 |      |      |      |        |      |           |                  |
| 2227 | 2346 |      |      |      |        |      |           | *-----           |
| 2228 | 2347 |      |      |      |        |      |           | *disp cluster    |
| 2229 | 2348 |      |      |      |        |      |           | *return:clusterL |
| 2230 | 2349 | 6EE3 |      |      | DSPCLS | EQU  | *         |                  |
| 2231 | 2350 | 6EE3 | 34   | 10   |        | PSHS | X         |                  |
| 2232 | 2351 | 6EE5 | BD   | 6D73 |        | JSR  | OUTSPC    |                  |
| 2233 | 2352 | 6EE8 | 8E   | 604A |        | LDX  | #CLSTL    |                  |
| 2234 | 2353 | 6EEB | C6   | 02   |        | LDB  | #2        |                  |
| 2235 | 2354 | 6EED | A6   | 80   | DCL0P1 | LDA  | ,X+       |                  |
| 2236 | 2355 | 6EEF | 8D   | 86   |        | BSR  | DB2ASC    |                  |
| 2237 | 2356 | 6EF1 | 5A   |      |        | DECB |           |                  |
| 2238 | 2357 | 6EF2 | 26   | F9   |        | BNE  | DCL0P1    |                  |
| 2239 | 2358 | 6EF4 | 35   | 90   |        | PULS | X,PC      |                  |
| 2240 | 2359 |      |      |      |        |      |           |                  |
| 2241 | 2360 |      |      |      |        |      |           | *-----           |
| 2242 | 2361 | 6EF6 |      |      | FNMIN  | EQU  | *         |                  |
| 2243 | 2362 |      |      |      |        |      |           | *clr buf         |
| 2244 | 2363 | 6EF6 | 8E   | 605A |        | LDX  | #FNAME+11 |                  |
| 2245 | 2364 | 6EF9 | C6   | 0B   |        | LDB  | #8+3      |                  |
| 2246 | 2365 | 6EFB | 86   | 20   |        | LDA  | #SPC      |                  |
| 2247 | 2366 | 6EFD | A7   | 82   | FNIN0  | STA  | , -X      |                  |
| 2248 | 2367 | 6EFF | 5A   |      |        | DECB |           |                  |
| 2249 | 2368 | 6F00 | 26   | FB   |        | BNE  | FNIN0     |                  |
| 2250 | 2369 |      |      |      |        |      |           |                  |
| 2251 | 2370 |      |      |      |        |      |           | *fname in        |
| 2252 | 2371 | 6F02 | 34   | 24   |        | PSHS | B,Y       |                  |
| 2253 | 2372 | 6F04 | 108E | 604F |        | LDY  | #FNAME    |                  |
| 2254 | 2373 | 6F08 | C6   | 08   |        | LDB  | #8        |                  |

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2255 2374 6F0A 8D 0D          BSR      LININ
2256 2375 6F0C 86 20          LDA      #$20
2257 2376 6F0E A7 3F          STA      -1,Y
2258 2377 6F10 35 A4          PULS     B,Y,PC
2259 2378                *
2260 2379                *-----
2261 2380                *
2262 2381 6F12 BD  FBB8  INOUTC  JSR      INCHA
2263 2382 6F15 BD  FBBB          JSR      OUTCHA
2264 2383 6F18 39              RTS
2265 2384                *
2266 2385                *-----
2267 2386                * line input
2268 2387                * input: B:count, Y:buf
2269 2388                * output: FILBUF (terminate $00)
2270 2389                * temp buf $043D - $053C
2271 2390 6F19          LININ  EQU      *
2272 2391 6F19 0F BF          CLR      $BF
2273 2392 6F1B BD  D807          JSR      LININP
2274 2393
2275 2394 6F1E 8E 043D          LDX      #$043D
2276 2395 6F21 A6 80  LINLOP  LDA      ,X+
2277 2396 6F23 A7 A0          STA      ,Y+
2278 2397 6F25 5A              DECB
2279 2398 6F26 27 03          BEQ      LEND
2280 2399 6F28 4D              TSTA
2281 2400 6F29 26 F6          BNE      LINLOP
2282 2401 6F2B 39      LEND    RTS
2283 2402
2284 2403                *
2285 2404                *****
2286 2405                *===== parameter =====
2287 2406 6F2C 40      CMD0    FCB      $40,$00,$00,$00,$00,$95 '@
2288 2407 6F32 41      CMD1    FCB      $41,$00,$00,$00,$00,$FF 'A
2289 2408 6F38 51      CMD17   FCB      $51,$00,$00,$00,$00,$FF 'Q
2290 2409 6F3E 58      CMD24   FCB      $58,$00,$00,$00,$00,$FF 'X
2291 2410 6F44 48      CMD8     FCB      $48,$00,$00,$01,$AA,$87 'H
2292 2411 6F4A 7A      CMD58    FCB      $7A,$00,$00,$00,$00,$FF 'z
2293 2412 6F50 77      CMD55    FCB      $77,$00,$00,$00,$00,$FF 'w

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|      |      |      |    |         |     |                                                       |
|------|------|------|----|---------|-----|-------------------------------------------------------|
| 2294 | 2413 | 6F56 | 69 | ACMD41  | FCB | \$69,\$40,\$FF,\$80,\$00,\$FF 'i                      |
| 2295 | 2414 | 6F5C | 49 | CMD9    | FCB | \$49,\$00,\$00,\$00,\$00,\$FF 'I                      |
| 2296 | 2415 |      | *  |         |     |                                                       |
| 2297 | 2416 | 6F62 | 43 | MCMD0   | FCC | /CMD0/,CR,LF                                          |
| 2298 | 2417 | 6F68 | 04 |         | FCB | 4                                                     |
| 2299 | 2418 | 6F69 | 43 | MCMD8   | FCC | /CMD8/,CR,LF                                          |
| 2300 | 2419 | 6F6F | 04 |         | FCB | 4                                                     |
| 2301 | 2420 | 6F70 | 43 | MCMD58  | FCC | /CMD58/,CR,LF                                         |
| 2302 | 2421 | 6F77 | 04 |         | FCB | 4                                                     |
| 2303 | 2422 | 6F78 | 43 | MCMD55  | FCC | /CMD55/,CR,LF                                         |
| 2304 | 2423 | 6F7F | 04 |         | FCB | 4                                                     |
| 2305 | 2424 | 6F80 | 41 | MCMD41  | FCC | /ACMD41/,CR,LF                                        |
| 2306 | 2425 | 6F88 | 04 |         | FCB | 4                                                     |
| 2307 | 2426 | 6F89 | 43 | MCMD9   | FCC | /CMD9/,CR,LF                                          |
| 2308 | 2427 | 6F8F | 04 |         | FCB | 4                                                     |
| 2309 | 2428 |      | *  |         |     |                                                       |
| 2310 | 2429 | 6F90 | 0D | MSMENU1 | FCC | CR,LF,CR,LF,/Main MENU/                               |
| 2311 | 2430 | 6F9D | 0D |         | FCC | CR,LF,/F)iles, L)oad, saveB)asic,                     |
| 2312 |      |      |    |         |     | saveM)code, K)ill, reS)etSD, 0)ption, Q)uit: /        |
| 2313 | 2431 | 6FE8 | 04 |         | FCB | 4                                                     |
| 2314 | 2432 | 6FE9 | 0D | MSMENU2 | FCC | CR,LF,CR,LF,/Option MENU/                             |
| 2315 | 2433 | 6FF9 | 0D |         | FCC | CR,LF,/1:Read Sct, 2:Disp SD                          |
| 2316 |      |      |    |         |     | data, 3:Disp MBR, 4:Disp BPB, 5:Disp FAT, 0:Return: / |
| 2317 | 2434 | 7045 | 04 |         | FCB | 4                                                     |
| 2318 | 2435 | 7046 | 53 | MSSCT   | FCC | /SctNo: /                                             |
| 2319 | 2436 | 704D | 04 |         | FCB | 4                                                     |
| 2320 | 2437 | 704E | 0D | MSSD0   | FCC | CR,LF,/SD Card/                                       |
| 2321 | 2438 | 7057 | 04 |         | FCB | 4                                                     |
| 2322 | 2439 | 7058 | 0D | MSSD1   | FCC | CR,LF,/SDHC Card/                                     |
| 2323 | 2440 | 7063 | 04 |         | FCB | 4                                                     |
| 2324 | 2441 | 7064 | 0D | MSINIT  | FCC | CR,LF,CR,LF,/SD Card                                  |
| 2325 |      |      |    |         |     | setting.../                                           |
| 2326 | 2442 | 707A | 04 |         | FCB | 4                                                     |
| 2327 | 2443 | 707B | 53 | MERR    | FCC | /Setting error!/,CR,LF                                |
| 2328 | 2444 | 708B | 04 |         | FCB | 4                                                     |
| 2329 | 2445 | 708C | 49 | MSERR   | FCC | /Input error!/,CR,LF                                  |
| 2330 | 2446 | 709A | 04 |         | FCB | 4                                                     |
| 2331 | 2447 | 709B | 0D | MSOFS   | FCC | CR,LF,/Sector no is:/                                 |
| 2332 | 2448 | 70AA | 04 |         | FCB | 4                                                     |

|      |       |      |       |        |     |                                    |
|------|-------|------|-------|--------|-----|------------------------------------|
| 2333 | 2449  | 70AB | 0D    | MSBPB  | FCC | CR,LF,/ BPB: /                     |
| 2334 | 2450  | 70B3 | 04    |        | FCB | 4                                  |
| 2335 | 2451  | 70B4 | 0D    | MSFAT  | FCC | CR,LF,/ FAT: /                     |
| 2336 | 2452  | 70BC | 04    |        | FCB | 4                                  |
| 2337 | 2453  | 70BD | 0D    | MSDIR  | FCC | CR,LF,/ DIR: /                     |
| 2338 | 2454  | 70C5 | 04    |        | FCB | 4                                  |
| 2339 | 2455  | 70C6 | 0D    | MSCL2  | FCC | CR,LF,/ 1stCLUSTER: /              |
| 2340 | 2456  | 70D5 | 04    |        | FCB | 4                                  |
| 2341 | 2457  | 70D6 | 0D    | MSINFN | FCC | CR,LF,CR,LF,/ FileName: /          |
| 2342 | 2458  | 70E5 | 04    |        | FCB | 4                                  |
| 2343 | 2459  | 70E6 | 0D    | MSMSV  | FCC | CR,LF,CR,LF,/ FileName & Param:    |
| 2344 | /     |      |       |        |     |                                    |
| 2345 | 2460  | 70FD | 04    |        | FCB | 4                                  |
| 2346 | 2461  | 70FE | 0D    | MSLDNG | FCC | CR,LF,/ Loading /                  |
| 2347 | 2462  | 7109 | 04    |        | FCB | 4                                  |
|      | 2463  | 710A | 0D    | MSWTNG | FCC | CR,LF,/ Writing /                  |
|      | 2464  | 7115 | 04    |        | FCB | 4                                  |
|      | 2465  | 7116 | 0D    | MSLDED | FCC | CR,LF,/ Load end./                 |
|      | 2466  | 7122 | 04    |        | FCB | 4                                  |
|      | 2467  | 7123 | 0D    | MSWTED | FCC | CR,LF,/ Write end./                |
|      | 2468  | 7130 | 04    |        | FCB | 4                                  |
|      | 2469  | 7131 | 0D    | MSN0FL | FCC | CR,LF,/ Not exist! /               |
|      | 2470  | 713E | 04    |        | FCB | 4                                  |
|      | 2471  | 713F | 0D    | MSNXTF | FCC | CR,LF,"Next page (SP:Next): "      |
|      | 2472  | 7156 | 04    |        | FCB | 4                                  |
|      | 2473  | 7157 | 3E    | MSOVR  | FCC | />64k/                             |
|      | 2474  | 715B | 04    |        | FCB | 4                                  |
|      | 2475  | 715C | 0D    | MSKILL | FCC | CR,LF," Kill OK?(Y/N): "           |
|      | 2476  | 716E | 04    |        | FCB | 4                                  |
|      | 2477  | 716F | 0D    | MSEXST | FCC | CR,LF," Exist! Over Write?(Y/N): " |
|      | 2478  | 718B | 04    |        | FCB | 4                                  |
|      | 2479  | 718C | 0D    | MSMCOF | FCC | CR,LF," Offset -\$2000?(Y/N): "    |
|      | 2480  | 71A4 | 04    |        | FCB | 4                                  |
|      | 2481  | 71A5 | 0D    | MSMCAS | FCC | CR,LF," Store ADR,SIZE: "          |
|      | 2482  | 71B8 | 04    |        | FCB | 4                                  |
| 2483 | ***** |      |       |        |     |                                    |
| 2484 |       |      |       |        |     |                                    |
| 2485 | END   |      | START |        |     |                                    |

0 error(s), 0 warning(s) is detected.

Label address

|        |      |         |      |          |      |        |      |          |      |
|--------|------|---------|------|----------|------|--------|------|----------|------|
| ABLOP  | 69D3 | ACMD41  | 6F56 | AD1      | 6DF8 | AD2    | 6E09 | AD3      | 6E13 |
| AD4    | 6E19 | ADD0    | 6C3E | ADD1     | 6C4C | ADD2   | 6C54 | ADD3     | 6C5C |
| ADD40H | 6C3C | ADD42   | 6DF2 | ADX0     | 6C2D | ADX1   | 6C33 | ADX2     | 6C37 |
| ADX3   | 6C3B | APLUSB  | 6A07 | ASC2BIN  | 6896 | ASC411 | 6218 | ASCMD41  | 620B |
| BASADP | 0033 | BASAEP  | 0035 | BASEXP   | 605D | BI21   | 6D4F | BI22     | 6D55 |
| BUSY   | 6BA5 | CALCHC  | 6D15 | CALCSD   | 6D0A | CALLOP | 6A3D | CEND     | 66D0 |
| CKRS   | 6B9B | CL1     | 6D9F | CLEND    | 6DAF | CLLOP  | 6DAA | CLOP     | 6D7D |
| CLRDIR | 6821 | CLSTH   | 6048 | CLSTL    | 604A | CMD0   | 6F2C | CMD1     | 6F32 |
| CMD17  | 6F38 | CMD24   | 6F3E | CMD55    | 6F50 | CMD58  | 6F4A | CMD8     | 6F44 |
| CMD9   | 6F5C | CMPEND  | 66EC | CMPFN    | 66D1 | CMPLOP | 66D9 | CNT      | 604C |
| CNV1   | 68E6 | CNV2    | 68F2 | CNVBUF   | 607F | CNVINT | 6631 | CNVLOP   | 6631 |
| CNVSUB | 68DE | CR      | 000D | CSRON    | FBF7 | CVBEND | 6DEC | CVBIN    | 6DDA |
| CVERR  | 6DEF | CVT1    | 6DB8 | CVTLEN   | 6067 | DB2    | 6E86 | DB2ASC   | 6E77 |
| DBLOP  | 6E7D | DCLOP1  | 6EED | DELMK    | 00E5 | DIRENT | 601F | DIRGET   | 698A |
| DIROFS | 605B | DLOP1   | 696C | DPGNO    | 605A | DSEND  | 6EDE | DSL0P1   | 6EAD |
| DSL0P2 | 6EC6 | DSPB1   | 6CA1 | DSPB2    | 6CA3 | DSPB3  | 6CB4 | DSPBI0   | 6D3B |
| DSPBI2 | 6D5F | DSPBIN  | 6D35 | DSPBPB   | 63E2 | DSPBUF | 6C97 | DSPC1    | 6473 |
| DSPCKE | 6433 | DSPCLS  | 6EE3 | DSPDIR   | 6406 | DSPED  | 6989 | DSPEND   | 64A9 |
| DSPFAT | 63F4 | DSPFL0  | 6412 | DSPFL1   | 6423 | DSPFL2 | 643D | DSPFL3   | 643F |
| DSPFL4 | 644E | DSPFL5  | 6491 | DSPFL6   | 649D | DSPMBR | 638E | DSPSDDAT |      |
| 62FE   |      |         |      |          |      |        |      |          |      |
| DSPSIZ | 6E92 | DSPYMD  | 6E33 | DSS1     | 6EA0 | DSS12  | 6EB6 | DSS13    | 6EBB |
| DSS14  | 6EC4 | DSS2    | 6ECF | DSS21    | 6ED2 | DSS3   | 6ED6 | DYMD1    | 6E45 |
| DYMD2  | 6E54 | EEND    | 6DD9 | EREND0   | 6DD2 | EREND1 | 6DD4 | ESC      | 001B |
| EXIST  | 694B | EXSTFL  | 6065 | FILBUF   | 6083 | FILL   | 6C24 | FIL0P    | 6C26 |
| FILSIZ | 6046 | FIXPTR  | C730 | FMDAT    | 02FA | FNAME  | 604F | FNIN0    | 6EFD |
| FNMIN  | 6EF6 | GET1    | 6399 | GET2     | 639B | GET8CH | 6D79 | GETBPB   | 63C9 |
| GETD1  | 6CD5 | GETDAT  | 6CD1 | GETFDT   | 66EE | GETMBR | 63AC | GR00     | 6185 |
| GR2580 | 6245 | GR2581  | 625C | GR2582   | 625E | GR410  | 6220 | GR550    | 6201 |
| GR580  | 61D7 | GR581   | 61E1 | GR582    | 61E3 | GR80   | 61A7 | GR81     | 61B9 |
| GR90   | 6311 | GR91    | 631D | GTBL0P   | 63D4 | GTF1   | 670D | GTF2     | 6710 |
| IL0P1  | 6281 | INCHA   | FBB8 | INCX4    | 6C2D | INEND  | 6D8B | INIT     | 60C7 |
| INL10  | 6647 | INL11   | 666B | INL12    | 66B2 | INL2   | 6677 | INLOP    | 6638 |
| INOUTC | 6F12 | INPBUF  | 6077 | INPFLDAT | 687C | INSCT  | 6CEB | ISTT     |      |
| 616B   |      |         |      |          |      |        |      |          |      |
| KILEND | 67B8 | KILFILE | 6764 | KILSUB   | 677D | KL1    | 67A2 | KL2      | 67B0 |
| LD1    | 6740 | LD10    | 65AC | LD2      | 6754 | LD3    | 64CF | LD41     | 64F4 |

|         |      |         |      |        |      |         |      |         |      |
|---------|------|---------|------|--------|------|---------|------|---------|------|
| LD5     | 650D | LD8     | 654F | LD80   | 6594 | LD81    | 6598 | LD82    | 65A1 |
| LD9     | 65AA | LDE0    | 65CF | LDE1   | 65D5 | LDEND   | 65BC | LDFILE  | 64AA |
| LDFL0   | 671C | LDFL1   | 6728 | LDFL5  | 674C | LDFLE   | 6763 | LDL0P   | 651D |
| LDNEND  | 65D9 | LEND    | 6F2B | LF     | 000A | LINBUF  | 043D | LININ   | 6F19 |
| LININP  | D807 | LININT  | FBF1 | LINLOP | 6F21 | LIN01   | 004B | LIN02   | 033A |
| MAIN    | 60A3 | MATCH   | 66E6 | MCADP  | 605F | MCAEP   | 6061 | MCEXP   | 6063 |
| MCFLG   | 604D | MCMD0   | 6F62 | MCMD41 | 6F80 | MCMD55  | 6F78 | MCMD58  | 6F70 |
| MCMD8   | 6F69 | MCMD9   | 6F89 | MEM2ED | 6626 | MEMSET  | 65E1 | MEMSET2 | 6611 |
| MENU    | 60CA | MENU1   | 60E4 | MENU2  | 60ED | MENU3   | 60F6 | MENU4   | 60FF |
| MENU5   | 6108 | MENU6   | 6110 | MENU7  | 6118 | MERR    | 707B | MKF1    | 69A9 |
| MKF2    | 69AD | MKFLDT  | 6990 | MM21   | 6621 | MS0     | 65F8 | MS1     | 65FC |
| MS3     | 660B | MSBPB   | 70AB | MSCL2  | 70C6 | MSDIR   | 70BD | MSERR   | 708C |
| MSEXST  | 716F | MSFAT   | 70B4 | MSINFN | 70D6 | MSINIT  | 7064 | MSKILL  | 715C |
| MSLDED  | 7116 | MSLDNG  | 70FE | MSLOP  | 65FE | MSMCAS  | 71A5 | MSMCOF  | 718C |
| MSMENU1 | 6F90 | MSMENU2 | 6FE9 | MSMSV  | 70E6 | MSNOFL  | 7131 | MSNXTP  | 713F |
| MSOFS   | 709B | MSOVR   | 7157 | MSSCT  | 7046 | MSSD0   | 704E | MSSD1   | 7058 |
| MSWTED  | 7123 | MSWTNG  | 710A | NOEXST | 694E | NOFEND  | 67C6 | NOTM    | 66EA |
| OPMEN10 | 614F | OPMEN6  | 612F | OPMEN7 | 6137 | OPMEN8  | 613F | OPMEN9  | 6147 |
| OPMENU  | 6120 | OUTCHA  | FBBB | OUTSPC | 6D73 | PBCTL   | FD9F | PBDAT   | FD9E |
| PCRLF   | 6D68 | PSEND   | 6E30 | PSLOP  | 6E25 | PSTRNG  | 6E23 | RD1     | 6C74 |
| RD1SCT  | 6C61 | RD2     | 6C7B | RD3    | 6C8B | RDLIN0  | 9162 | RDSCT   | 637B |
| RDTOP   | 6069 | RESDAT  | 6008 | RPT41  | 61EC | RWBUF   | 5E00 | SC01    | 617C |
| SC2581  | 623D | SC551   | 61F9 | SC581  | 61CF | SC80    | 619F | SC91    | 6309 |
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| SCTN0   | 6004 | SCTN1   | 6005 | SCTN2  | 6006 | SCTN3   | 6007 | SCTPCL  | 601E |
| SD1     | 6272 | SDERR   | 6157 | SDHC   | 6009 | SDINIT  | 615E | SDW1    | 6B67 |
| SDW2    | 6B6F | SETCKH  | 6D2C | SETCKL | 6D23 | SETCL   | 6936 | SETFNM  | 6805 |
| SETSCT  | 6CF4 | SETSN0  | 63BC | SETX   | 6925 | SFATED  | 6941 | SFL0P   | 6D98 |
| SKP1    | 62B8 | SKP2    | 62F5 | SNDD1  | 6CBC | SNDDAT  | 6CB5 | SNDDT1  | 6B8A |
| SPC     | 0020 | SRC0    | 6905 | SRC00  | 6942 | SRCDIR  | 6951 | SRCDLM  | 6889 |
| SRCFAT  | 68F9 | SRCFL0  | 6960 | SRCFL1 | 6963 | SRCFL5  | 697E | SRCL0P  | 690E |
| SRDLM1  | 688B | SRL0P1  | 68A7 | SRL0P2 | 68BF | START   | 6000 | STDATE  | 69CA |
| STTADR  | 6000 | SVTOP   | 606B | TPGFLG | 604E | TRANS   | 6C19 | TRL0P   | 6C1B |
| TXTADR  | 2000 | TXTBUF  | 033C | VN     | 6003 | WFSIZ   | 6042 | WK0     | 6075 |
| WK1     | 606D | WK2     | 606F | WK3    | 6071 | WK4     | 6073 | WMDY    | 603F |
| WR2     | 6A78 | WR3     | 6AA2 | WR4    | 6AD8 | WR5     | 6ADA | WR6     | 6ADC |
| WRB0    | 67CD | WRB1    | 67EB | WRBEND | 6804 | WRBFILE | 67B9 | WRDL0P  | 6BC3 |
| WRDSET  | 6B47 | WRDSKP  | 6BC8 | WREND  | 6AE7 | WRF1    | 6BFC | WRF2    | 6BFF |

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## SDRWFM

I use this card to create programs for the FM-7 using a familiar editor on my Windows PC and transfer them to the FM-7, or conversely, to save programs running on the FM-7 to my Windows PC. I feel it's a fairly useful tool.

Also, according to an idea from an acquaintance who tested it, even though folders on the SD card are not visible to the FM-7, you could use this to your advantage by putting multiple FM-7 files inside a folder (hiding them), and then taking the necessary files out of the folder when needed to use them with the FM-7.

I've also revised the instructions for use, so I'll upload them all together to [OneDrive](#).